Symplectic Topology

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I. Foundations

Material from the book "Introduction to Symplectic Topology" *by Dusa McDuff & Dietmar Salamon*

1 Symplectic Linear Algebra

1.1 Symplectic spaces

Definition 1.1.1.1

Let V be a finite-dimensional vector space. The **kernel** of a 2-form $\omega \in \Lambda^2(V^*)$ is the subspace

$$\ker \omega = \{v \in V \mid \omega(v, \bullet) = 0\}.$$

A 2-form is **nondegenerate** if its kernel is zero.

A nondegenerate 2-form is called a **symplectic form** and the pair (V, ω) a **symplectic vector space**.

Example 1.1.1.2

The archetypal symplectic vector space is $\mathbb{R}^{2n}$ equipped with the 2-form

$$\omega_0 = \sum_{i=1}^{n} dx_i \wedge dy_i$$

where $x_1, \dots, x_n, y_1, \dots, y_n$ forms the standard basis and dx_i, dy_i form its dual basis.

It is clear that a 2-form ω is nondegenerate iff its representation matrix relative to any basis is non-singular. The representation matrix relative to $B = (v_1, \dots, v_n)$ is $\omega_{ij} = \omega(v_i, v_j)$. Equivalently, if the induced map $\widehat{\omega}\colon V \to V^*$ given by $\widehat{\omega}(v)(u) = \omega(v, u)$ is injective, equivalently an isomorphism.

Definition 1.1.1.3

Let (V, ω_V) and (W, ω_W) be two symplectic spaces. A **symplectomorphism** is a linear isomorphism $\varphi\colon V \to W$ such that the pullback of ω_W is ω_V: $\omega_V = \varphi^*\omega_W$. Recall that the pullback is defined by

$$(\varphi^*\omega_W)(v, u) = \omega_W(\varphi v, \varphi u).$$

It is clear that the composition and inverse of symplectomorphisms are symplectomorphisms, so the category of symplectic spaces with symplectomorphisms forms a groupoid.

The collection of symplectomorphisms $V \to V$ is a group denoted by $\mathrm{Sp}(V, \omega)$.

Example 1.1.1.4

In the case of the standard symplectic structure on $\mathbb{R}^{2n}$, we use the notation $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$. Notice that the representation of ω_0 relative to the standard basis is

$$J_0 = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Thus $\mathrm{Sp}(2n)$ can be viewed as the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ defined by

$$\mathrm{Sp}(2n) = \left\{ A \in \mathrm{GL}_{2n}(\mathbb{R}) \;\middle|\; A^\top J_0 A = J_0 \right\}.$$

Proposition 1.1.1.5

$\mathrm{Sp}(2n)$ is a Lie group of dimension $2n^2+n$ with Lie algebra

$$\mathfrak{sp}(2n) = \{-J_0 S \mid S \in \mathrm{Mat}_{2n} \text{ symmetric}\}.$$

Proof

Let $\mathfrak{so}(2n) \leq \mathrm{Mat}_{2n}$ be the linear subspace of skew-symmetric matrices, and define the map

$$f\colon \mathrm{Mat}_{2n} \to \mathfrak{so}(2n), \qquad f(A) = A^\top J_0 A.$$

Then $\mathrm{Sp}(2n) = f^{-1}J_0$, so to show that $\mathrm{Sp}(2n)$ is a Lie subalgebra of $\mathrm{GL}_{2n}(\mathbb{R})$, it is sufficient to show that J_0 is a regular value of f.
Note that f's differential at $A \in \mathrm{Mat}_{2n}$ is

$$df_A(X) = X^\top J_0 A + A^\top J_0 X.$$

If $f(A) = J_0$ and $B \in \mathfrak{so}(2n)$ (which is equal to $\mathfrak{so}(2n)$'s tangent space), then $X = -\frac{1}{2}AJ_0B$ is a preimage of B. Thus df_A is surjective so that J_0 is a regular value as desired. And we see that

$$\dim \mathrm{Sp}(2n) = \dim \mathrm{Mat}_{2n} - \dim \mathfrak{so}(2n) = (2n)^2 - \frac{(2n)^2 - 2n}{2} = 2n^2 + n.$$

Now,

$$\mathfrak{sp}(2n) = T_I \mathrm{Sp}(2n) = \ker df_I = \left\{X \in \mathrm{Mat}_{2n} \;\middle|\; X^\top J_0 + J_0 X = 0\right\}.$$

Note that if $X \in \mathfrak{sp}(2n)$ then $X = -J_0 S$ where $S = J_0 X$, and this is symmetric:

$$S^\top = -X^\top J_0 = J_0 X = S.$$

Conversely for S symmetric, $-J_0 S \in \mathfrak{sp}(2n)$. ∎

Definition 1.1.1.6

Let (V, ω) be a symplectic space and $W \leq V$ a subspace. Its **symplectic complement** is the subspace

$$W^\omega = \{v \in V \mid \omega(v, w) = 0 \text{ for all } w \in W\}.$$

Note that unlike the case for inner products, W^ω and W need not be transversal. The subspace W is called

(1) **isotropic** if $W \leq W^\omega$,

(2) **coisotropic** if $W^\omega \leq W$,

(3) **Lagrangian** if $W = W^\omega$,

(4) **symplectic** if $W \cap W^\omega = 0$.

Note that W is isotropic iff ω vanishes on W, and symplectic iff $(W, \omega|_W)$ is a symplectic space.

Lemma 1.1.1.7

For a subspace $W \leq V$,

$$\dim W + \dim W^\omega = \dim V, \qquad W^{\omega\omega} = W.$$

Proof

Consider $\widehat{\omega}\colon V \to V^*$ which is an isomorphism by non-degeneracy. Let us define

$$\operatorname{ann} W = \{\varphi \in V^* \mid \varphi(W) = 0\}.$$

Because $\widehat{\omega}$ is an isomorphism, we have

$$\operatorname{ann} W = \{\widehat{\omega}(v) \mid \widehat{\omega}(v)(W) = 0\} = \widehat{\omega}(W^\omega)$$

as desired.

Notice that a functional in $\operatorname{ann} W$ is the same as a functional over V/W. Indeed, $\varphi \in \operatorname{ann} W$ iff $W \leq \ker \varphi$ iff we can quotient φ by W. Thus $\operatorname{ann} W \cong (V/W)^*$ which has dimension $\dim V - \dim W$ as desired. ■

Let V be a $2n$-dimensional vector space with basis $u_1, \ldots, u_n, v_1, \ldots, v_n$. Then we can define a symplectic form on V by

$$\omega = \sum_{i=1}^{n} du_i \wedge dv_i,$$

equivalently

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \quad \omega(u_i, v_j) = \delta_{ij} \qquad \iff \qquad [\omega] = -J_0.$$

Conversely any basis of a symplectic space which represents the symplectic form like this is called a **symplectic basis**.

Notice that the existence of a symplectic basis is equivalent to the existence of a symplectomorphism $(\mathbb{R}^{2n}, \omega_0) \to (V, \omega)$.

Theorem 1.1.1.8

Let (V, ω) be a symplectic space. Then V has even dimension, $\dim V = 2n$, and there exists a symplectic basis.

Proof

We induct on $m = \dim V$. For $m = 0$ this is trivial, so suppose $m > 0$. Because ω is non-degenerate there are $u_1, v_1 \in V$ such that $\omega(u_1, v_1) \neq 0$. We can normalize this and assume $\omega(u_1, v_1) = 1$.

Because ω is alternating, this means that u_1, v_1 are linearly independent. Furthermore the subspace W spanned by u_1, v_1 is symplectic as ω is nondegenerate there. Thus $V = W \oplus W^\omega$, and by induction W^ω has even dimension and a symplectic basis. By adding u_1, v_1 to W^ω's symplectic basis we get a symplectic basis of V, and V has even dimension. ■

An immediate corollary of this theorem is the following.

Corollary 1.1.1.9

Two symplectic spaces V, W are symplectomorphic iff $\dim V = \dim W$.

Corollary 1.1.1.10

Let V be a $2n$-dimensional vector space and ω a 2-form on V. Then ω is nondegenerate iff its n-fold exterior power ω^n is nonzero.

Proof

If ω is degenerate then there is a $v \neq 0$ such that $v \lrcorner \omega = 0$. Because interior multiplication is an antiderivation:

$$v \lrcorner \omega^n = \omega \wedge (v \lrcorner \omega^{n-1}) + (v \lrcorner \omega) \wedge \omega^{n-1} = \omega \wedge (v \lrcorner \omega^{n-1}).$$

Thus inductively we have $v \lrcorner \omega^n = 0$. Because ω^n is a top-form this means $\omega^n = 0$ (we can see this by extending v to a basis, then ω on that basis is zero).

Conversely if ω is nondegenerate, then by the previous theorem there is a symplectic basis

$$\omega = \sum_i dx_i \wedge dy_i.$$

Then

$$\omega^n = \sum_I dx_{i_1} \wedge dy_{i_1} \wedge \cdots \wedge dx_{i_n} \wedge dy_{i_n},$$

and we may remove Is with repeated indices, so the sum is over permutations of $1, \ldots, n$. Each permutation gives the same product as 2-forms commute, thus

$$\omega^n = n!\, dx_1 \wedge dy_1 \wedge \cdots dx_n \wedge dy_n \neq 0.$$ ■

Thus ω^n is an orientation form of V when ω is a symplectic form. In the proof above we showed that

$$\omega^n = n!\, dx_1 \wedge dy_1 \wedge \cdots dx_n \wedge dy_n,$$

which means that it corresponds to the oriented basis $(x_1, y_1, \ldots, x_n, y_n)$.

Proposition 1.1.1.11

The Lagrangian subspaces of V are precisely the maximal isotropic subspaces. Moreover, every basis of a Lagrangian subspace can be extended to a symplectic basis of V.

Proof

If W is isotropic then $W \leq W^\omega$ meaning $\dim W \leq \frac{1}{2} \dim V$. Since Lagrangian subspaces have dimension equal to $\frac{1}{2} \dim V$, Lagrangian subspaces are maximal. Conversely if W if isotropic and $W \neq W^\omega$, then there is a $v \in W^\omega - W$. Adding v to W results in an isotropic subspace as ω vanishes on $W + v$. Thus maximal isotropic subspaces must have $W = W^\omega$ so that W is Lagrangian.

For the second point it is sufficient to consider $V = \mathbb{R}^{2n}$ with the standard symplectic structure. Let $\Lambda \leq V$ be Lagrangian, then $\Lambda' = J_0\Lambda$ is symplectic as well. Notice that $\Lambda' \cap \Lambda = 0$ because for $u \in \Lambda$ we have $\omega_0(u, J_0 u) = -u^\top u \neq 0$. Thus $\mathbb{R}^{2n} = \Lambda' \oplus \Lambda$.

Let $u_1, \ldots, u_n$ be a basis for Λ. Consider the map $\widehat{\omega}\colon \Lambda' \to \Lambda^*$. This is injective as its kernel is $\Lambda \cap \Lambda^\omega = 0$, and thus defines an isomorphism. Let $\varphi_1, \ldots, \varphi_n$ be the dual basis of $u_1, \ldots, u_n$, and thus there are $v_1, \ldots, v_n \in \Lambda'$ such that $\varphi_i = \widehat{\omega}(v_i)$.

Because Λ, Λ' are Lagrangian we have that

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \qquad \omega(v_i, u_j) = \varphi_i(u_j) = \delta_{ij}$$

as desired. ■

Proposition 1.1.1.12

Let $W \leq V$ be coisotropic.

(1) The quotient $\overline{W} = W/W^\omega$ has a unique symplectic structure $\overline{\omega}$ whose pullback under projection $W \to \overline{W}$ agrees with $\omega|_W$.

(2) If $\Lambda \leq V$ is Lagrangian, then

$$\overline{\Lambda} = \frac{(\Lambda \cap W) + W^\omega}{W^\omega} \leq \overline{W}$$

is Lagrangian.

Proof

Let $[w] \in \overline{W}$ denote the coset of w. Notice that $\overline{\omega}$ must satisfy

$$\omega(w, w') = \overline{\omega}([w], [w']),$$

which uniquely determines $\overline{\omega}$, showing uniqueness. Because W is coisotropic, W^ω is an isotropic subspace of W, and so this is well-defined: for $v, v' \in W^\omega$ and $w, w' \in W$,

$$\omega(w + v, w' + v') = \omega(w, w') + \omega(v, w') + \omega(w, v') + \omega(v, v') = \omega(w, w').$$

The middle two terms are zero because $v, v' \in W^\omega$, and $v \in W^\omega \leq W$ so the last term is zero. Furthermore $\overline{\omega}$ is nondegenerate as if $\overline{\omega}([w], \bullet) = 0$, then $w \in W^\omega$ so $[w] = 0$. So we have shown (1).

For (2) we show that

$$\widetilde{\Lambda} = (\Lambda \cap W) + W^\omega \leq V$$

is Lagrangian. Indeed,

$$\widetilde{\Lambda}^\omega = (\Lambda \cap W)^\omega \cap W = (\Lambda^\omega + W^\omega) \cap W = \Lambda \cap W + W^\omega = \widetilde{\Lambda}.$$

And $\widetilde{\Lambda}$ quotients to $\overline{\Lambda}$, and $\widetilde{\Lambda}^\omega$ quotients to $\overline{\Lambda}^\omega$, which are then equal. ■

1.2 The symplectic group

Recall that we have defined a groupoid whose objects are symplectic spaces, and whose isomorphisms are the symplectomorphisms. And corollary 1.1.1.9 says that the dimension of a symplectic space is its only significant invariant. Let us denote by

$$\mathrm{Sp}(V, \omega) = \{\Psi \in \mathrm{GL}(V) \mid \Psi^* \omega = \omega\}$$

the automorphism group of (V, ω), called the **symplectic group**.

By proposition 1.1.1.5, $\mathrm{Sp}(V, \omega)$ is a Lie group. Following this proposition, we have the following result.

Theorem 1.1.2.1

The symplectic group $\mathrm{Sp}(V, \omega)$ is a Lie group with Lie algebra

$$\mathfrak{sp}(V, \omega) = \{T \in \mathrm{end}(V) \mid \omega(A\bullet, \bullet) + \omega(\bullet, A\bullet) = 0\}.$$

We can understand this group by focusing on the case of $(\mathbb{R}^{2n}, \omega_0)$. Then $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$ may be thought of as a Lie group of matrices $\Psi \in \mathrm{Mat}_{2n}$ satisfying $\Psi^\top J_0 \Psi = J_0$. Such matrices are called **symplectic**.

Lemma 1.1.2.2

Symplectic matrices have determinant one.

Proof

By corollary 1.1.1.10, ω_0^n is nonzero and thus a volume form on $\mathbb{R}^{2n}$. Furthermore a symplectic matrix Ψ satisfies $\Psi^*\omega_0 = \omega_0$ and thus $\Psi^*\omega_0^n = \omega_0^n$. Because this is a volume form, $\det \Psi = 1$. ■

We identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ by identifying $(x, y) \in \mathbb{R}^{2n}$ with $x + iy \in \mathbb{C}^n$. Then J_0 corresponds to multiplication by i in $\mathbb{C}^n$. With this identification, $\mathrm{GL}_n(\mathbb{C})$ is the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ consisting of matrices which commute with J_0:

$$\mathrm{GL}_n(\mathbb{C}) = \{A \in \mathrm{GL}_{2n}(\mathbb{R}) \mid J_0 A = A J_0\}.$$

In this identification, a matrix $X + iY \in \mathrm{Mat}_n(\mathbb{C})$ corresponds to the matrix

$$X + iY \in \mathrm{Mat}_n(\mathbb{C}) \leftrightarrow \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Mat}_{2n})(\mathbb{R}).$$

Consider the determinant of these matrices. Letting

$$P = \begin{pmatrix} I & iI \\ I & -iI \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} I & I \\ -iI & iI \end{pmatrix},$$

then

$$P \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} P^{-1} = \begin{pmatrix} X + iY & \\ & X - iY \end{pmatrix},$$

so that

$$\det \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} = \det(X + iY)\det(X - iY) = \det(X + iY) \cdot \overline{\det(X + iY)} = |\det(X + iY)|^2.$$

Notice how the conjugate transpose acts on $\mathrm{Mat}_n(\mathbb{C})$:

$$(X + iY)^\dagger = (X - iY)^\top = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix} = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^\top.$$

Thus $X + iY \in \mathrm{U}(n)$ iff

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^{-1} = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix},$$

which is iff

$$XX^\top + YY^\top = I, \qquad XY^\top = YX^\top.$$

Notice that

$$i(X + iY)^\dagger (X + iY) = i \iff (X + iY)^\dagger = (X + iY)^{-1} \iff X + iY \in \mathrm{U}(n).$$

Thus we see

$$\mathrm{U}(n) = \mathrm{GL}_n(\mathbb{C}) \cap \mathrm{Sp}(2n).$$

In fact we can extend this result.

Lemma 1.1.2.3

$$\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{O}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n).$$

Proof

A matrix $\Psi \in \mathrm{GL}_{2n}(\mathbb{R})$ satisfies

$$\Psi \in \mathrm{GL}_n(\mathbb{C}) \iff \Psi J_0 = J_0 \Psi, \qquad \Psi \in \mathrm{Sp}(2n) \iff \Psi^\top J_0 \Psi = J_0, \qquad \Psi \in \mathrm{O}(2n) \iff \Psi^\top \Psi = I.$$

Note that any of these two conditions imply the third, so we have the first two equalities. By the above discussion, we also get the final. ■

For a linear operator Ψ we denote its spectrum by $\sigma(\Psi)$; this is its set of eigenvalues.

Lemma 1.1.2.4

Let $\Psi \in \mathrm{Sp}(2n)$, then $\lambda \in \sigma(\Psi)$ iff $\lambda^{-1} \in \sigma(\Psi)$, and the multiplicities of λ and λ^{-1} are equal. If ± 1 is an eigenvalue of Ψ then it has even multiplicity.

Finally, if $\lambda, \lambda' \in \sigma(\Psi)$ are two (not necessarily distinct) eigenvalues such that $\lambda\lambda' \neq 1$, then $\omega_0(z, z') = 0$ for any eigenvectors of eigenvalues λ, λ' respectively.

Proof

The first statement is due to the fact that Ψ^{-1} is similar to $\Psi^\top$:

$$\Psi^\top = J_0 \Psi^{-1} J_0^{-1},$$

Ψ and $\Psi^\top$ share the same eigenvalues (with the same multiplicities), and λ is an eigenvalue of Ψ iff λ^{-1} is of Ψ^{-1} (and the multiplicities agree).

Thus the total multiplicity of the eigenvalues not equal to ± 1 is even (as they come in pairs). The determinant of Ψ is the product of eigenvalues, as they come in pairs with their inverses this is equal to $(-1)^m$ where m is -1's multiplicity. By lemma 1.1.2.2 this is equal to 1 so m is even. Because Ψ has dimension $2n$, the multiplicity of 1 must be even as well.

The last statement is because

$$\omega_0(z, z') = \omega_0(\Psi z, \Psi z') = \lambda\lambda' \omega_0(z, z')$$

and since $\lambda\lambda' \neq 1$, $\omega_0(z, z') = 0$. ■

Suppose D is a diagonal matrix, $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, then define

$$D^\alpha = \mathrm{diag}(\lambda_1^\alpha, \ldots, \lambda_n^\alpha)$$

for any real $\alpha \in \mathbb{R}$ where this is defined. Suppose $A \in \mathrm{Mat}_n(\mathbb{R})$ is a diagonalizable matrix with $A = P^{-1} D P$, D diagonal. Then define

$$A^\alpha = P^{-1} D^\alpha P$$

for real α where this is defined. Note that this agrees with A^n for n natural. It is easy to see that this is well-defined (i.e. not dependent on P, D).

If A^α exists and $Av = \lambda v$, then $A^\alpha v = \lambda^\alpha v$. This is clear for diagonal A, and extends to diagonalizable A by extending v to an eigenbasis and using that P.

Note that if A's eigenvalues are all positive, then this is defined for all α. In particular if S is symmetric and positive definite, then its powers exist for all real α.

Lemma 1.1.2.5

Exponentiation is continuous as a map

$$\mathrm{Sym}_+ \times \mathbb{R} \to \mathrm{Sym}_+, \qquad (S, \alpha) \mapsto S^\alpha$$

where $\mathrm{Sym}_+ \subseteq \mathrm{Mat}_n$ is the subspace of positive-definite symmetric matrices.

Proof

Let $S \in \mathrm{Sym}_+$, and let its minimum and maximum eigenvalues by $\lambda_{\min}, \lambda_{\max}$ respectively. Then we know for all vectors v,

$$\lambda_{\min} |v|^2 \leq v^\top S v \leq \lambda_{\max} |v|^2$$

and these bounds are tight. We claim that there is an open neighborhood $S \in U \subseteq \mathrm{Sym}_+$ and numbers $0 < m < M$ such that every $A \in U$ has eigenvalues bound by m, M.

Let $m = \lambda_{\min}/2$, then for every symmetric matrix A and v of unit norm:

$$v^\top A v = v^\top S v + v^\top (A - S) v \geq \lambda_{\min} - \left| v^\top (A - S) v \right| \geq \lambda_{\min} - |(A - S)v| \geq \lambda_{\min} - |A - S|$$

where $|A - S|$ denotes the operator norm. Thus if $|A - S| < m$ then the right-hand side is equal to $\lambda_{\min}/2$ and thus $v^\top A v \geq \lambda_{\min}/2$; so all of A's eigenvalues are bound from below by m. So all matrices in the ball of radius m around S have eigenvalues bound from below by m.

We can bound this from above too: for unit norm v

$$v^\top A v \leq |A| \leq |S| + |A - S|,$$

and so if we require $|A - S| < 1$ then

$$v^\top A v \leq |S| + 1 = \lambda_{\max} + 1,$$

so matrices in the ball of radius 1 around S have eigenvalues bound from above by $M = \lambda_{\max} + 1$. So the desired open set is

$$U = \{A \in \mathrm{Sym}_+ \mid |A - S| < 1, m\}.$$

Instead of considering exponentiating over arbitrary α, we show it is continuous for $a \leq \alpha \leq b$ for any arbitrary a, b; this is clearly sufficient. Consider the map $f : [m, M] \times [a, b] \to \mathbb{R}$ by $f(x, t) = t^x$, this is continuous as $m > 0$. By Stone-Weirestrass, because the codomain is compact, we can approximate f uniformly by a sequence of polynomials $p_k \in \mathbb{R}[x, t]$.

Now suppose $A \in U$, so $A = P^{-1} \operatorname{diag}(\lambda_1, \ldots, \lambda_n) P$, and because p_k are polynomial

$$p_k(x, A) = P^{-1} \operatorname{diag}(p_k(x, \lambda_1), \ldots, p_k(x, \lambda_n)) P.$$

Thus

$$A^\alpha - p_k(\alpha, A) = P^{-1} \operatorname{diag}(\ldots, \lambda_i - p_k(\alpha, \lambda_i), \ldots) P.$$

Because $A \in U$ the λ_i fall in $[m, M]$, and so under the operator norm

$$|A^\alpha - p_k(\alpha, A)| = \max_i |\lambda_i - p_k(\alpha, \lambda_i)| \leq \max_{t \in [m, M]} |t^\alpha - p_k(\alpha, t)|.$$

And so we see that the map $F : U \times [a, b] \to \mathrm{Sym}_+$, $F(A, \alpha) = A^\alpha$, satisfies for all A, α

$$|F(A, \alpha) - p_k(\alpha, A)| \leq \max_{t \in [m, M], \alpha \in [a, b]} |t^\alpha - p_k(\alpha, t)|,$$

so that $(\alpha, A) \mapsto p_k(\alpha, A)$ uniformly converges to F. Because p_k are all polynomial and thus continuous, so too is F. ∎

Lemma 1.1.2.6

If $S \in \mathrm{Sp}(2n)$ is symmetric and positive definite, then $P^\alpha \in \mathrm{Sp}(2n)$ for all $\alpha \in \mathbb{R}$.

Proof

As S is symmetric and positive definite we can decompose $\mathbb{R}^{2n}$ into eigenspaces

$$\mathbb{R}^{2n} = \bigoplus_{i=1}^{k} E_i, \qquad E_i = \ker(\lambda_i I - S),$$

for $\lambda_i > 0$ the eigenvalues. For $z, w \in \mathbb{R}^{2n}$ let their projections into E_i be denoted z_i, w_i respectively. Then because S is symplectic

$$\omega_0(z_i, w_j) = \omega_0(Pz_i, Pw_j) = \lambda_i\lambda_j\omega_0(z_i, w_j).$$

So either $\lambda_i\lambda_j = 1$ or $\omega_0(z_i, w_j) = 0$, and thus Then

$$\omega_0(P^\alpha z, P^\alpha w) = \sum_{i,j}^{k} (\lambda_i\lambda_j)^\alpha \omega_0(z_i, w_j) = \sum_{i,j}^{k} \omega_0(z_i, w_j) = \omega_0(z, w)$$

so that $P^\alpha \in \mathrm{Sp}(2n)$. ■

Proposition 1.1.2.7

(1) $\mathrm{U}(n)$ is a maximal compact subgroup of $\mathrm{Sp}(2n)$.

(2) The inclusion $\mathrm{U}(n) \to \mathrm{Sp}(2n)$ is a strong deformation retract. In particular $\mathrm{Sp}(2n)$ is connected.

Proof

We first prove (2). Define

$$H\colon [0,1] \times \mathrm{Sp}(2n) \to \mathrm{Sp}(2n), \qquad H(t, \Psi) = \Psi(\Psi^\top\Psi)^{-t/2}.$$

This is continuous by lemma 1.1.2.5, as $\Psi^\top\Psi$ is positive definite for any non-singular Ψ. This is well-defined by the previous lemma, $H(\Psi, t) \in \mathrm{Sp}(2n)$ for all t.

Now

$$H_0 = \mathrm{id}, \qquad H_t|_{\mathrm{U}(n)} = \mathrm{id}$$

where the latter is because $\Psi^\top\Psi = I$ for unitary Ψ. And for $\Psi \in \mathrm{Sp}(2n)$, $H_1(\Psi)$ is orthogonal:

$$H_1(\Psi)^\top H_1(\Psi) = (\Psi^\top\Psi)^{-1/2}\Psi^\top\Psi(\Psi^\top\Psi)^{-1/2} = I.$$

Because $\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{U}(n)$, we have $H_1(\mathrm{Sp}(2n)) = \mathrm{U}(n)$. Thus H is indeed a strong deformation retract. And so $\mathrm{Sp}(2n)$ and $\mathrm{U}(n)$ are homotopy equivalent, and $\mathrm{U}(n)$ is connected so therefore $\mathrm{Sp}(2n)$ is as well.

We show (1) now. Suppose G is a compact subgroup of $\mathrm{Sp}(2n)$ containing $\mathrm{U}(n)$, but not equal to $\mathrm{U}(n)$. Then let $\Psi \in G - \mathrm{U}(n)$, and so $H_1(\Psi) = \Psi(\Psi^\top\Psi)^{-1/2} \in \mathrm{U}(n) \subseteq G$. Thus the positive definite symmetric matrix $P = \Psi^\top\Psi$ belongs to $G - \mathrm{U}(n)$. Therefore it must have an eigenvalue greater than 1: its eigenvalues come in pairs λ, λ^{-1}, so otherwise $P = I$. But then the sequence $P, P^2, P^3, \ldots$ in G has no convergent subsequence, contradicting G's compactness. ■

Remark 1.1.2.8

Every symplectic matrix $\Psi \in \mathrm{Sp}(2n)$ can be uniquely written as $\Psi = UP$ where $U \in \mathrm{U}(n)$ and P is a symmetric, positive definite, symplectic matrix. Existence is by

$$U = \Psi(\Psi^\top\Psi)^{-1/2}, \qquad P = (\Psi^\top\Psi)^{1/2}.$$

Uniqueness is because the only symmetric positive definite unitary matrix is I (because its eigenvalues come in pairs, and thus must all be 1). If $U \in \mathrm{U}(n)$ can be written as $U = U'P'$, then $P = U(U')^{-1}$ is unitary as well, and thus $P = I$ so $U = U'$. So this is true for unitary matrices, and thus if $UP = U'P'$ we have $U = U'P'P^{-1}$ so $U = U'$ and $P'P^{-1} = I$ meaning $P = P'$. This shows uniqueness.

This decomposition is called **symplectic polar decomposition**.

Proposition 1.1.2.9

The complex determinant map $\det_{\mathbb{C}}\colon \mathrm{U}(n) \to S^1$ induces an isomorphism of fundamental groups $\pi_1(\mathrm{U}\, n) \cong \pi_1(S^1) \cong \mathbb{Z}$.

Proof

The determinant map is a fibration with typical fiber $\mathrm{SU}(n)$. Indeed for every $e^{i\theta} \in S^1$ we can consider the open subset W of S^1 consisting of numbers whose arguments are between $\theta - \varepsilon$ and $\theta + \varepsilon$. Then $\det_{\mathbb{C}}^{-1}(W) \to \mathrm{SU}(n) \times W$ given by $\Psi \mapsto (e^{i\varphi/n}\Psi, e^{i\varphi})$ where $e^{i\varphi} = \det \Psi$ is a homeomorphism with inverse $(\Phi, e^{i\varphi}) \to e^{i\varphi/n}\Phi$. It also commutes with $\det_{\mathbb{C}}$ because $\det(e^{i\varphi/n}\Phi) = e^{i\varphi}$.

Thus there is an exact sequence

$$\pi_1(\mathrm{SU}(n)) \to \pi_1(\mathrm{U}\, n) \to \pi_1(S^1) \to \pi_0(\mathrm{SU}(n)).$$

It is sufficient then to show that $\mathrm{SU}(n)$ is simply connected, and we get the desired result.

It is trivial that $\mathrm{SU}(n)$ is simply connected for $n = 1$, as $\mathrm{SU}(1) = \{1\}$. For $n \geq 2$ the map $\mathrm{SU}(n) \to S^{2n-1}$ which maps $U \in \mathrm{SU}(n)$ to its first column is a fibration with typical fiber $\mathrm{SU}(n-1)$. And so we have an exact sequence

$$\pi_1(\mathrm{SU}(n-1)) \to \pi_1(\mathrm{SU}\, n) \to \pi_1(S^{2n-1}) \to \pi_0(\mathrm{SU}(n-1)).$$

By induction this means $\pi_1(\mathrm{SU}\, n) \cong \pi_1(S^{2n-1}) = 1$ because $2n - 1 > 1$. ■

Because $\mathrm{U}(n)$ is a deformation retract of $\mathrm{Sp}(2n)$, they have the same fundamental group so $\pi_1(\mathrm{Sp}(2n)) \cong \mathbb{Z}$. We now construct an explicit isomorphism.

Recall that the **degree** of a loop $\gamma\colon \mathbb{R}/\mathbb{Z} \to S^1$ is defined to be its value in $\mathbb{Z}$ under the usual isomorphism $\pi_1(S^1) \cong \mathbb{Z}$. Equivalently, we lift γ to $\widetilde{\gamma}\colon [0,1] \to \mathbb{R}$ (because covering spaces have the path lifting property), and define

$$\deg \gamma = \widetilde{\gamma}(1) - \widetilde{\gamma}(0).$$

Theorem 1.1.2.10

There exists a unique function μ, called the **Maslov index**, which assigns an integer $\mu(\Psi) \in \mathbb{Z}$ to every loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$ and satisfies the following axioms:

(1) Homotopy: two loops are homotopic iff they have the same Maslov index.

(2) Product: the Maslov index of the product of two loops is the sum of their Maslov indices:

$$\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2).$$

Note here we are taking the pointwise product of loops of matrices: $(\Psi_1\Psi_2)(t) = \Psi_1(t)\Psi_2(t)$.

(3) Direct sum: if $n = n' + n''$, we may identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ as a subgroup of $\mathrm{Sp}(2n)$. Then for loops Ψ', Ψ'' in $\mathrm{Sp}(2n'), \mathrm{Sp}(2n'')$ respectively,

$$\mu(\Psi' \oplus \Psi'') = \mu(\Psi') + \mu(\Psi'').$$

(4) Normalization: the loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2)$ defined by

$$\Psi(t) = \begin{pmatrix} \cos(2\pi t) & -\sin(2\pi t) \\ \sin(2\pi t) & \cos(2\pi t) \end{pmatrix}$$

has Maslov index 1.

Note:
We identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ with a subgroup of $\mathrm{Sp}(2n)$ by

$$\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \oplus \begin{pmatrix} X' & Y' \\ Z' & W' \end{pmatrix} = \left(\begin{array}{cc|cc} X & & Y & \\ & X' & & Y' \\ \hline Z & & W & \\ & Z & & W' \end{array}\right).$$

Proof

Define $\rho\colon \mathrm{Sp}(2n) \to S^1$ by mapping $\Psi \in \mathrm{Sp}(2n)$ (not a loop) to

$$\rho(\Psi) = \det_{\mathbb{C}}(X + iY), \qquad \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} = \Psi(\Psi^\top \Psi)^{-1/2} \in \mathrm{U}(n).$$

Then define the Maslov index to be

$$\mu(\Psi) = \deg(\rho \circ \Psi).$$

Equivalently $\mu(\Psi) = \alpha(1) - \alpha(0)$ where $\alpha\colon I \to \mathbb{R}$ is a continuous lift of $\rho \circ \Psi$, meaning

$$\exp(2\pi i \alpha(t)) = \det_{\mathbb{C}}(X(t) + iY(t)).$$

Clearly this definition depends only on the homotopy class of Ψ, and by above we know $\rho\colon \mathrm{Sp}(2n) \to S^1$ defines an isomorphism of the fundamental groups. This isomorphism is μ, so we have axiom (1). The product axiom is true for loops of unitary matrices, because $\rho(\Psi) = \det_{\mathbb{C}} \Psi$, so if α_i is a lift of $\rho\Psi_i$ then

$$\det_{\mathbb{C}}(\Psi_1\Psi_2) = \det_{\mathbb{C}} \Psi_1 \cdot \det_{\mathbb{C}} \Psi_2 = \exp(2\pi i \alpha_1(t)) \exp(2\pi i \alpha_2(t)) = \exp(2\pi i(\alpha_1 + \alpha_2)(t)).$$

Thus $\alpha_1 + \alpha_2$ is a lift of $\rho(\Psi_1\Psi_2)$, so $\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2)$. Because a loop of symplectic matrices is homotopic to a loop of unitary matrices, we have the desired result.

The last two axioms are easily verified. Thus the existence is seen, and uniqueness can be shown as well. ■

1.3 Lagrangian subspaces

For a symplectic space (V, ω), let $\mathcal{L}(V, \omega)$ denote the set of its Lagrangian subspaces. We denote

$$\mathcal{L}(n) = \mathcal{L}(\mathbb{R}^{2n}, \omega_0).$$

We can characterize the Lagrangian subspaces of $\mathbb{R}^{2n}$ more explicitly. Firstly a Lagrangian subspace of $\mathbb{R}^{2n}$ has dimension n, and is thus the image of some full-rank linear transform $\mathbb{R}^n \to \mathbb{R}^{2n}$, or equivalently a $2n \times n$-matrix of rank n. We can then determine when such an image is Lagrangian.

Proposition 1.1.3.1

Let $X, Y \in \mathrm{Mat}_n$ be matrices, and define $\Lambda \leq \mathbb{R}^{2n}$ by

$$\Lambda = \operatorname{im} Z, \qquad Z = \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Then $\Lambda \in \mathcal{L}(n)$ iff Z has rank n and $X^\top Y = Y^\top X$.
In particular the graph $\Lambda = \{(x, Ax) \mid x \in \mathbb{R}^n\}$ of some matrix $A \in \mathrm{Mat}_n$ is Lagrangian iff A is symmetric.

Proof

It is sufficient to consider when ω_0 is trivial on Λ. Let $z = (Xu, Yu)$ and $z' = (Xu', Yu')$ in Λ, then

$$\omega_0(z, z') = \begin{pmatrix} (Xu)^\top & (Yu)^\top \end{pmatrix} \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} \begin{pmatrix} Xu' \\ Yu' \end{pmatrix} = u^\top (X^\top Y - Y^\top X) u'.$$

This is zero for all u, u' iff $X^\top Y = Y^\top X$, as desired.
The second claim is because $\Lambda = \operatorname{im}(I, A)$, which has rank n. Thus it is Lagrangian iff $A^\top = A$. ■

A matrix $Z \in \mathbb{R}^{2n \times n}$ of the form

$$Z = \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad X^\top Y = Y^\top X$$

is called a **Lagrangian frame**. Notice that Z's columns form an orthonormal basis for $\Lambda = \operatorname{im} Z$ iff $U = X + iY$ is unitary. Indeed Z's columns are orthonormal iff

$$I = Z^\top Z = X^\top X + Y^\top Y.$$

And U is unitary iff

$$I = U^\dagger U = (X^\top - iY^\top)(X + iY) = X^\top X + Y^\top Y + i(X^\top Y - Y^\top X) = X^\top X + Y^\top Y.$$

A Lagrangian frame Z for which $U = X + iY$ is unitary is called a **unitary Lagrangian frame**. Every Lagrangian subspace has a unitary Lagrangian frame as we can take an orthonormal basis and put it in a matrix.

Let us define the Lagrangian subspace

$$\Lambda_{\mathsf{hor}} = \left\{(x, 0) \in \mathbb{R}^{2n} \;\middle|\; x \in \mathbb{R}^n\right\} = \operatorname{im} \begin{pmatrix} I \\ 0 \end{pmatrix}.$$

Lemma 1.1.3.2

(1) If $\Lambda \in \mathcal{L}(n)$ and $\Psi \in \mathrm{Sp}(2n)$, then $\Psi\Lambda \in \mathcal{L}(n)$.

(2) For any two Lagrangian subspaces $\Lambda, \Lambda' \in \mathcal{L}(n)$, there is a symplectic matrix $\Psi \in \mathrm{U}(n)$ such that $\Lambda' = \Psi\Lambda$.

(3) $\mathcal{L}(n)$ can be seen to be naturally isomorphic to $\mathrm{U}(n)/\mathrm{O}(n)$, and thus carries the natural structure of a manifold of dimension $n(n+1)/2$.

Proof

(1) is clear. For (2) let $\Lambda \leq \mathbb{R}^{2n}$ be given by a unitary Lagrangian frame $Z = (X, Y)$ and define

$$\Psi = X + iY = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Sp}(2n) \cap \mathrm{O}(n).$$

Then

$$\Psi\Lambda_{\mathsf{hor}} = \Lambda,$$

and thus we have shown (2).

A unitary Lagrangian frame of Λ is uniquely determined up to choice of orthonormal basis. Altering the basis amounts to multiplication on the right by an orthonormal matrix, so we see that there is a bijection $\mathrm{U}(n)/\mathrm{O}(n) \cong \mathcal{L}(n)$. The dimension of $\mathrm{U}(n)$ is n^2, and the dimension of $\mathrm{O}(n)$ is $n(n-1)/2$, so $\dim \mathcal{L}(n) = n(n+1)/2$. ■

We can show that $\mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n)$ has a fundamental group isomorphic to the integers. This isomorphism is also called the **Maslov index**.

Theorem 1.1.3.3

There exists a unique function, the **Maslov index**, μ which assigns to each loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ an integer and satisfies the following axions:

(1) homotopy: two loops in $\mathcal{L}(n)$ are homotopic iff they have the same Maslov index.

(2) product: for any two loops $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ and $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$, we have

$$\mu(\Psi\Lambda) = \mu(\Lambda) + 2\mu(\Psi).$$

(3) direct sum: if $n = n' + n''$, then we can identify $\mathcal{L}(n') \oplus \mathcal{L}(n'')$ as a submanifold of $\mathcal{L}(n)$ and then

$$\mu(\Lambda' \oplus \Lambda'') = \mu(\Lambda') + \mu(\Lambda'').$$

(4) zero: a constant loop $\Lambda(t) = \Lambda_0$ has Maslov index zero.

(5) normalization: the loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(1)$ given by

$$\Lambda(t) = e^{\pi i t}\mathbb{R} \leq \mathbb{C} = \mathbb{R}^2$$

has Maslov index one.

Proof

An orthonormal matrix has determinant ± 1, thus the map

$$\rho\colon \mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n) \to S^1, \qquad \rho([U]) = \det_{\mathbb{C}} U^2$$

is well-defined. Here $[U] \in \mathrm{U}(n)/\mathrm{O}(n)$ denotes the coset of the unitary matrix $U = X + iY$, which in turn represents the Lagrangian subspace $\Lambda = \mathrm{im}(X, Y)$.

Define the Maslov index to be the induced map $\pi_1(\mathcal{L}(n)) \to \pi_1(S^1) \xrightarrow{\sim} \mathbb{Z}$. That is for a loop Λ,

$$\mu(\Lambda) = \deg(\rho \circ \Lambda) = \alpha(1) - \alpha(0)$$

where $\alpha\colon I \to \mathbb{R}$ is a lift of $\rho \circ \Lambda$, meaning

$$e^{\pi i \alpha(t)} = \det_{\mathbb{C}}(U(t)), \qquad \Lambda = [U].$$

The Maslov index then depends only on the homotopy class of Λ. We show it satisfies the product axiom: first note that if Ψ is unitary $\rho(\Psi\Lambda) = \rho(\Psi)^2\rho(\Lambda)$ ($\rho(\Psi)$ corresponds to the ρ in theorem 1.1.2.10). And so the degree of this is

$$\deg(\rho(\Psi\Lambda)) = \deg \rho(\Lambda) + 2\deg \rho(\Psi) = \mu(\Lambda) + 2\mu(\Psi).$$

If Ψ is not unitary, it is homotopic to a loop of unitary matrices, and the result follows.

We now show the converse direction in the homotopy axiom. Suppose that Λ_0, Λ_1 are two loops of Lagrangian subspaces with the same Maslov index, we must show Λ_0, Λ_1 are homotopic. By the previous lemma and the product axiom, we may assume $\Lambda_j(0) = \Lambda_j(1) = \Lambda_{\mathsf{hor}}$. Lift $\Lambda_j(t)$ to $U_j(t) = X_j(t) + iY_j(t) \in \mathrm{U}(n)$ such that $U_0(0) = U_1(0) = I$ (as these represent Λ_{hor}). We may multiply U_j by right multiplication with orthogonal matrices to get $U_0(1) = U_1(1) = I$ as well. Then $U(t) = U_1(t)U_0(t)^{-1}$ forms a loop of unitary matrices.

Note that $\mu(\Lambda_0) = \mu(\Lambda_1)$ means $2\mu(U_0) = 2\mu(U_1)$, and thus U_0, U_1 are homotopic. Thus Λ_0, Λ_1 are homotopic.

The normalization axiom follows from the other axioms. Note that the loop $\Lambda(t) = e^{\pi i t}\mathbb{R}$ satisfies

$$\Lambda(t) \oplus \Lambda(t) = \Psi(t)\Lambda_0$$

where $\Lambda_0 = \mathbb{R}^2 \leq \mathbb{C}^2$ is constant and Ψ is the loop

$$\Psi(t) = e^{2\pi i t}X(t) = e^{\pi i t}\begin{pmatrix} \cos(\pi t) & -\sin(\pi t) \\ \sin(\pi t) & \cos(\pi t) \end{pmatrix}.$$

Here $X(t)$ has constant determinant 1 and so $\det \Psi = e^{2\pi i t}$ lifts to $\alpha(t) = t$ and thus $\mu(\Psi) = 1$. Thus we have

$$2\mu(\Lambda) = \mu(\Lambda \oplus \Lambda) = \mu(\Lambda_0) + 2\mu(\Psi) = 2 \Longrightarrow \mu(\Lambda) = 1. \qquad \blacksquare$$

1.4 The affine nonsqueezing theorem

Define an **affine symplectomorphism** of $\mathbb{R}^{2n}$ to be a map $\psi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ of the form

$$\psi(z) = \Psi(z) + z_0, \qquad \Psi \in \mathrm{Sp}(2n),\ z_0 \in \mathbb{R}^{2n}.$$

Let $\mathrm{ASp}(2n)$ denote the set of affine symplectomorphisms, this is a group as

$$\psi \circ \psi'(z) = \Psi \circ \Psi'(z) + \Psi(z_0') + z_0, \qquad \psi^{-1}(z) = \Psi^{-1}(z) - \Psi^{-1}(z_0)$$

are affine symplectomorphisms.

The affine nonsqueezing theorem says that a ball in $\mathbb{R}^{2n}$ can only be embedded into a symplectic cylinder if the ball's radius is smaller than the cylinder. If we let $e_1, \dots, e_n, f_1, \dots, f_n$ denote the standard basis of $\mathbb{R}^{2n}$, then the symplectic cylinder of radius $R > 0$ is

$$Z^{2n}(R) = B^2(R) \times \mathbb{R}^{2n-2} = \left\{ z \in \mathbb{R}^{2n} \;\middle|\; \langle z, e_1\rangle^2 + \langle z, f_1\rangle^2 \leq R^2 \right\} \subseteq \mathbb{R}^{2n}.$$

Theorem 1.1.4.1 (The affine nonsqueezing theorem)

Let $\psi \in \mathrm{ASp}(2n)$ be an affine symplectomorphism such that $\psi(B^{2n}(r)) \subseteq Z^{2n}(R)$. Then $r \leq R$.

Proof

We may assume that $r = 1$ and let $\psi(z) = \Psi(z) + z_0$. Then define

$$u = \Psi^\top e_1, \qquad v = \Psi^\top f_1, \qquad a = \langle e_1, z_0\rangle, \qquad b = \langle f_1, z_0\rangle.$$

Because $\Psi \in \mathrm{Sp}(2n)$ we have

$$\omega_0(u, v) = \omega_0(\Psi^\top e_1, \Psi^\top f_1) = \omega_0(e_1, f_1) = 1.$$

The condition $\psi(B^{2n}(1)) \subseteq Z^{2n}(R)$ is equivalent to

$$\sup_{|z|=1}\left(\langle e_1, \psi(z)\rangle^2 + \langle f_1, \psi(z)\rangle^2\right) = \sup_{|z|=1}\left((\langle u, z\rangle + a)^2 + (\langle v, z\rangle + b)^2\right) \leq R^2.$$

Because $1 = \omega_0(u,v) \leq |u|\,|v|$ (as $\omega_0(x,y) = \langle x, J_0 y\rangle$ and J_0 is norm-preserving), one of u, v must have norm at least 1. Without loss of generality $|u| \geq 1$, and then let $z = \pm u/|u|$ whose sign agrees with a's. Then

$$(\langle u, z\rangle + a)^2 \geq |u|^2 \geq 1,$$

which then implies that $1 \leq R^2$ and so $1 \leq R$ as desired. ■

Because $B^{2n}(R) \subseteq Z^{2n}(R)$, we have that

$$R = \sup\left\{r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R)\right\}.$$

Either affine or linear in the above supremum gives the same result.

Call a set $B \subseteq \mathbb{R}^{2n}$ an **linear** (resp. **affine**) **symplectic ball** of radius r if it is linearly (resp. affinely) symplectomorphic to $B^{2n}(r)$. Similarly $Z \subseteq \mathbb{R}^{2n}$ is a **linear** (resp. **affine**) **symplectic cylinder** of radius R if it is linearly (resp. affinely symplectomorphic to $Z^{2n}(R)$.

The radius of a symplectic ball or cylinder is invariant. This follows from the above theorem: if B is symplectomorphic to $B^{2n}(r)$ and $B^{2n}(r')$, then $B^{2n}(r), B^{2n}(r')$ are symplectomorphic. By the above theorem this means that $r = r'$. And the radius of a symplectic cylinder is

$$R = \sup\left\{r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R)\right\}.$$

A matrix $\Psi \in \mathrm{Mat}_{2n}$ has the **linear nonsqueezing property** if for every linear symplectic ball B of radius r and linear symplectic cylinder Z of radius R

$$\Psi B \subseteq Z \Longrightarrow r \leq R.$$

By The affine nonsqueezing theorem any symplectic matrix has the linear nonsqueezing property. Call a matrix **anti-symplectic** if $\Psi^*\omega_0 = -\omega_0$. The affine nonsqueezing theorem clearly holds for anti-symplectic matrices as well.

Theorem 1.1.4.2 (The affine rigidity theorem)

Let $\Psi \in \mathrm{Mat}_{2n}$ be nonsingular such that Ψ, Ψ^{-1} have the linear nonsqueezing property. Then Ψ is either symplectic or anti-symplectic.

Proof

Suppose Ψ is neither symplectic nor anti-symplectic. Then neither is $\Psi^\top$ so there are $u, v \in \mathbb{R}^{2n}$ such that

$$\omega_0(\Psi^\top u, \Psi^\top v) \neq \pm\omega_0(u,v).$$

We may assume that neither side is zero, and potentially by replacing Ψ with Ψ^{-1}, $\left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < |\omega_0(u,v)|$. We can rescale u so that $\omega_0(u,v) = 1$ so that

$$0 < \lambda^2 = \left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < \omega_0(u,v) = 1.$$

Thus $\mathrm{span}\,\{u,v\}$ is a symplectic subspace of which u, v is a symplectic basis, so we can extend it to a symplectic basis of all of $\mathbb{R}^{2n}$ by considering its symplectic complement. So let $u_1, v_1, \ldots, u_n, v_n$ be a symplectic basis of $\mathbb{R}^{2n}$ with $u = u_1, v = v_1$. Now let $u_1' = \lambda^{-1}\Psi^\top u$ and $v_1' = \pm\lambda^{-1}\Psi^\top v$ where the sign is so that $\omega_0(u_1', v_1') = 1$. Extend this to a symplectic basis $u_1', v_1', \ldots, u_n', v_n'$ of $\mathbb{R}^{2n}$.

Define $\Phi, \Phi' \in \mathrm{Sp}(2n)$ to be the symplectic matrices which map the standard basis $e_1, \ldots, f_n$ to $u_1, \ldots, v_n$ and $u_1', \ldots, v_n'$ respectively. Then $A = (\Phi')^{-1}\Psi^\top\Phi$ satisfies

$$Ae_1 = \lambda e_1, \qquad Af_1 = \pm\lambda f_1.$$

Thus $A^\top$ maps the unit ball $B^{2n}(1)$ in $Z^{2n}(\lambda)$: if $|z| \leq 1$ then

$$\left\langle A^\top z, e_1\right\rangle^2 + \left\langle A^\top z, f_1\right\rangle^2 = \langle z, Ae_1\rangle^2 + \langle z, Af_1\rangle^2 = \lambda^2(\langle z, e_1\rangle^2 + \langle z, f_1\rangle^2) \leq \lambda^2.$$

Because $\lambda < 1$, this means that $A^\top$ does not have the linear squeezing property. Because Φ, Φ' are symplectic this means that Ψ does not have the linear squeezing property. ■

Let $A \subseteq \mathbb{R}^{2n}$ be an arbitrary subset. Define its **linear symplectic width** to be

$$w_L(A) = \sup\left\{\pi r^2 \;\middle|\; \psi(B^{2n}(r)) \subseteq A \text{ for some } \psi \in \mathrm{ASp}(\mathbb{R}^{2n})\right\}.$$

The linear symplectic width has the following properties:

(1) **monotonicity**: if $\psi(A) \subseteq B$ for $\psi \in \mathrm{ASp}(\mathbb{R}^{2n})$ then $w_L(A) \leq w_L(B)$.

(2) **conformality**: $w_L(\lambda A) = \lambda^2 w_L(A)$.

(3) **nontriviality**: $w_L(B^{2n}(r)) = w_L(Z^{2n}(r)) = \pi r^2$.

(1) is obvious, (2) is because $\psi(B^{2n}(r)) \subseteq A$ iff $\psi(B^{2n}(\lambda r)) \subseteq \lambda A$. (3) is due to The affine nonsqueezing theorem: $B^{2n}(r')$ embeds into $B^{2n}(r), Z^{2n}(r)$ iff $r' \leq r$ so $w_L(B^{2n}(r)) = \pi r^2$.

Lemma 1.1.4.3

Anti-symplectic linear maps have determinant $(-1)^n$. Furthermore anti-symplectic affine maps preserve linear symplectic width.

Proof

If $\Psi^*\omega_0 = -\omega_0$ then $\Psi^*\omega_0^n = (-1)^n\omega_0$. Because ω_0^n is a volume form, this means that $\det\Psi = (-1)^n$ as desired.

For the second point note that the composition of two anti-symplectic maps is symplectic, of a symplectic with an anti-symplectic is anti-symplectic. Define $D = \mathrm{diag}(I, -I)$; this is anti-symplectic and $D^2 = I$. Thus Ψ is anti-symplectic iff ΨD is symplectic. Furthermore D is orthonormal and thus $D(B^{2n}(r)) = B^{2n}(r)$.

This shows that $w_L(A)$ can be defined by taking the supremum over $\psi(B^{2n}(r)) \subseteq A$ where ψ is an anti-symplectic affine map. Thus w_L is monotonic with respect to anti-symplectic affine maps, and we have shown the second claim. ■

An **ellipsoid** is a subset $E \subseteq \mathbb{R}^{2n}$ of the form $\left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(z) \leq 1\right\}$ where Q is any positive definite quadratic form. That is $Q(z) = z^\top \overline{Q} z$ for a symmetric positive-definite $\overline{Q} \in \mathrm{Sym}_+(2n)$. Notice that if $\Psi \in \mathrm{Mat}_{2n}$ is nonsingular then ΨE is an ellipsoid as well:

$$\Psi(E) = \left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(\Psi^{-1}z) \leq 1\right\},$$

and $Q \circ \Psi^{-1}$ is a positive definite quadratic form.

Theorem 1.1.4.4

Let $\Psi \in \mathrm{Mat}_{2n}$ be a linear map. Then the following are equivalent.

(1) Ψ preserves the linear symplectic width of all ellipsoids.

(2) Ψ is either symplectic or anti-symplectic.

Proof

As shown above, symplectic and anti-symplectic maps preserve all widths, so (2) implies (1). So assume (2) and we will show that Ψ has the linear nonsqueezing property. Let B be a linear symplectic ball of radius r and Z a linear symplectic cylinder of radius R such that $\Psi B \subseteq Z$. Then by hypothesis

$$\pi r^2 = w_L(B) = w_L(\Psi B) \leq w_L(Z) = \pi R^2 \Longrightarrow r \leq R$$

as desired.

A symplectic ball cannot be embedded into a subspace of dimension $< 2n$, so if Ψ were singular then $w_L(\Psi B^{2n}(1)) = 0$, contradicting (1). Thus Ψ is nonsingular, and Ψ^{-1} satisfies (1) because

$$w_L(\Psi^{-1}E) = w_L(\Psi\Psi^{-1}E) = w_L(E)$$

as $\Psi^{-1}E$ is an ellipsoid.

Thus both Ψ and Ψ^{-1} have the linear nonsqueezing property, so by The affine rigidity theorem Ψ is symplectic or anti-symplectic. ■

Lemma 1.1.4.5

Let $g: V \times V \to \mathbb{R}$ be an inner product on a symplectic space. Then there exists a basis $u_1, \dots, u_n, v_1, \dots, v_n$ of V which is both g-orthogonal and ω-standard. More so, the basis can be chosen so that $g(u_j, u_j) = g(v_j, v_j)$ for all j.

Proof

Consider $V = \mathbb{R}^{2n}$ with the Euclidean inner product and assume that

$$\omega(z, w) = \langle z, Aw \rangle .$$

Then A is nondegenerate and antisymmetric: $A^\top = -A$. Thus $iA \in \mathrm{Mat}_{2n}(\mathbb{C})$ is Hermitian, which means its eigenvalues are all real, so A's eigenvalues are all pure imaginary. Furthermore the eigenvalues come in pairs $\pm i\alpha_j$ for $j = 1, \dots, n$, with $\alpha_j > 0$.

As iA is Hermitian, we can find $z_j = u_j + iv_j \in \mathbb{C}^{2n}$ an orthonormal basis of eigenvectors:

$$Az_j = i\alpha_j z_j, \qquad \overline{z_j}^\top z_k = \delta_{jk}.$$

And $\overline{z}_j$ fill out the basis of eigenvectors, so we have that $z_j^\top z_k = 0$. Thus

$$Au_j = -\alpha_j v_j, \qquad Av_j = \alpha_j u_j, \qquad u_j^\top v_k = u_j^\top u_k = v_j^\top v_k = 0 \text{ for } j \neq k.$$

And so

$$\omega(u_j, v_j) = u_j^\top A v_j = \alpha_j |u_j|^2 > 0,$$

and similarly $\omega(u_j, v_k) = \omega(u_j, u_k) = \omega(v_j, v_k) = 0$ for $j \neq k$. We can thus rescale u_j, v_k to get the desired basis.

Let $r = (r_1, \dots, r_n)$ be an n-tuple with $0 < r_1 \leq \cdots \leq r_n$, and define the ellipsoid

$$E(r) = \left\{ z \in \mathbb{C}^n \;\middle|\; \sum_{i=1}^n \left| \frac{z_i}{r_i} \right|^2 \leq 1 \right\}.$$

Lemma 1.1.4.6

Any ellipsoid

$$E = \left\{ w \in \mathbb{R}^{2n} \;\middle|\; w^\top A w \leq 1 \right\}$$

is symplectomorphic to a unique $E(r)$.

Proof

We can write E as $\left\{w \in \mathbb{R}^{2n} \;\middle|\; g(w,w) \leq 1\right\}$ for the inner product

$$g(u,v) = u^\top A v.$$

By the above lemma there is a basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ of $\mathbb{R}^{2n}$ which is both symplectic and orthogonal for g. Define r_j so that $g(u_j, u_j) = g(v_j, v_j) = 1/r_j^2$. We may rearrange the u_j, v_j so that the r_j are ascending; this does not affect them forming a symplectic basis.

Let $\Phi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be the symplectomorphism mapping the standard basis to this new basis:

$$\Phi(z) = \sum_{j=1}^{n} (x_j u_j + y_j v_j), \qquad z = (x_1, \ldots, x_n, y_1, \ldots, y_n).$$

Thus

$$g(\Psi z, \Psi z) = \sum_{j=1}^{n} \frac{x_j^2 + y_j^2}{r_j^2},$$

which means that

$$\Psi^{-1} E = E(r),$$

as desired.

Now to show uniqueness of $r = (r_1, \ldots, r_n)$, define the matrix

$$\Delta(r) = \operatorname{diag}(1/r_1^2, \ldots, 1/r_n^2, 1/r_1^2, \ldots, 1/r_n^2)$$

so that

$$E(r) = \left\{z \in \mathbb{R}^{2n} \;\middle|\; z^\top \Delta(r) z \leq 1\right\}.$$

Because $\Psi E(r)$ maps to the ellipsoid defined by $\Psi^\top \Delta(r) \Psi$, we want to show that if $\Psi^\top \Delta(r) \Psi = \Delta(r')$, then $r = r'$. We know that $J_0 \Psi^\top = \Psi^{-1} J_0$ so

$$\Psi^{-1} J_0 \Delta(r) \Psi = J_0 \Delta(r').$$

And so $J_0\Delta(r)$ and $J_0\Delta(r')$ have the same eigenvalues. But these matrices correspond to the complex matrices $i \operatorname{diag}(1/r_1^2, \ldots, 1/r_n^2)$ and $i \operatorname{diag}(1/{r'_1}^2, \ldots, 1/{r'_n}^2)$. Thus we must have $r = r'$ as desired. ■

Note that $E(r, \ldots, r)$ is just $B^{2n}(r)$.

Note:
In the case of $n = 1$, symplectic forms are volume forms and symplectomorphisms are the same as area-preserving maps. Thus the above lemma states that every ellipse in $\mathbb{R}^2$ can be mapped into a circle by an area-preserving linear transformation, and that this circle is unique.

Define the **symplectic spectrum** of an ellipsoid E to be the unique n-tuple $r = (r_1, \ldots, r_n)$ with $0 < r_1 \leq \cdots \leq r_n$ such that E is symplectomorphic to $E(r)$. This is an invariant which uniquely determines the ellipsoid up to symplectomorphism. So if E has symplectic spectrum r, we see that

$$\operatorname{vol} E = \operatorname{vol} E(r) = \frac{\pi^n}{n!} \prod_{j=1}^{n} r_j^2$$

where the last equality is just an explicit integral.

Theorem 1.1.4.7
Let $E \subseteq \mathbb{R}^{2n}$ be an ellipsoid, then

$$w_L(E) = \sup_{B \subseteq E} w_L(B) = \inf_{Z \supseteq E} w_L(E),$$

where the supremum runs over all linear (or affine) symplectic balls contained in E, and the infimum over all linear symplectic cylinders containing E. Let $r = (r_1, \dots, r_n)$ be E's symplectic spectrum, then

$$w_L(E) = \pi r_1^2.$$

Proof

Suppose $\Psi \in \mathrm{Sp}(2n)$ is a symplectomorphism such that $\Psi E = E(r)$. Then we know

$$B^{2n}(r_1) \subseteq E(r) \subseteq Z^{2n}(r_1) \Longrightarrow \Psi^{-1}B^{2n}(r_1) \subseteq E \subseteq \Psi^{-1}Z^{2n}(r_1).$$

The widths of the bounds are both πr_1^2 so we have

$$w_L(E) = \pi r_1^2.$$

And furthermore we then get

$$\inf_{Z \supseteq E} w_L(Z) \leq \pi r_1^2 \leq \sup_{B \subseteq E} w_L(B)$$

because $\Psi^{-1}Z^{2n}(r_1)$ is in the infimum on the left, and $\Psi^{-1}B^{2n}(r_1)$ is in the supremum on the right. But if $B \subseteq E$ is a symplectic ball and $E \subseteq Z$ is a symplectic cylinder, then $w_L(B) \leq w_L(Z)$ by monotonicity, and so every element in the supremum is less than every element in the infimum, so we have equality. ■

For an ellipsoid E, define its **dual** to be

$$E^* = \left\{ v \in \mathbb{R}^{2n} \;\middle|\; \langle v, e \rangle \leq 1 \text{ for all } e \in E \right\}.$$

Let us temporarily denote the ellipsoid determined by the positive definite quadratic form A by $E(A)$, and the inner product $g_A(v,u) = v^\top A u$. Then we claim $E(A)^* = E(A^{-1})$. Notice that $v \in E(A^{-1})$ iff $v^\top A^{-1} v \leq 1$ iff $A^{-1}v \in E(A)$. Thus for all $e \in E(A)$ we have by Cauchy-Schwarz

$$\langle v, e \rangle = g_A(A^{-1}v, e) \leq 1$$

so $v \in E(A)^*$. Conversely if $v \in E(A)^*$ then

$$\langle v, e \rangle = g_A(A^{-1}v, e) \leq 1$$

for all $e \in E(A)$. Let $\alpha = \sqrt{g_A(A^{-1}v, A^{-1}v)}$ be $A^{-1}v$'s norm, so that $A^{-1}v/\alpha$ has g_A-norm 1 and thus is in $E(A)$. But

$$g_A(A^{-1}v, A^{-1}v/\alpha) = \alpha \leq 1$$

meaning $A^{-1}v \in E(A)$ and thus $v \in E(A^{-1})$ as desired.

Note that A being positive definite means A^{-1} is too, so $E(A^{-1}) = E(A)^*$ is an ellipsoid. Furthermore it is immediate from this that $E^{**} = E$ and $(\Psi E)^* = (\Psi^\top)^{-1}E^*$, the latter being because

$$(\Psi E(A))^* = E((\Psi^\top)^{-1}A\Psi^{-1})^* = E(\Psi A^{-1}\Psi^\top) = (\Psi^\top)^{-1}E(A)^*.$$

1.5 Linear complex structures

Definition 1.1.5.1

A **(linear) complex structure** on a real vector space is an automorphism

$$J\colon V \to V \qquad \text{such that} \qquad J^2 = -I.$$

Denote the space of linear complex structures over V by

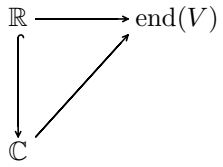
$$\mathcal{J}(V) \subseteq \operatorname{end}(V).$$

The archetypal linear complex structure is

$$J_0 = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$$

over $\mathbb{R}^{2n}$.

V having a complex structure is equivalent to it having the structure of a $\mathbb{C}$-vector space compatible with its real vector space structure. This is equivalent to the existence of a ring map $\mathbb{C} \to \operatorname{end}(V)$ which is a lifting of the ring map $\mathbb{R} \to \operatorname{end}(V)$ which defines V's real linear structure:



That is, the existence of a map of $\mathbb{R}$-algebras $\mathbb{C} \to \operatorname{end}(V)$. Because $\mathbb{C}$ is generated by i as a real algebra, this is equivalent to the existence of $J \in \operatorname{end}(V)$ such that $J^2 = -I$. Spelling this out, if V has a complex structure then it becomes a complex vector space with multiplication

$$\mathbb{C} \times V \to V, \qquad (a + ib, v) \mapsto av + bJv.$$

Notice that this necessitates that V has even real dimension. This can be seen because

$$\dim_{\mathbb{R}} V = \dim_{\mathbb{C}} V \cdot \dim_{\mathbb{R}} \mathbb{C} = 2 \dim_{\mathbb{C}} V$$

and because $J^2 = -I$ means that $\det J^2 = (-1)^{\dim V}$ is positive so $\dim V$ is even.

Proposition 1.1.5.2

Let V be a $2n$-dimensional real vector space with complex structure J. Then there is a vector space isomorphism $\Phi\colon \mathbb{R}^{2n} \to V$ such that $J\Phi = \Phi J_0$.

Proof

Viewing V as a complex vector space, there is a basis $v_1, \ldots, v_n$. Then $v_1, \ldots, v_n, Jv_1, \ldots, Jv_n$ is a basis of the real vector space V. Mapping the standard basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ to this basis gives the desired result. ■

Note that the map $\Phi\colon \operatorname{end}(V) \to \operatorname{end}(V)$ given by $A \mapsto A^2$ is smooth, and its differential at A is $X \mapsto AX + XA$. For $A \in \mathcal{J}(V)$ we have that

$$\ker d\Phi_A = \{X \in \operatorname{end}(V) \mid AX + XA = 0\},$$

which we claim has dimension $2n^2$. We can take $V = \mathbb{R}^{2n}$ and since there is an automorphism mapping A to J_0, we may assume $A = J_0$. Then

$$\begin{pmatrix} X & Y \\ Z & W \end{pmatrix}\begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} + \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} = \begin{pmatrix} Y & -X \\ W & -Z \end{pmatrix} + \begin{pmatrix} -Z & -W \\ X & Y \end{pmatrix} = \begin{pmatrix} Y - Z & -X - W \\ X + W & Y - Z \end{pmatrix}.$$

And this is zero iff $Z = Y$ and $W = -X$, so we have $2n^2$ degrees of freedom as desired.

Thus Φ has constant rank $(2n)^2 - 2n^2 = 2n^2$ on $\mathcal{J}(V) = \Phi^{-1}(-I)$, which means that $\mathcal{J}(V)$ is an embedded submanifold of $\operatorname{end}(V)$ of dimension $2n^2$.

A linear complex structure J on V induces an orientation as follows. Let $v_1, \dots, v_n \in V$ form a complex basis so that $v_1, Jv_1, \dots, v_n, Jv_n$ forms a real basis. This real basis is the orientation. This orientation is well-defined, as in it is not dependent on v_i. Let $u_1, \dots, u_n$ be another complex basis, and let Φ be the complex map $v_i \mapsto u_i$. Let us write

$$u_i = \sum_j a_{ij} v_j + b_{ij} J v_j, \qquad Ju_i = \sum_j a_{ij} J v_j - b_{ij} v_j$$

so that the representation of the base transfer matrix (of $v_1, \dots, v_n, Jv_1, \dots, Jv_n \to u_1, \dots, u_n, Ju_1, \dots, Ju_n$) is

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix} \leftrightarrow A + iB,$$

and we know that this has determinant $|\det(A+iB)|^2 = |\det \Phi|^2 > 0$, so they are compatibly oriented. Note here that we permuted the oriented basis, but we did this for both using the same permutation, so the transfer matrix has the same determinant.

A linear complex structure is **compatible** with an orientation if it induces that orientation. For example J_0 induces the standard orientation on $\mathbb{R}^{2n}$.

Proposition 1.1.5.3

(1) $\mathcal{J}(\mathbb{R}^{2n})$ is diffeomorphic to $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ and thus has two connected components.

(2) The connected component $\mathcal{J}^+(\mathbb{R}^{2n})$ containing J_0 is the space of all complex structures which are compatible with the standard orientation. It is thus diffeomorphic to $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ and is homotopy equivalent to

$$\mathcal{J}^+(\mathbb{R}^{2n}) \cap \mathrm{SO}(2n) \cong \frac{\mathrm{SO}(2n)}{\mathrm{U}(n)}$$

of orthogonal complex structures which are compatible with the standard orientation.

(3) $\mathcal{J}^+(\mathbb{R}^4)$ is homotopy equivalent to S^2.

Proof

Consider the smooth map

$$\mathrm{GL}_{2n}(\mathbb{R}) \to \mathcal{J}(\mathbb{R}^{2n}), \qquad \Psi \mapsto \Psi J_0 \Psi^{-1}.$$

By the above lemma this is a surjection, and it quotients to a bijective smooth $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C}) \to \mathcal{J}(\mathbb{R}^{2n})$. Both of these manifolds have dimension $2n^2$ and the above map is a submersion, so its quotient is a submersion as well, thus the quotient is a bijective submersion between manifolds of the same dimension, and thus a diffeomorphism.

The connected components of $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ correspond to the connected components of $\mathrm{GL}_{2n}(\mathbb{R})$, which are determined by the sign of the determinant, equivalently the oriented and non-oriented matrices. As $\det J = 1$, it lies in $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$.

Now $\mathrm{O}(2n)/\mathrm{U}(n) \hookrightarrow \mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ is a strong deformation retract, via the homotopy

$$H\colon [0,1] \times \mathrm{GL}_{2n}(\mathbb{R}) \to \mathrm{GL}_{2n}(\mathbb{R}), \qquad H(t,\Psi) = \Psi(\Psi^\top \Psi)^{-t/2}.$$

This homotopy preserves the quotients, so it descends into them giving the desired homotopy. The connected component $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ corresponds to $\mathrm{SO}(2n)/\mathrm{U}(n) \cong \mathcal{J}^+(\mathbb{R}^{2n}) \cap \mathrm{SO}(2n)$.

For (3) let $n = 2$ and let e_0, e_1, e_2, e_3 be the standard basis of $\mathbb{R}^4$ and denote a vector by $x = (x_0, x_1, x_2, x_3)$. Then a matrix

$$J \in \mathcal{J}(\mathbb{R}^4) \cap \mathrm{SO}(4)$$

is determined by Je_0. And J satisfies $\langle v, Jv \rangle = 0$ because J is orthogonal and similar to J_0, they are orthogonally similar, and because this identity holds for J_0 it holds for J. Thus e_0, Je_0 form an orthonormal basis of a 2-plane $E \subseteq \mathbb{R}^4$.

Now for any unit $v \in E^\perp$, v, Jv form an orthonormal basis of $E^\perp$. And because J is compatible with the standard orientation, e_0, Je_0, v, Jv is positively oriented. There are only two unit vectors in $E^\perp$ orthogonal to v, and only one makes the basis oriented. So Jv is uniquely defined.

Thus $\mathcal{J}^+(\mathbb{R}^4) \cap \mathrm{SO}(2n) \to S^2$ given by $J \mapsto Je_0$ is a continuous bijection, and is seen to be a homeomorphism. Thus $\mathcal{J}^+(\mathbb{R}^4)$ is homotopy equivalent to S^2. ■

Definition 1.1.5.4

Let (V, ω) be a symplectic space. Then a complex structure $J \in \mathcal{J}(V)$ is **compatible** with ω if

$$J^*\omega = \omega, \qquad \omega(v, Jv) > 0 \text{ for all } v \in V.$$

Denote the space of compatible complex structures on (V, ω) by $\mathcal{J}(V, \omega)$.

If J is compatible with ω then

$$g_J(v, u) = \omega(v, Ju)$$

is an inner product on V, and J is skew-adjoint with respect to this inner product: $g_J(v, Ju) = -g_J(Jv, u)$.

Proposition 1.1.5.5

Let (V, ω) be a symplectic space and J a complex structure on V. The following are then equivalent.

(1) J is compatible with ω.

(2) (V, ω) has a symplectic basis of the form

$$v_1, \dots, v_n, Jv_1, \dots, Jv_n.$$

(3) There exists a vector space isomorphism $\Psi\colon \mathbb{R}^{2n} \to V$ such that

$$\Psi^*\omega = \omega_0, \qquad \Psi^* J = J_0.$$

(4) J satisfies $\omega(v, Jv) > 0$ and

$$\Lambda \in \mathcal{L}(V, \omega) \implies J\Lambda \in \mathcal{L}(V, \omega).$$

Proof

Suppose (1), and let $\Lambda \subseteq V$ be a Lagrangian subspace. Choose an orthonormal basis $v_1, \dots, v_n$ of Λ with respect to g_J, then

$$\omega(v_i, Jv_j) = g_J(v_i, v_j) = \delta_{ij}, \qquad \omega(Jv_i, Jv_j) = \omega(v_i, v_j) = 0$$

(remember that $v_i \in \Lambda$ and ω is trivial on Λ). Thus $v_1, \dots, v_n, Jv_1, \dots, Jv_n$ forms a symplectic basis as desired.

(2) then implies (3) by mapping

$$\Psi(z) = \sum_{i=1}^{n} (x_i v_i + y_i J v_i), \qquad z = (x_1, \dots, x_n, y_1, \dots, y_n).$$

(3) is easily seen to imply (1).

Now (1) implies (4), because $J\Lambda$ has the same dimension of Λ and for $v, u \in \Lambda$,

$$\omega(Jv, Ju) = \omega(u, v) = 0$$

so that $J\Lambda$ is isotropic and thus Lagrangian.

Now (4) implies (1): $g_J(v, u) = \omega(v, Ju)$ is positive definite by assumption, so we just need to show that it is symmetric. Suppose that g_J is not symmetric, then there are $v, u \in V$ such that

$$\omega(v, Ju) \neq \omega(u, Jv).$$

Then v is nonzero so $\omega(v, Jv) > 0$, and define

$$w = u - frac\omega(v, Ju)\omega(v, Jv)v.$$

Then

$$\omega(v, Jw) = 0, \qquad \omega(w, Jv) = \omega(u, Jv) - \omega(v, Ju) \neq 0.$$

Thus w, Jv are linearly independent and thus so too are Jw, v. Because $\omega(Jw, v) = 0$, we can extend v, Jw to a Lagrangian subspace Λ. Then $J\Lambda$ is Lagrangian, but $Jv, w \in J\Lambda$ have nonzero ω. ■

We now provide three proofs of the fact that $\mathcal{J}(V, \omega)$ is contractible. We do this in three different ways:

(1) The first method identifies $\mathcal{J}(V, \omega)$ with the space of symmetric, positive definite, and symplectic matrices. Then this is shown to be contractible.

(2) The second method constructs an explicit homotopy equivalence between $\mathcal{J}(V, \omega)$ and the space $\mathrm{Met}(V)$ of all inner products on V. This is based on proposition 1.1.5.7 which shows that $\mathcal{J}(V)$ is homotopy equivalent to the space of nondegenerate skew-symmetric bilinear forms on V. This method will show that every compact subgroup of $\mathrm{Sp}(2n)$ is conjugate to a subgroup of $\mathrm{U}(n)$.

(3) The third method identifies $\mathcal{J}(V, \omega)$ with the Siegel upper half space.

Proposition 1.1.5.6

Let (V, ω) be a $2n$-dimensional symplectic space. Then $\mathcal{J}(V, \omega)$ is diffeomorphic to $\mathrm{Sym}_+(2n) \cap \mathrm{Sp}(2n)$, which is contractible.

Proof

We may assume that $V = \mathbb{R}^{2n}$ and $\omega = \omega_0$ in lieu of the previous theorem. Then $J \in \mathcal{J}(\mathbb{R}^{2n}, \omega_0)$ iff

$$J^2 = -1, \qquad J^\top J_0 J = J_0, \qquad \langle v, -J_0 Jv\rangle = v^\top(-J_0 J)v > 0.$$

Thus

$$(J_0 J)^\top = -J^\top J_0 = -J_0 J^{-1} = J_0 J,$$

and so

$$P = -J_0 J$$

is symmetric, positive definite, and symplectic (because J_0, J are symplectic). Conversely if P has these properties then $J = J_0 P$ is a compatible complex structure. These maps are both smooth inverses, so we have the desired diffeomorphism.

And $\mathrm{Sym}_+(2n) \cap \mathrm{Sp}(2n)$ is contractible by the null-homotopy

$$H(t, P) = P^t,$$

which is continuous by lemma 1.1.2.5 and well-defined by lemma 1.1.2.6. This is a null homotopy to the point I as $H_0(\bullet) = I$. ■

For the second proof we show that $\mathcal{J}(V)$ is homotopy equivalent to the space of symplectic forms on V:

$$\mathcal{S}(V) = \left\{\omega \in \Lambda^2(V^*) \;\middle|\; \omega \text{ is nondegenerate}\right\}.$$

The homotopy equivalence is given explicitly, depending only on a choice of inner product on V. Define

$$\mathrm{Met}(V) = \left\{g \in \mathrm{Sym}^2(V^*) \;\middle|\; g \text{ is positive definite}\right\},$$

the space of all inner products on V. $\mathrm{Met}(V)$ is an open convex subspace of $\mathrm{Sym}^2(V^*)$: it is convex because the positive multiples and sums of inner products are inner products, it is open because $\mathrm{Sym}_+(n) \subseteq \mathrm{Sym}(n)$ is (a similar agument to the proof of lemma 1.1.2.5 shows that for a positive definite matrix A there is $0 < m$ and a neighborhood U in Sym such that all matrices in U have eigenvalues bound below by m, thus are positive definite). Because $\mathrm{Met}(V)$ is convex, it is contractible.

For $g \in \mathrm{Met}(V)$ define

$$\mathcal{J}(V,g) = \{J \in \mathcal{J}(V) \mid J^*g = g\}, \qquad \mathcal{S}(V,g) = \{g(J\bullet,\bullet) \mid J \in \mathcal{J}(V,g)\}.$$

Notice that $J^*g = g$ iff $\omega_g = g(J\bullet,\bullet) \in \mathcal{S}(V)$: if $J^*g = g$ then

$$\omega_g(v,u) = g(Jv,u) = g(-v,Ju) = -g(Ju,v) = -\omega_g(u,v),$$

and if $\omega_g \in \mathcal{S}(V)$ then

$$g(Ju,Jv) = \omega_g(u,Jv) = -\omega_g(Jv,u) = -g(-v,u) = g(v,u) = g(u,v).$$

And so $\mathcal{S}(V,g) \subseteq \mathcal{S}(V)$.

Elements of $\mathcal{J}(V,g)$ and $\mathcal{S}(V,g)$ are said to be **compatible with** g.

Now consider the map

$$\mathrm{Met}(V) \times \mathcal{J}(V) \to \mathcal{S}(V), \qquad (g,J) \mapsto \omega_{g,J} = \frac{1}{2}(g(J\bullet,\bullet) - g(\bullet,J\bullet)).$$

This is well-defined as $\omega_{g,J}$ is clearly antisymmetric bilinear, and it is nondegenerate

$$\omega_{g,J}(v,Jv) = \frac{1}{2}(g(Jv,Jv) + g(v,v)) > 0.$$

Furthermore this map is smooth as it can be seen as evaluating a polynomial at a matrix. Recall that $\mathrm{GL}(V)$ acts on $T^2(V^*)$ (bilinear forms on V) by $(\Psi^*\alpha)(v,u) = \alpha(\Psi^{-1}v, \Psi^{-1}u)$, and this is a smooth action which restricts to $\mathrm{Met}(V)$ and $\mathcal{S}(V)$. The map above is then $\mathrm{GL}(V)$-equivariant:

$$\begin{aligned}\omega_{\Psi^*g,\Psi^*J}(v,u) = \frac{1}{2}((\Psi^*g)(\Psi^*Jv,u) - (\Psi^*g)(v,\Psi^*Ju)) &= \frac{1}{2}(g(J\Psi^{-1}v,\Psi^{-1}u) - g(\Psi^{-1}v,J\Psi^{-1}u)) \\ &= \omega_{g,J}(\Psi^{-1}v,\Psi^{-1}u) = (\Psi^*\omega_{g,J})(v,u).\end{aligned}$$

For a fixed $g \in \mathrm{Met}(V)$ this map restricts to $f_g\colon \mathcal{J}(V) \to \mathcal{S}(V)$ by $f_g(J) = \omega_{g,J}$. And for $J \in \mathcal{J}(V,g)$ we have that $g(J\bullet,\bullet) \in \mathcal{S}(V,g)$ meaning $g(\bullet,J\bullet) = -g(J\bullet,\bullet)$ so that $f_g(J) = g(J\bullet,\bullet) \in \mathcal{S}(V,g)$. Thus $f_g\colon \mathcal{J}(V,g) \to \mathcal{S}(V,g)$ is a bijective $\mathrm{GL}(V)$-equivariant smooth map, and thus a diffeomorphism. And so we have the following commutative diagram

$$\begin{array}{ccc} \mathcal{J}(V,g) & \xrightarrow[\sim]{f_g} & \mathcal{S}(V,g) \\ \downarrow & & \downarrow \\ \mathcal{J}(V) & \xrightarrow{f_g} & \mathcal{S}(V) \end{array}$$

Proposition 1.1.5.7

Let V be an even-dimensional real vector space. Then there exists a $\mathrm{GL}(V)$-equivariant smooth map

$$\mathrm{Met}(V) \times \mathcal{S}(V) \to \mathcal{J}(V), \qquad (g, \omega) \mapsto J_{g,\omega}$$

with the following properties.

(1) For $g \in \mathrm{Met}(V), \omega \in \mathcal{S}(V), J \in \mathcal{J}(V)$, we have

$$J_{g,\omega} \in \mathcal{J}(V, \omega) \cap \mathcal{J}(V, g), \qquad J \in \mathcal{J}(V, \omega_{g,J})$$

and if $g = \omega(\bullet, J\bullet)$ then

$$J_{g,\omega} = J, \qquad \omega_{g,J} = \omega.$$

Thus for $g \in \mathrm{Met}(V)$, $\mathcal{J}(V, g) \to \mathcal{S}(V, g)$ mapping $J \mapsto \omega_{g,J}$ has inverse $\mathcal{S}(V, g) \to \mathcal{J}(V, g)$ mapping $\omega \mapsto J_{g,\omega}$.

(2) For every $g \in \mathrm{Met}(V)$, the map

$$\mathcal{S}(V) \to \mathcal{J}(V), \qquad \omega \mapsto J_{g,\omega}$$

is a homotopy equivalence with inverse

$$\mathcal{J}(V) \to \mathcal{S}(V), \qquad J \mapsto \omega_{g,J}.$$

(3) For every $g \in \mathrm{Met}(V)$, the inclusions $\mathcal{J}(V, g) \hookrightarrow \mathcal{J}(V)$ and $\mathcal{S}(V, g) \hookrightarrow \mathcal{S}(V)$ are homotopy equivalences with inverses

$$\begin{aligned} \mathcal{J}(V) &\to \mathcal{J}(V, g), \qquad J \mapsto J_{g,\omega_{g,J}} \\ \mathcal{S}(V) &\to \mathcal{S}(V, g), \qquad \omega \mapsto \omega_{g,J_{g,\omega}}. \end{aligned}$$

Proof

First we construct this map. For $g \in \mathrm{Met}(V)$ and $\omega \in \mathcal{S}(V)$ define $A_{g,\omega} \in \mathrm{GL}(V)$ by

$$g(A_{g,\omega} v, u) = \omega(v, u).$$

So $A_{g,\omega}$ is g-skew-adjoint and thus $-A_{g,\omega}^2$ is g-self-adjoint and g-positive definite. This means there exists a unique g-self-adjoint and g-positive definite automorphism $Q_{g,\omega}: V \to V$ such that $Q_{g,\omega}^2 = -A_{g,\omega}^2$ (consider the Euclidean case). By proposition 1.1.2.5, this is a smooth map. Note that $Q_{g,\omega}$ and $A_{g,\omega}$ commute (because $A_{g,\omega}^{-1} Q_{g,\omega} A_{g,\omega}$ is also g-self-adjoint and g-positive definite, so it is equal to $Q_{g,\omega}$).

So define

$$J_{g,\omega} = Q_{g,\omega}^{-1} A_{g,\omega} \in \mathcal{J}(V, \omega) \cap \mathcal{J}(V, g).$$

This is indeed a complex structure as

$$J_{g,\omega}^2 = A_{g,\omega}^2 Q_{g,\omega}^{-2} = -I,$$

and it is ω- and g-compatible as

$$g(J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega}\bullet, Q_{g,\omega}^{-2} A_{g,\omega}\bullet) = g(\bullet, \bullet)$$

and

$$\omega(J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega} J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_\omega \bullet, \bullet) = \omega(\bullet, \bullet),$$

finally

$$\omega(\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega}\bullet, J_{g,\omega}\bullet) = -g(Q_{g,\omega}^{-1} A_{g,\omega}^2 \bullet, \bullet) > 0$$

as both $-A_{g,\omega}^2$ and $Q_{g,\omega}^{-1}$ are g-positive definite.

The map $(g,\omega)\mapsto J_{g,\omega}$ is smooth, we show that it is also $\mathrm{GL}(V)$-equivariant. For $\Psi\in\mathrm{GL}(V)$, first we see that $A_{\Psi^*g,\Psi^*\omega}=\Psi^*A_{g,\omega}=\Psi A_{g,\omega}\Psi^{-1}$:

$$(\Psi^*g)(\Psi A_{g,\omega}\Psi^{-1}v,u)=g(A_{g,\omega}\Psi^{-1}v,\Psi^{-1}u)=\omega(\Psi^{-1}v,\Psi^{-1}u)=(\Psi^*\omega)(v,u).$$

And $\Psi^*Q_{g,\omega}$ is Ψ^*g-self adjoint and positive definite and squaring gives $-\Psi^*A_{g,\omega}^2=-(\Psi^*A_{g,\omega})^2$. Thus

$$J_{\Psi^*g,\Psi^*\omega}=(\Psi^*Q_{g,\omega})^{-1}(\Psi^*A_{g,\omega})=\Psi^*(Q_{g,\omega}^{-1}A_{g,\omega})=\Psi^*J_{g,\omega}$$

as desired.

Now we show that this construction satisfies (1): we already showed that $J_{g,\omega}\in\mathcal{J}(V,\omega)\cap\mathcal{J}(V,g)$. And for $g=\omega(\bullet,J\bullet)$, we have that $A_{g,\omega}=J$ as

$$g(Jv,u)=g(u,Jv)=\omega(u,J^2v)=-\omega(u,v)=\omega(v,u),$$

then $Q=I$ is g-symmetric and positive definite and satisfies $Q^2=-A^2$, so $J_{g,\omega}=J$ as desired.

Now we need to show (2). For a set $g\in\mathrm{Met}(V)$, the composite

$$\psi\colon\mathcal{S}(V)\xrightarrow{\omega\mapsto J_{g,\omega}}\mathcal{J}(V)\xrightarrow{J\mapsto\omega_{g,J}}\mathcal{S}(V),\qquad\omega\mapsto\omega_{g,J_{g,\omega}}$$

is homotopic to the identity via the homotopy

$$\psi_\bullet\colon[0,1]\times\mathcal{S}(V)\to\mathcal{S}(V),\qquad\psi_t(\omega)=(1-t)\omega+t\omega_{g,J_{g,\omega}}.$$

This is well-defined as $J_{g,\omega}\in\mathcal{J}(V,\omega)\cap\mathcal{J}(V,\omega_{g,J_{g,\omega}})$, so it is compatible with both of these symplectic structures, thus $\psi_t(\omega)(v,J_{g,\omega}v)>0$ and so $\psi_t(\omega)$ is symplectic for all t. As $\psi_0=\mathrm{id}$ and $\psi_1=\psi$ we have the desired homotopy.

And consider the composite

$$\varphi\colon\mathcal{J}(V)\xrightarrow{J\mapsto\omega_{g,J}}\mathcal{S}(V)\xrightarrow{\omega\mapsto J_{g,\omega}}\mathcal{J}(V),\qquad J\mapsto J_{g,\omega_{g,J}}.$$

This homotopic to the identity via $\varphi_\bullet\colon[0,1]\times\mathcal{J}(V)\to\mathcal{J}(V)$ by

$$\varphi_t(J)=J_{g_t,\omega_{g_t,J}},\qquad g_t=\frac{1+t}{2}g+\frac{1-t}{2}J^*g.$$

Because $g_1=g$ we see that $\varphi_1(J)=J_{g,\omega_{g,J}}=\varphi(J)$. And $g_0=\frac{1}{2}(g+J^*g)$ is compatible with J (because $(-I)^*g=g$), and thus also with

$$\omega_0=\omega_{g_0,J}=\frac{1}{2}(g_0(J\bullet,\bullet)-g_0(\bullet,J\bullet))=\frac{1}{2}(g(J\bullet,\bullet)-g(J\bullet,\bullet))=\omega_{g,J}$$

so that

$$\psi_0(J)=J_{g_0,\omega_0}=J$$

as desired.

For (3), for $\omega\in\mathcal{S}(V,g)$ there is a $J\in\mathcal{J}(V)$ such that $g=\omega(\bullet,J\bullet)$, thus

$$J_{g,\omega}=J,\qquad\omega_{g,J_{g,\omega}}=\frac{1}{2}(g(J\bullet,\bullet)-g(\bullet,J\bullet))=\omega.$$

And so the composition $\mathcal{S}(V,g)\hookrightarrow\mathcal{S}(V)\to\mathcal{S}(V,g)$ is the identity. The other direction of the composition is ψ, which is homotopic to the identity as we already showed.

Similarly for $J\in\mathcal{J}(V,g)$, we have that

$$\omega_{g,J}=g(J\bullet,\bullet)\Longrightarrow g=\omega_{g,J}(\bullet,J\bullet)$$

and so

$$J_{g,\omega_{g,J}}=J$$

so that $\mathcal{J}(V,g)\hookrightarrow\mathcal{J}(V)\to\mathcal{J}(V,g)$ is the identity, and the other direction is φ, which is homotopic to the identity. ∎

Corollary 1.1.5.8

For every $\omega \in \mathcal{S}(V)$, there is a homotopy equivalence

$$\mathcal{J}(V,\omega) \underset{g \mapsto J_{g,\omega}}{\overset{J \mapsto g_J}{\rightleftarrows}} \mathrm{Met}(V), \qquad g_J = \omega(\bullet, J\bullet).$$

Thus $\mathcal{J}(V,\omega)$ is contractible.

Proof

By the above proposition, $J_{g_J,\omega} = J$ so one composition is the identity. The other composition is $\mathrm{Met}(V) \to \mathrm{Met}(V)$ given by $g \mapsto g_{J_{g,\omega}}$, which is homotopic to the identity because $\mathrm{Met}(V)$ is convex. ■

Note:
If $X \subseteq V$ is a convex subset of a topological vector space V, then every map $f\colon X \to X$ is homotopic to the identity via linear interpolation:

$$\varphi_t(x) = tf(x) + (1-t)x.$$

We now need to briefly explain Haar measures. Let V, W be normed real vector spaces of potentially infinite dimension. A linear map $F\colon V \to W$ is **bounded** if there is some real $C \geq 0$ such that

$$|F(v)|_W \leq C\,|v|_V \qquad \text{for all } v \in V.$$

This is equivalent to the following quantity being finite:

$$|F|_{\mathsf{op}} = \sup_{0 \neq v \in V} \frac{|Fv|_W}{|v|_V} = \sup_{|v|_V = 1} |Fv|_W < \infty.$$

This makes the space of bounded linear maps $\hom(V,W)$ into a normed vector space, as is easily verified.
The **Haar measure** on a compact topological group G is the unique bounded linear functional

$$C(G,\mathbb{R}) \to \mathbb{R}, \qquad f \mapsto \int_G f\,d\mu$$

from the space of continuous maps $C(G,\mathbb{R})$ (with the supremum norm), which is invariant under both left and right translations, and the Haar measure of the constant function 1 is 1. By invariance under left and right translations, we mean

$$\int_G f(\Phi\bullet)\,d\mu = \int_G f(\bullet\Phi)\,d\mu = \int_G f(\bullet)\,d\mu \qquad \text{for all } \Phi \in G.$$

The reason for calling this the Haar *measure* is because it can be, and commonly is, constructed as a measure μ over the Borel subsets of G which is unique in that it has certain properties, and the linear functional is just the Lebesgue integral of f relative to μ. Thus we can utilize results regarding the Lebesgue integral when discussing the Haar measure.
We will now prove the following proposition with the help of some lemmas.

Proposition 1.1.5.9

Every compact subgroup $G \subseteq \mathrm{Sp}(2n)$ is conjugate to a subgroup of $\mathrm{U}(n)$.

Lemma 1.1.5.10

There exists a symmetric positive definite $S \in \mathrm{Mat}_{2n}$ such that $\Psi^\top S \Psi = S$ for all $\Psi \in G$.

Proof

For all pairs of indices i, j, define the continuous linear functional $\pi_{i,j}\colon \mathrm{Mat}_{2n} \to \mathbb{R}$ mapping a matrix to its value at that coordinate, $A \mapsto A_{ij}$. Then define

$$S_{ij} = \int_{\Psi \in G} \pi_{ij}(\Psi^\top \Psi)\, d\mu.$$

Because $\Psi^\top \Psi$ is symmetric, we see immediately that S is. And S is positive definite: for $v = (v_1, \ldots, v_{2n})$ we have

$$v^\top S v = \sum_{i,j} v_i S_{ij} v_j = \sum_{i,j} \int_{\Psi \in G} v_i \pi_{ij}(\Psi^\top \Psi) v_j = \int_{\Psi \in G} v^\top \Psi^\top \Psi v.$$

Because $\Psi^\top \Psi$ is positive-definite, this is the integral of a positive function and is thus positive, so S is positive-definite.

A similar computation shows that for every $\Phi \in G$ we have

$$\Phi^\top S \Phi = \int_{\Psi \in G} \Phi^\top \Psi^\top \Psi \Phi \, d\mu = \int_{\Psi \in G} (\Psi\Phi)^\top (\Psi\Phi)\, d\mu = S$$

because of the left invariance of the Haar measure. ■

Lemma 1.1.5.11

There is an ω_0-compatible complex structure $J \in \mathcal{J}(\mathbb{R}^{2n}, \omega_0)$ which commutes with every element of G.

Proof

Let S be as in the above lemma, and let $g = \langle \bullet, S\bullet \rangle \in \mathrm{Met}(\mathbb{R}^{2n})$ be the inner product determined by S. Then both g and ω_0 are G-invariant, and thus so is $J_{g,\omega}$ from proposition 1.1.5.7: for $\Psi \in G$,

$$\Psi^* J_{g,\omega} = J_{\Phi^* g, \Phi^* \omega} = J_{g,\omega}.$$ ■

Proof (1.1.5.9)

Let J be as in the previous lemma, so there is a vector space isomorphism $\Phi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ such that $\Phi^* \omega_0 = \omega_0$ and $\Phi^* J = J_0$. Thus $\Phi \in \mathrm{Sp}(2n)$ and $\Phi J \Phi^{-1} = J_0$. For every $\Psi \in G$, $\Phi\Psi\Phi^{-1} \in \mathrm{Sp}(2n)$ commutes with J_0 and thus is in $\mathrm{U}(n)$. So we have

$$\Phi G \Phi^{-1} \subseteq \mathrm{U}(n)$$

as desired. ■

Remark 1.1.5.12

Let V be even-dimensional and define

$$\mathcal{C}(V) = \{(\omega, J) \mid \omega \in \mathcal{S}(V),\ J \in \mathcal{J}(V, \omega)\},$$

the space of compatible pairs. The projections $\pi_{\mathcal{S}}\colon \mathcal{C}(V) \to \mathcal{S}(V)$ and $\pi_{\mathcal{J}}\colon \mathcal{C}(V) \to \mathcal{J}(V)$ are then homotopy

equivalences. Indeed for $g \in \mathrm{Met}(V)$, the maps

$$\begin{aligned} \iota_{\mathcal{S}} &: \mathcal{S}(V) \to \mathcal{C}(V), \qquad \iota_{\mathcal{S}}(\omega) = (\omega, J_{g,\omega}) \\ \iota_{\mathcal{J}} &: \mathcal{J}(V) \to \mathcal{C}(V), \qquad \iota_{\mathcal{J}}(J) = (\omega_{g,J}, J) \end{aligned}$$

are homotopy inverses of $\omega_{\mathcal{S}}, \omega_{\mathcal{J}}$ respectively. Indeed we see that $\pi_{\mathcal{S}} \circ \iota_{\mathcal{S}} = \mathrm{id}$ and $\pi_{\mathcal{J}} \circ \iota_{\mathcal{J}} = \mathrm{id}$. And $\iota_{\mathcal{S}} \circ \pi_{\mathcal{S}}$ maps (ω, J) to $(\omega, J_{g,\omega})$ which is homotopic to the identity via

$$[0,1] \times \mathcal{C}(V) \to \mathcal{C}(V), \qquad (t, J, \omega) \mapsto (\omega, J_{tg+(1-t)\omega(\bullet,J\bullet),\omega}),$$

and $\iota_{\mathcal{J}} \circ \pi_{\mathcal{J}}$ maps (ω, J) to $(\omega_{g,J}, J)$ which is homotopic to the identity via

$$[0,1] \times \mathcal{C}(V) \to \mathcal{C}(V), \qquad (t, J, \omega) \mapsto (t\omega_{g,J} + (1-t)\omega, J).$$

Remark 1.1.5.13

We can understand the above remark by focusing on the case $V = \mathbb{R}^{2n}$ with the standard symplectic and complex structures. We then get the following diagram

$$\begin{array}{ccccccccc} \frac{\mathrm{O}(2n)}{\mathrm{U}(n)} & \hookrightarrow & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{GL}_n(\mathbb{C})} & \hookleftarrow & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{U}(n)} & \twoheadrightarrow & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{Sp}(2n)} & \hookleftarrow & \frac{\mathrm{O}(2n)}{\mathrm{U}(n)} \\ \downarrow \sim & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ \mathcal{J}(V,g) & \hookrightarrow & \mathcal{J}(V) & \xleftarrow{\pi_{\mathcal{J}}} & \mathcal{C}(V) & \xrightarrow{\pi_{\mathcal{S}}} & \mathcal{S}(V) & \hookleftarrow & \mathcal{S}(V,g) \end{array}$$

The horizontal arrows are the homotopy equivalences discussed thus far, and the vertical arrows are diffeomorphisms. The outer vertical arrows are obtained by quotienting $\Psi \mapsto \Psi^* J_0$ or $\Psi^* \omega_0$ and quotienting. The inner vertical arrow is defined as follows. Recall that by proposition 1.1.5.5 $(\omega, J) \in \mathcal{C}(V)$ iff there is a linear isomorphism $\Psi : \mathbb{R}^{2n} \to V$ such that $\Psi^* \omega_0 = \omega_0$ and $\Psi J_0 \Psi^{-1} = J$. So define the surjection

$$\mathrm{GL}_{2n}(\mathbb{R}) \to \mathcal{C}(\mathbb{R}^{2n}), \qquad \Psi \mapsto \Psi^*(\omega_0, J_0) = (\Psi^* \omega_0, \Psi J_0 \Psi^{-1}).$$

The kernel is $\mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n)$, giving the desired diffeomorphism.

Definition 1.1.5.14

Let (V, ω) be a symplectic space. Then a complex linear structure $J \in \mathcal{J}(V)$ is called **ω-tame** if

$$\omega(v, Jv) > 0 \text{ for all } v \neq 0.$$

Denote the space of ω-tame complex structures by $\mathcal{J}_\tau(V, \omega)$.

Note that $\mathcal{J}_\tau(V, \omega) \subseteq \mathcal{J}(V, \omega)$ is open, and for $J \in \mathcal{J}_\tau(V, \omega)$,

$$g_{\omega,J}(v, u) = \frac{1}{2}(\omega(v, Ju) - \omega(Jv, u))$$

is an inner product.

Proposition 1.1.5.15

The space $\mathcal{J}_\tau(V,\omega)$ is contractible.

Proof

Define the space of tame pairs

$$\mathcal{C}_\tau(V) = \{(\omega, J) \mid \omega \in \mathcal{S}(V),\ J \in \mathcal{J}_\tau(V,\omega)\}.$$

Now for $(\omega, J) \in \mathcal{C}_\tau(V)$ define the 2-form

$$\pi_1(\omega, J) = \frac{1}{2}(\omega + J^*\omega).$$

Then

$$\frac{1}{2}(\omega + J^*\omega)(\bullet, J\bullet) = \frac{1}{2}(\omega(\bullet, J\bullet) - \omega(J\bullet, \bullet)) = g_{\omega,J},$$

which means that $\pi_1(\omega, J)$ is non-degenerate and thus symplectic. Furthermore $\pi_1(\omega, J)$ is compatible with J and defines the same inner product as ω, $g_{\omega,J}$.

The fiber of the projection

$$\text{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{C}(V), \qquad (\omega, J) \mapsto \left(\frac{1}{2}(\omega + J^*\omega), J\right)$$

over $(\omega, J) \in \mathcal{C}(V)$ consists of all $(\omega + \tau, J) \in \mathcal{C}_\tau(V)$ for $\tau \in \Lambda^2(V^*)$ such that $\tau + J^*\tau = 0$. Any such pair is tame, as

$$\tau(v, Jv) = -\tau(Jv, J^2v) = \tau(v, Jv) \Longrightarrow \tau(v, Jv) = 0$$

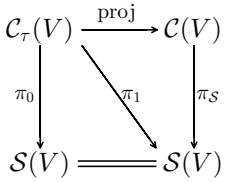
and thus $(\omega + \tau)(v, Jv) > 0$ as ω is compatible.

Note that the set of all 2-forms τ such that $\tau + J^*\tau = 0$ is a vector space. Thus we see that $\text{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{C}(V)$ is a vector bundle, and thus is a homotopy equivalence. By remark 1.1.5.12 so is $\pi_{\mathcal{S}}\colon \mathcal{C}(V) \to \mathcal{S}(V)$, and their composition $\pi_{\mathcal{S}} \circ \text{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V)$ is the map π_1.

Define

$$\pi_0\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V), \qquad \pi_0(\omega, J) = \omega$$

and consider the diagram



The top triangle commutes, but the bottom does not. Despite this, π_1 and π_0 are homotopic via $\pi_t = (1-t)\pi_0 + t\pi_1$: because both $\pi_0(\omega, J)$ and $\pi_1(\omega, J)$ are compatible with J, their convex combinations are symplectic. Thus $\pi_0\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V)$ is a homotopy equivalence, and as it is a fibration, its fibers $\pi_0^{-1}\omega \cong \mathcal{J}_\tau(V,\omega)$ are contractible. ■

We now construct a smooth map

$$\text{Met}(V) \times \mathcal{S}(V) \times \Lambda^2(V^*) \to \text{end}(V), \qquad (g, \omega, \widehat{\omega}) \mapsto A_g(\omega, \widehat{\omega})$$

which has the following properties:

(1) $\widehat{\omega} + \omega(A_g(\omega,\widehat{\omega})\bullet, \bullet) + \omega(\bullet, A_g(\omega,\widehat{\omega})\bullet) = 0,$

(2) For every $B \in \text{end}(V)$: if $\omega(B\bullet, \bullet) + \omega(\bullet, B\bullet) = 0$ then

$$\langle A_g(\omega,\widehat{\omega}), B\rangle_g = 0,$$

where $\langle \bullet, \bullet \rangle_g$ is the induced inner product on $\operatorname{end}(V)$.

(3) If $\Phi\colon V' \to V$ is an isomorphism,

$$A_{\Phi^* g}(\Phi^*\omega, \Phi^*\widehat{\omega}) = \Phi^{-1} A_g(\omega, \widehat{\omega})\Phi.$$

The conditions (1) and (2) uniquely define A_g. Indeed, consider the map

$$F\colon \operatorname{end}(V) \to \Lambda^2(V^*), \qquad F(A) = \omega(A\bullet, \bullet) + \omega(\bullet, A\bullet).$$

The kernel of this map is $\ker F = \mathfrak{sp}(V, \omega)$, which by proposition 1.1.1.5 has dimension $2n^2 + n$. And so we have that

$$\dim \ker F + \dim \Lambda^2(V^*) = 2n^2 + n + 2n^2 - n = (2n)^2 = \dim \operatorname{end}(V),$$

meaning F is surjective. Thus there are $A \in \operatorname{end}(V)$ such that $F(A) = -\widehat{\omega}$, i.e. satisfying (1).

Condition (2) requires that A be orthogonal to $\ker F$. In general if we have a direct sum $V = U \oplus W$, then for every $v \in V$, the intersection $(v + U) \cap W$ is a singleton. This is because the canonical map $W \to V/U$ is an isomorphism, so every coset $v + U \in V/U$ has a single element of W. In this case we consider $\operatorname{end}(V) = \ker F \oplus (\ker F)^\perp$, then $F^{-1}(-\widehat{\omega})$ is an affine subset of the form $A + \ker F$, and thus has a unique intersection with $(\ker F)^\perp$; this gives us the unique $A_g(\omega, \widehat{\omega})$.

Point (3) follows directly from the existence and uniqueness imposed by (1) and (2).

We also construct a smooth map

$$\operatorname{Met}(V) \times [0, 1] \times \mathcal{C}_\tau(V) \to \operatorname{GL}(V), \qquad (g, t, \omega, J) \mapsto \Psi_g(t, \omega, J)$$

satisfying the following:

(1) $\Psi_g(0, \omega, J) = \mathrm{id}_V$.

(2) If $\omega = J^*\omega$ (i.e. $(\omega, J) \in \mathcal{C}(V)$) then $\Psi_g(t, \omega, J) = \mathrm{id}_V$.

(3) For all g, t, ω, J,

$$\Psi_g(t, \omega, J)^* \left(\omega + \frac{t}{2}(J^*\omega - \omega)\right) = \omega.$$

(4) For any isomorphism $\Phi\colon V' \to V$,

$$\Psi_{\Phi^* g}(t, \Phi^*\omega, \Phi^* J) = \Phi^{-1}\Psi_g(t, \omega, J)\Phi.$$

For $g \in \operatorname{Met}(V), \omega \in \mathcal{S}(V), J \in \mathcal{J}_\tau(V, \omega)$, define $t \mapsto \Psi_g(t, \omega, J)$ to be the unique solution of the ODE

$$\frac{d}{dt}\Psi_g(t, \omega, J) = A_g\left(\omega + \frac{t}{2}(J^*\omega - \omega), \frac{1}{2}(J^*\omega - \omega)\right)\Psi_g(t, \omega, J),$$

with the initial condition $\Psi_g(0, \omega, J) = \mathrm{id}$. Abbreviate

$$\omega_t = \omega + \frac{t}{2}(J^*\omega - \omega), \qquad \widehat{\omega} = \frac{1}{2}(J^*\omega - \omega), \qquad \Psi_t = \Psi_g(t, \omega, J), \qquad A_t = A_g(\omega_t, \widehat{\omega}).$$

Then the ODE becomes

$$\partial_t \Psi_t = A_t \Psi_t, \qquad \Psi_0 = \mathrm{id}, \qquad \partial_t \omega = \widehat{\omega}.$$

Thus

$$\begin{aligned}
\frac{d}{dt}\Psi_t^*\omega_t &= \Psi_t^*\partial_t\omega_t + \omega_t(\partial_t\Psi_t\bullet, \Psi_t\bullet) + \omega_t(\Psi_t\bullet, \partial_t\Psi_t\bullet) \\
&= \Psi_t^*\widehat{\omega} + \omega_t(A_t\Psi_t\bullet, \bullet) + \omega_t(\Psi_t\bullet, A_t\Psi_t\bullet) \\
&= \Psi_t^*(\widehat{\omega} + \omega_t(A_g(\omega_t, \widehat{\omega})\bullet, \bullet) + \omega_t(\bullet, A_g(\omega_t, \widehat{\omega})\bullet)) \\
&= 0.
\end{aligned}$$

Thus we see $\Psi_t^*\omega_t$ is constant, and thus equal to $\Psi_0^*\omega_0 = \omega$. Thus Ψ_t satisfies (3).

When $\omega = J^*\omega$ then $\omega_t = \omega$ and $\widehat{\omega} = 0$, thus $A_t = A_g(\omega, 0) = 0$ (because $A_g(\omega, 0)$ is in the kernel of the map defined above, but also orthogonal to it, thus it must be zero). And so Ψ_t is constant, and so equal to id, showing (2).

Equivariance, point (4), follows from equivariance of A_g and uniqueness of solutions to ODEs. Smoothness follows from smoothness of solutions to ODEs.

These definitions allow us to define a homotopy inverse to $\mathcal{J}(V,\omega) \hookrightarrow \mathcal{J}_\tau(V,\omega)$ by choosing an inner product g and defining

$$\mathcal{J}_\tau(V,\omega) \to \mathcal{J}(V,\omega), \qquad J \mapsto \Psi_g(1,\omega,J)^{-1} J \Psi_g(1,\omega,J).$$

Now define

$$f_\bullet\colon [0,1] \times \mathcal{J}_\tau(V,\omega) \to \mathcal{J}_\tau(V,\omega), \quad f_t(J) = \Psi_g(t,\omega,J)^* J, \qquad \omega_t = \omega + \frac{t}{2}(J^*\omega - \omega).$$

Because $\Psi_g(t,\omega,J)^*\omega_t = \omega$ and $J \in \mathcal{J}_\tau(V,\omega_t)$, we have that $f_t(J) = \Psi_g(t,\omega,J)^* J \in \mathcal{J}_\tau(V,\omega)$ for all t, so this is well-defined. If $J \in \mathcal{J}(V,\omega)$ then $\Psi_g(t,\omega,J) = \mathrm{id}$ and so $f_t(J) = J$. And for $t = 1$, J is compatible with ω_1 so $f_1(\bullet) \in \mathcal{J}(V,\omega)$.

Thus $f_\bullet$ forms a strong homotopy retract $\mathcal{J}(V,\omega) \hookrightarrow \mathcal{J}_\tau(V,\omega)$, so they are homotopy equivalent, and thus $\mathcal{J}_\tau(V,\omega)$ is contractible. This gives an alternative proof of the above proposition.

Proposition 1.1.5.16

Let ω_t be a family of symplectic forms on V, depending smoothly on $t \in \mathbb{R}$. Then there exists a smooth family of isomorphisms $\Phi_t\colon V \to V$ such that $\Phi_0 = \mathrm{id}$ and $\Phi_t^*\omega_t = \omega_0$ for all t.

Proof

Fix $g \in \mathrm{Met}(V)$ and define, similar to above,

$$\Psi_t = \Psi_g(t,\omega_0,J), \qquad A_t = A_g(\omega_t, \widehat{\omega}), \qquad \widehat{\omega} = \partial_t \omega.$$

Then let Ψ_t be the unique solution to the ODE with initial condition

$$\partial_t \Psi_t = A_t \Psi_t, \qquad \Psi_0 = \mathrm{id}\,.$$

The same computation as above shows that $\partial_t \Psi_t^* \omega_t = 0$ and thus $\Psi_t^*\omega_t = \omega_0$ for all t. ∎

1.6 Symplectic vector bundles

Definition 1.1.6.1

A **symplectic vector bundle** over a manifold M is a pair (E,ω) where $\pi\colon E \to M$ is a real vector bundle and $\omega \in \Gamma(\Lambda^2(E^*))$ is a smooth tensor field such that for all $p \in M$, ω_p is a symplectic form on $E_q = \pi^{-1}q$.

Two symplectic vector bundles (E_i,ω_i), for $i = 1,2$, are isomorphic if there is a vector bundle isomorphism $\Psi\colon E_1 \to E_2$ such that $\Psi^*\omega_2 = \omega_1$.

Example 1.1.6.2

If $E \to M$ is a vector bundle, $E \oplus E^*$ has a canonical structure as a symplectic vector bundle via

$$\Omega_{\mathsf{can}}(v \oplus \varphi, u \oplus \psi) = \psi(v) - \varphi(u).$$

Lemma 1.1.6.3

If (E, ω) is a symplectic vector bundle over M, $F \subseteq E$ a vector subbundle, then $F^\omega \subseteq E$ is also a vector subbundle defined pointwise by

$$(F^\omega)_p = (F_p)^{\omega_p}.$$

Proof

Define

$$\widehat{\omega}\colon E \to F^*, \qquad \widehat{\omega}(\varepsilon)(\eta) = \omega(\varepsilon, \eta) \quad \text{for } \varepsilon \in E, \eta \in F,$$

then $\widehat{\omega}$ is a smooth vector bundle morphism and $F^\omega = \ker \widehat{\omega}$, which is a subbundle iff F has constant rank. And indeed it does, its rank at each point is $\dim(F_p)^\omega = \dim E_p - \dim F_p = \operatorname{rank} E - \operatorname{rank} F$. ■

Proposition 1.1.6.4

Let (E, ω) be a symplectic vector bundle of rank $2n$ over M, then:

(1) E has a system of local trivializations whose transition maps take values in $\mathrm{Sp}(2n)$.

(2) Every point $p \in E$ has a neighborhood $U \subseteq M$ such that $(\pi^{-1}U, \omega) \subseteq (E, \omega)$ is isomorphic to the symplectic bundle $(U \times \mathbb{R}^{2n}, \omega_0)$.

(3) Every point $p \in E$ has a local frame which forms a symplectic basis at each point (called a **symplectic frame**).

Furthermore (2) and (3) are equivalent to (E, ω) being symplectic, and (1) gives E a symplectic structure.

Proof

Clearly (2) and (3) are equivalent, as a symplectic frame $\varepsilon_1, \dots, \varepsilon_n, \delta_1, \dots, \delta_n$ over U defines an isomorphism from $(U \times \mathbb{R}^{2n}, \omega_0)$ by

$$(p, (x_1, \dots, x_n, y_1, \dots, y_n)) \mapsto \sum_{i=1}^{n} x_i \varepsilon_i(p) + y_i \delta_i(p),$$

and conversely such an isomorphism induces a symplectic frame by considering the image of the symplectic frame $x_1, \dots, x_n, y_1, \dots, y_n$ of $U \times \mathbb{R}^{2n}$. Clearly (2) implies that (E, ω) is symplectic, as through these isomorphisms we see that it is pointwise antisymmetric, bilinear, and nondegenerate.

We show that (E, ω) being symplectic implies (3). Let ε_1, δ_1 be local sections around $p \in E$ such that $\omega(\varepsilon_1, \delta_1) \neq 0$. We can find them by choosing $e_1, f_1 \in E_p$ such that $\omega_p(e_1, f_1) \neq 0$ and extending them to local sections, and then by continuity there is a neighborhood where $\omega(\varepsilon_1, \delta_1) \neq 0$. Then ε_1, δ_1 generate a symplectic subbundle of $E|_U$, and taking its symplectic complement we obtain a symplectic subbundle of rank $2n - 2$. By induction we get a local symplectic frame.

Now we show (3) implies (1). If $\{\varepsilon_i^\alpha, \delta_i^\alpha\}_{i \leq n}$ and $\left\{\varepsilon_i^\beta, \delta_i^\beta\right\}_{i \leq n}$ are local symplectic frames over U_α, U_β respectively, then the transition maps at each point corresponds to a transition map between symplectic bases. This is a symplectic matrix, and so the transition map takes values in $\mathrm{Sp}(2n)$ as desired.

Now assume that $(U_\alpha, \Phi_\alpha)_\alpha$ is a system of local trivializations whose transition maps take values in $\mathrm{Sp}(2n)$, then they induce a symplectic structure on E. Indeed, taking an α, let $\varepsilon_i^\alpha = \Phi_\alpha^{-1} \circ x_i$ and $\delta_i^\alpha = \Phi_\alpha^{-1} \circ y_i$ be the pullbacks of the symplectic frames of $U_\alpha \times \mathbb{R}^{2n}$. Then define ω_α on $\pi^{-1}U_\alpha$ via this local frame. Because the transition maps take values in $\mathrm{Sp}(2n)$, this means that ω_α and ω_β agree on their overlaps, and thus glue together to form a symplectic structure on E. ■

Definition 1.1.6.5

A **complex structure** on a vector bundle $E \to M$ is an automorphism J of E such that $J^2 = -\mathrm{id}$. If (M, ω) is symplectic, J is said to be **compatible** with ω if J_p is compatible with ω_p for all $p \in M$. Any compatible pair (ω, J) defines a metric g_J on E bt

$$g_J(v, u) = \omega(v, Ju).$$

Such a triple (ω, J, g) is called a **Hermitian structure** on E.

Proposition 1.1.6.6

Let $E \to M$ be a rank $2n$ vector bundle.

(1) For every symplectic structure ω on E, the space of ω-compatible and ω-tame complex structures, denoted $\mathcal{J}(E, \omega)$ and $\mathcal{J}_\tau(E, \omega)$ respectively, are nonempty and contractible.

(2) Let J be a complex structure on E, then the space of symplectic structures on E compatible with J (or which tame J) is nonempty and contractible.

(3) Let g be a metric on E, then

$$\mathcal{S}(E) \underset{J \mapsto \omega_{g,J}}{\overset{\omega \mapsto J_{g,\omega}}{\rightleftarrows}} \mathcal{J}(E)$$

form a homotopy equivalence.

Proof

For (1) consider the maps

$$\mathcal{J}(E, \omega) \underset{g \mapsto J_{g,\omega}}{\overset{J \mapsto g_J}{\rightleftarrows}} \mathrm{Met}(E).$$

These are homotopy inverses, by corollary 1.1.5.8. And $\mathrm{Met}(E) \subseteq \mathrm{Sym}^2(V^*)$ is a convex subset of a vector space, and thus contractible.

Now we need to show that $\mathcal{J}_\tau(E, \omega)$ is contractible. Let $J_0 \in \mathcal{J}(E, \omega)$, and denote by $\langle \bullet, \bullet \rangle_0 = g_{J_0}$ its associated metric on E; and denote its associated norm by $|\bullet|_0$. This induces the operator norm $\mathrm{end}(E_p)$ for $p \in M$, which we denote by $|\bullet|_0$ as well. Now define

$$\mathcal{A} = \left\{ A \in \mathrm{end}(E) \;\middle|\; AJ_0 + J_0 A = 0,\ |A_p|_0 < 1 \text{ for all } p \in M \right\}.$$

Then define

$$\mathcal{F} \colon \mathcal{J}_\tau(E, \omega) \to \mathcal{A}, \qquad \mathcal{F}(J) = (\mathrm{id} + J_0 J)(\mathrm{id} - J_0 J)^{-1},$$

notice that $\mathrm{id} - J_0 J$ is invertible because if $J_0 J v = v$ then $Jv = -J_0 v$, but then

$$\omega(v, Jv) = -\omega(v, J_0 v) < 0,$$

contradicting tameness. It is then (non-trivial) linear algebra to show that this is well-defined.

$\mathcal{F}$ is a homeomorphism with inverse

$$\mathcal{F}^{-1} \colon \mathcal{A} \to \mathcal{J}_\tau(E, \omega), \qquad \mathcal{F}^{-1}(A) = J_0(\mathrm{id} + A)^{-1}(\mathrm{id} - A).$$

Thus $\mathcal{J}_\tau(E, \omega)$ is homeomorphic to the convex $\mathcal{A}$, so it is contractible.

In (2), the existence of a symplectic structure compatible with J is equivalent to the existence of a metric g with respect to which J is skew-adjoint. Take any metric g and consider $\frac{1}{2}(g + J^* g)$, giving the desired result. Contractibility follows from the fact that the space in question is convex.

(3) follows from proposition 1.1.5.7; the same definitions form the desired homotopy inverses. ∎

Lemma 1.1.6.7

Suppose J_t is a smooth family of complex structures on E. Then there is a smooth family of isomorphisms $\Phi_t \colon E \to E$ such that $\Phi_t^* J_t = J_0$.

Proof

Define

$$B_t = \frac{1}{2}\partial_t J_t J_t,$$

and then consider

$$[B_t, J_t] = B_t J_t - J_t B_t = \frac{1}{2}\partial_t J_t J_t^2 - \frac{1}{2} J_t \partial_t J_t J_t = -\frac{1}{2}\partial_t J_t - \frac{1}{2}\partial_t J_t = \partial_t J_t,$$

here we use the fact that $J_t \partial_t J_t J_t = \partial_t J_t$, this is because differentiating $J_t^2 = -\operatorname{id}$ gives

$$\partial_t J_t J_t + J_t \partial_t J_t = 0 \implies J_t \partial_t J_t J_t = \partial_t J_t.$$

Then consider the linear ODE

$$\partial_t \Phi_t = B_t \Phi_t, \qquad \Phi_0 = \operatorname{id},$$

which has a unique smooth solution. Furthermore Φ_t is an isomorphism (this is a classical result in ODE theory).

Now define $D_t = J_t \Phi_t - \Phi_t J_0$, then

$$\partial_t D_t = \partial_t J_t \Phi_t + J_t B_t \Phi_t - B_t \Phi_t J_0 = (\partial_t J_t + J_t B_t - B_t J_t)\Phi_t + B_t D_t = B_t D_t.$$

Thus D_t solves the ODE $\partial_t D_t = B_t D_t$ with the initial value $D_0 = 0$. Because $D_t = 0$ solves this problem, by uniqueness $D_t = 0$ as desired. ■

Theorem 1.1.6.8

Let (E_i, ω_i) be symplectic vector bundles over M, and let J_i be ω_i-tame for $i = 1, 2$. Then (E_1, ω_1) and (E_2, ω_2) are isomorphic iff the complex vector bundles (E_1, J_1) and (E_2, J_2) are isomorphic.

Proof

Suppose there is an isomorphism $\Psi \colon E_1 \to E_2$ with $\omega_1 = \Psi^* \omega_2$. Then both J_1 and $\Psi^* J_2$ are ω_1-tame, so by (1) in proposition 1.1.6.6, there is a smooth family of complex structures $J_t \colon E_1 \to E_2$ which are ω_1-tame and $J_0 = \Psi^* J_2$. By the above lemma there is a smooth family of isomorphisms $\Phi_t \colon E_1 \to E_1$ such that $\Phi_t^* J_t = J_1$. Then $\Psi \circ \Phi_0 \colon E_1 \to E_2$ is an isomorphism and

$$(\Psi \circ \Phi_0)^* J_2 = \Phi_0^* \Psi^* J_2 = \Phi_0^* J_0 = J_1,$$

as desired.

Conversely suppose there is a bundle isomorphism $\Psi \colon E_1 \to E_2$ such that $\Psi^* J_2 = J_1$. Then $\Psi^* \omega_2, \omega_1$ are symplectic forms on E_1 which tame J_1, and therefore so is

$$\omega_t = t\omega_1 + (1-t)\Psi^* \omega_2,$$

and so $\Psi^* \omega_2, \omega_1$ are connected by a smooth family of symplectic forms. By proposition 1.1.5.16 there is a smooth family of isomorphisms $\Phi_t \colon E_1 \to E_1$ such that $\Phi_1 = \operatorname{id}$ and $\Phi_t^* \omega_t = \omega_1$. Thus $\Phi_0^* \Psi^* \omega_2 = \omega_1$, meaning $\Psi \circ \Phi_0$ is the desired isomorphism. ■

2 Symplectic Manifolds

2.1 Basic concepts

We will assume, unless otherwise specified, that the manifolds in question are smooth, connected, and without boundary. A **closed** manifold is a compact manifold without boundary.

Definition 1.2.1.1

A **symplectic structure** on a manifold M is a 2-form $\omega \in \Omega^2(M)$ which is closed, and for every $p \in M$, ω_p is nondegenerate. A **symplectic manifold** is a pair (M, ω) where ω is a symplectic structure on M.

If M is a symplectic manifold of dimension $2n$, then $\omega^n = \omega \wedge \cdots \wedge \omega$ is a non-vanishing top-form, so that M is oriented. Thus we see that not all even-dimensional manifolds have a symplectic structure.

Example 1.2.1.2

Euclidean space $\mathbb{R}^{2n}$ has a standard symplectic structure

$$\omega_0 = \sum_{i=1}^{n} dx_i \wedge dy_i.$$

Note that this symplectic structure is closed (it is in fact exact); this explains why we want our symplectic structures in general to be closed. We will show that symplectic manifolds are locally symplectomorphic to $(\mathbb{R}^{2n}, \omega_0)$, and this implies that the symplectic form is closed (as it is locally closed).

Example 1.2.1.3

A symplectic structure on a 2-manifold is the same as a volume form. Indeed any 2-form on a 2-manifold is closed, and any nonvanishing 2-form is nondegenerate.

So for example S^2 has a symplectic structure induced from its standard volume form. Recall that its standard volume form is defined by as

$$\begin{aligned}\omega_{(x_1,x_2,x_3)} = \left(\sum_{i=1}^{3} x_i \partial_i\right) \lrcorner\, dx_1 \wedge dx_2 \wedge dx_3 &= \sum_{i=1}^{3} x_i\, dx_1 \wedge dx_2 \wedge dx_3(\partial_i, \bullet, \bullet) \\ &= x_1\, dx_2 \wedge dx_3(\bullet, \bullet) - x_2\, dx_1 \wedge dx_3(\bullet, \bullet) + x_3\, dx_1 \wedge dx_2(\bullet, \bullet).\end{aligned}$$

We can rewrite this as

$$\omega_x(u, v) = \det \begin{pmatrix} — & x & — \\ — & u & — \\ — & v & — \end{pmatrix} = \langle x, u \times v \rangle .$$

Note:

Henceforth, we will denote interior multiplication by one of the two

$$X \lrcorner\, \omega = \iota(X)\omega = \omega(X, \bullet).$$

By nondegeneracy, a symplectic form induces an isomorphism of vector bundles

$$\widehat{\omega}: TM \to T^*M, \qquad X \mapsto \iota(X)\omega,$$

which is equivalent to forming an isomorphism $\mathfrak{X}(M) \to \Omega^1(M)$ (much like the musical isomorphisms from Riemannian geometry).

Example 1.2.1.4

There exist orientable $2n$-manifolds which have no symplectic structure. If (M,ω) is symplectic, then as ω is closed, it represents an element in the cohomology $[\omega] \in H^2(M;\mathbb{R})$. The n-fold cup product of $[\omega]$ corresponds to $[\omega^n]$. Because ω^n is a volume form, its integral is nonzero, and thus it cannot be exact (by Stokes'). This is equivalent to $[\omega]^n = [\omega^n]$ being nonzero.

So any closed manifold whose 2nd cohomology class is zero cannot have a symplectic. For instance S^{2n} with $n \geq 2$.

There is an obvious notion of symplectomorphism between symplectic manifolds.

Definition 1.2.1.5

A **symplectomorphism** is a diffeomorphism $\psi\colon (M_1,\omega_1) \to (M_2,\omega_2)$ such that $\omega_1 = \psi^*\omega_2$. Denote the group of symplectomorphisms of (M,ω) by

$$\mathrm{Symp}(M,\omega) = \{\psi \in \mathrm{Diff}(M) \mid \psi^*\omega = \omega\}\,.$$

Definition 1.2.1.6

A vector field $X \in \mathfrak{X}(M)$ is **symplectic** if $\iota(X)\omega$ is closed. Denote the space of symplectic vector spaces by $\mathfrak{X}(M,\omega)$.

By Cartan's formula,

$$\mathcal{L}_X\omega = \iota(X)\,d\omega + d(\iota(X)\omega) = d(\iota(X)\omega),$$

so X is symplectic iff $\mathcal{L}_X\omega = 0$.

If $\varphi\colon M \to N$ is a diffeomorphism, it defines an isomorphism $\varphi^*\colon \mathfrak{X}(N) \to \mathfrak{X}(M)$ by mapping X to

$$(\varphi^*X)_p = d\varphi_{\varphi p}^{-1} X_{\varphi p}.$$

Proposition 1.2.1.7

Let M be a closed manifold, and $t \mapsto \psi_t \in \mathrm{Diff}(M)$ a smooth family of diffeomorphisms. Suppose that this family is generate by a smooth family of vector fields $X_t \in \mathfrak{X}(M)$, i.e.

$$\frac{d}{dt}\psi_t = X_t \circ \psi_t, \qquad \psi_0 = \mathrm{id}\,.$$

Then $\psi_t \in \mathrm{Symp}(M,\omega)$ for all t iff $X_t \in \mathfrak{X}(M,\omega)$ for all t.

Furthermore, if $X, Y \in \mathfrak{X}(M,\omega)$ then $[X,Y] \in \mathfrak{X}(M,\omega)$ as

$$\iota([X,Y])\omega = dH, \qquad H = \omega(X,Y).$$

Thus $\mathfrak{X}(M,\omega)$ is a Lie subalgebra of $\mathfrak{X}(M)$.

Proof

The first claim is due to

$$\frac{d}{dt}\psi_t^*\omega = \psi_t^*(\mathcal{L}_{X_t}\omega) = \psi_t^*\, d(\iota(X_t)\omega).$$

And ψ_t are symplectomorphisms iff $\psi_t^*\omega$ is constant, iff the above equation is zero, if $\iota(X_t)\omega$ is closed as desired.

Now let $X, Y \in \mathfrak{X}(M,\omega)$ have flows φ_t, ψ_t respectively. Then

$$[X,Y] = \mathcal{L}_Y X = \left.\frac{d}{dt}\right|_{t=0} \psi_t^* X,$$

so that

$$\iota([X,Y])\omega = \left.\frac{d}{dt}\right|_{t=0} \iota(\psi_t^* X)\omega = \mathcal{L}_Y(\iota(X)\omega) = \iota(Y)\,d\iota(X)\omega + d\iota(Y)\iota(X)\omega = d\omega(X,Y).$$

The second equality here is because, by the first claim, $\psi_t^*\omega = \omega$ for all t, and thus

$$\iota(\psi_t^* X)\omega = \omega(d\psi_t^{-1} X \circ \psi_t, \bullet) = \omega(X(\psi_t), d\psi_t) = \psi_t^*\omega(X, \bullet) = \psi_t^*(\iota(X)\omega).$$

■

Definition 1.2.1.8

Let $H\colon M \to \mathbb{R}$ be smooth. Because $\widehat{\omega}\colon \mathfrak{X}(M) \to \Omega^1(M)$ is an isomorphism, there is a unique $X_H \in \mathfrak{X}(M)$ such that

$$\omega(X_H, \bullet) = dH.$$

This vector field is called the **Hamiltonian vector field** associated to the **Hamiltonian function** H. The flow of X_H is called the **Hamiltonian flow** of H, and is commonly denoted φ_H^t.

It is immediate that Hamiltonian vector fields are symplectic. Hamiltonian vector fields are those for which $\iota(X)\omega$ is exact, and symplectic vector fields are those for which $\iota(X)\omega$ is only closed. Furthermore, $X_H = X_{H'}$ iff $H - H'$ is constant, as this is only when $dH = dH'$.

Notice that

$$dH(X_H) = \omega(X_H, X_H) = 0$$

which means that X_H is tangent to H's regular level sets (recall that the tangent space of $H^{-1}x$ is $\ker dH$).

For two smooth real-valued functions $F, G \in C^\infty(M)$, define their **Poisson bracket** to be

$$\{F,G\} = \omega(X_F, X_G) = dF(X_G) = X_G F.$$

Notice that the Poisson bracket is bilinear and antisymmetric. The former is because $X_{aH} = aX_H$ and $X_{F+G} = X_F + X_G$ clearly, and the latter is because ω is antisymmetric.

Proposition 1.2.1.9

The Poisson bracket turns $C^\infty(M)$ into a Lie algebra, and

$$X_{\{F,G\}} = [X_F, X_G].$$

Proof

We first prove the second statement. It is sufficient to show that that for all $Y \in \mathfrak{X}(M)$,

$$\omega(X_{\{F,G\}}, Y) = \omega([X_F, X_G], Y).$$

The left-hand side is

$$\omega(X_{\{F,G\}}, Y) = d\{F,G\}\,Y = Y\{F,G\} = YX_G F.$$

Because Hamiltonian vector fields are symplectic, we see that

$$0 = \mathcal{L}_{X_G}\omega(X_F, Y) = -X_G\omega(X_F, Y) - \omega([X_G, X_F], Y) - \omega(X_F, [X_G, Y]).$$

The first term is

$$X_G\omega(X_F, Y) = X_G Y F,$$

and the third is

$$\omega(X_F, [X_G, Y]) = [X_G, Y]F = YX_GF - X_GYF = \omega(X_{\{F,G\}}, Y) - X_GYF.$$

Thus

$$\omega(X_{\{F,G\}}, Y) = \omega([X_F, X_G], Y)$$

as desired.

Now the Jacobi identity for the Poisson bracket is because

$$\begin{aligned}\{F, \{G, H\}\} = X_{\{G,H\}}F = [X_G, X_H]F = X_HX_GF - X_GX_HF = X_H\{F, G\} - X_G\{F, H\} \\ = \{\{F, G\}, H\} - \{\{F, H\}, G\} = -\{H, \{F, G\}\} - \{G, \{H, F\}\}.\end{aligned}$$

Thus the Poisson bracket defines a Lie algebra as desired. ■

In lieu of this result, we have the following Lie algebra morphism

$$C^\infty(M) \to \mathfrak{X}(M, \omega), \qquad H \mapsto X_H.$$

The image of this is the Lie subalgebra of Hamiltonian vector fields, and its kernel are the constant maps.

Note that if $F, G\colon M \to \mathbb{R}$ are smooth, then

$$\{F, G\}\frac{\omega^n}{n!} = dF \wedge dG \wedge \frac{\omega^{n-1}}{(n-1)!}.$$

Indeed, recall that

$$\iota(X)\omega^n = n\iota(X)\omega \wedge \omega^{n-1},$$

and so

$$dG \wedge \omega^{n-1} = \iota(X_G)\omega \wedge \omega^{n-1} = \frac{1}{n}\iota(X_G)\omega^n.$$

Thus

$$dF \wedge dG \wedge \omega^{n-1} = \frac{1}{n}\, dF \wedge \iota(X_G)\omega^n,$$

and recall that

$$0 = \iota(X_G)(dF \wedge \omega^n) = (\iota(X_G)\, dF)\omega^n - dF \wedge \iota(X_G)\omega^n = \{F, G\}\,\omega^n - dF \wedge \iota(X_G)\omega^n,$$

and thus

$$dF \wedge dG \wedge \omega^{n-1} = \frac{1}{n}\{F, G\}\,\omega^n$$

giving the desired result.

Note that $dF \wedge dG \wedge \omega^{n-1} = d(F\, dG \wedge \omega^{n-1})$ is exact, and thus if M is closed (or has empty boundary and either F or G has compact support), then

$$\int_M \{F, G\}\,\omega^n = n\int_M dF \wedge dG \wedge \omega^{n-1} = 0.$$

In other words the mean value of $\{F, G\}$ is zero.

Proposition 1.2.1.10

Let (M, ω) be symplectic.

(1) The Hamiltonian flow φ_H^t is a symplectomorphism for all t. Furthermore H is constant along $t \mapsto \varphi_H^t(p)$.

(2) For every Hamiltonian function $H\colon M \to \mathbb{R}$ and every symplectomorphism $\psi \in \mathrm{Symp}(M, \omega)$, $X_{H\circ\psi} = \psi^* X_H$.

Proof

(1) is due to proposition 1.2.1.7, and

$$\dot{\varphi}_H^t(H) = X_H H = \{H, H\} = 0,$$

so that $H \circ \varphi_H^t$ is constant. (2) is due to

$$\iota(X_{H\circ\psi})\omega = d(H \circ \psi) = \psi^* \, dH = \psi^* \iota(X_H)\omega = \iota(\psi^* X_H)\omega. \qquad \blacksquare$$

Definition 1.2.1.11

An **isotopy** of a manifold M is a smooth family of maps $\psi_t\colon M \to M$ such that ψ_t is a diffeomorphism and $\psi_0 = \mathrm{id}$. An isotopy is a **symplectic isotopy** if ψ_t is a symplectomorphism for all t.

Every isotopy induces a time-dependent vector field X_t by

$$\frac{d}{dt}\psi_t = X_t \circ \psi_t, \qquad \psi_0 = \mathrm{id},$$

i.e. we define $X_t(p) = \frac{d}{dt}\psi_t(\psi_t^{-1}(p))$. Conversely every time-dependent vector field X_t determines a unique isotopy (called its **flow**), although the isotopy is only globally defined in certain cases (e.g. if X_t is compactly supported). By proposition 1.2.1.7 if ψ_t is X_t's flow, ψ_t is a symplectic isotopy iff X_t is a symplectic vector space for all t.

Definition 1.2.1.12

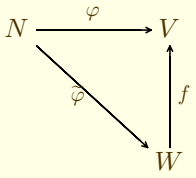
Let ψ_t be an isotopy with associated time-dependent vector field X_t. Then ψ_t is a **Hamiltonian isotopy** if $\iota(X_t)\omega$ is exact for all t (i.e. X_t is a Hamiltonian vector field). In such a case there is a smooth family of functions $H_t \in C^\infty(M)$ such that $\iota(X_t)\omega = dH_t$. Then H is called a **time-dependent Hamiltonian**, and it is determined by Hamiltonian isotopy up to addition with a smooth function $c_t \in \mathbb{R}$.

If M is simply connected then $H^1(M; \mathbb{R}) = 0$ as well, and thus a 1-form is closed iff it is exact. Thus all symplectic isotopies are Hamiltonian isotopies when M is simply connected.

Also note that a compactly-supported time-dependent Hamiltonian defines a Hamiltonian isotopy via $X_t = \widehat{\omega}^{-1}(dH_t)$

Note:

The existence of H_t amounts to a linear algebra problem. We restate it as follows: suppose $f\colon W \to V$ is a surjective linear map and $\varphi\colon N \to V$ is a smooth map. Then there is a smooth lift $\widetilde{\varphi}\colon N \to W$ such that $\varphi = f \circ \widetilde{\varphi}$:



Because f is surjective, it has a right inverse $\widetilde{f}$: $f \circ \widetilde{f} = \mathrm{id}$. Then define $\widetilde{\varphi} = \widetilde{f} \circ \varphi$.
In our case V is the space of exact 1-forms

$$V = \mathcal{B}^1(M), \qquad W = C^\infty(M), \qquad N = \mathbb{R}, \qquad f = d, \qquad \varphi = \iota(X_t)\omega.$$

Definition 1.2.1.13

A symplectomorphism $\psi \in \mathrm{Symp}(M,\omega)$ is **Hamiltonian** if there exists a Hamiltonian isotopy from $\psi_0 = \mathrm{id}$ to $\psi_1 = \psi$. Denote the space of Hamiltonian symplectomorphisms by $\mathrm{Ham}(M,\omega)$.

Let H_t be a compactly supported time-dependent Hamiltonian function. This then determines a compactly supported Hamiltonian isotopy ψ_t, and define $\varphi_H = \psi_1$. Such a Hamiltonian symplectomorphism is called **compactly supported** and we denote the space of compactly supported Hamiltonian symplectomorphisms by

$$\mathrm{Ham}_c(M,\omega) = \{\varphi_H \mid H \in C_c^\infty([0,1] \times M)\}.$$

Note that if M is closed then $\mathrm{Ham}_c(M,\omega) = \mathrm{Ham}(M,\omega)$.

Proposition 1.2.1.14

$\mathrm{Ham}(M,\omega)$ is a normal subgroup of $\mathrm{Symp}(M,\omega)$.

Proof

First we need to show that $\mathrm{Ham}(M,\omega)$ is a subgroup; we begin by showing it is closed under composition. Let φ_t, ψ_t be Hamiltonian isotopies generated by X_t, Y_t respectively. Then by the chain rule

$$\partial_t(\varphi_t \circ \psi_t) = (\partial_t\varphi_t) \circ \psi_t + d\varphi_t \circ \partial_t\psi_t, \tag{1}$$

here $d\varphi_t : TM \to TM$ is φ_t's differential. This equality is obtained by considering the partial derivatives of $\varphi_t \circ \psi_s$ and adding them according to the chain rule. Continuing this computation, by assumption we have

$$\partial_t(\varphi_t \circ \psi_t) = (X_t \circ \varphi_t) \circ \psi_t + d\varphi_t \circ Y_t \circ \psi_t = (X_t + d\varphi_t Y_t \circ \varphi_t^{-1}) \circ (\varphi_t \circ \psi_t).$$

Thus we must show that $(X\#Y)_t = X_t + \varphi_t^* Y_t$ is Hamiltonian. This is clear, as φ_t is a symplectomorphism and so by proposition 1.2.1.10 $\varphi_t^* Y_t$ is Hamiltonian as well. The sum of Hamiltonian vector fields is Hamiltonian, and thus $X\#Y$ is.
Note that by (1), we have

$$0 = \partial_t(\varphi_t^{-1} \circ \varphi_t) = (\partial_t\varphi_t^{-1}) \circ \varphi_t + d\varphi_t^{-1} \circ \partial_t\varphi_t,$$

and thus

$$\partial_t\varphi_t^{-1} = -d\varphi_t^{-1} \circ \partial_t\varphi_t \circ \varphi_t^{-1} = -d\varphi_t^{-1} \circ X_t \circ \varphi_t \circ \varphi_t^{-1},$$

and so φ_t^{-1} is generated by $-(\varphi_t^{-1})^* X_t$, so it is Hamiltonian. Thus $\mathrm{Ham}(M,\omega)$ is a subgroup.
Now if $\chi \in \mathrm{Symp}(M,\omega)$ and φ_t is a Hamiltonian isotopy generated by X_t, then

$$\partial_t(\chi^{-1} \circ \varphi_t \circ \chi) = d\chi^{-1} \circ \partial_t\varphi_t \circ \chi = d\chi^{-1} X_t \circ \varphi_t \circ \chi.$$

And so $\chi^{-1} \circ \varphi_t \circ \chi$ is generated by $(\chi^{-1})^* X_t$, which is Hamiltonian. Thus $\mathrm{Ham}(M,\omega)$ is closed under conjugation, and thus a normal subgroup of $\mathrm{Symp}(M,\omega)$ as desired. ■

Remark 1.2.1.15

While not every manifold admits a symplectic structure, be it due to dimension, orientation, or otherwise, every manifold induces a symplectic structure on an associated manifold. More exactly, every manifold L induces a canonical symplectic structure on its cotangent bundle T^*L. We explain how to define this symplectic structure.

Consider the projection

$$\pi\colon T^*L \to L, \qquad (p,\varphi) \mapsto p,$$

whose differential is a map

$$d\pi(p,\varphi)\colon T_{(p,\varphi)}T^*L \to T_pL.$$

Now consider the composition

$$\lambda_{\mathsf{can},(p,\varphi)} = T_{(p,\varphi)}T^*L \xrightarrow{d\pi(p,\varphi)} T_pL \xrightarrow{\varphi} \mathbb{R} = \varphi \circ d\pi_{(p,\varphi)}.$$

This defines a rough (potentially non-smooth) 1-form λ_{can} on T^*L. By considering local coordinates, we will show that λ_{can} is a smooth 1-form.

Let (x_i) be a coordinate chart for L, which then induces a coordinate chart (x_i, dx_i) for T^*L. Then in terms of these coordinates π has the representation $\pi(x,\varphi) = x$ where $\varphi = \sum_i \varphi_i\, dx_i$. Thus

$$d\pi(x,\varphi)\frac{\partial}{\partial x_j} = \frac{\partial}{\partial x_j}, \qquad d\pi(x,\varphi)\frac{\partial}{\partial \varphi_j} = 0,$$

and so

$$\lambda_{\mathsf{can},(x,\varphi)}\left(\sum_i \xi_i \frac{\partial}{\partial x_i} + \eta_i \frac{\partial}{\partial \varphi_i}\right) = \varphi\left(\sum_i \xi_i \frac{\partial}{\partial x_i}\right) = \sum_i \xi_i \varphi_i,$$

which is clearly smooth in all parameters. Equivalently we may write

$$\lambda_{\mathsf{can}} = \sum_{i=1}^{n} \varphi_i\, dx_i.$$

Now we define the symplectic form

$$\omega_{\mathsf{can}} = -\, d\lambda_{\mathsf{can}},$$

which is nondegenerate because locally it has the form

$$\omega_{\mathsf{can}} = \sum_{i=1}^{n} d\varphi_i \wedge dx_i.$$

Notice that ω_{can} is locally symplectomorphic to $(\mathbb{R}^{2n}, \omega_0)$; we will see this is true of all symplectic manifolds.

2.2 Moser isotopy and Darboux's theorem

Lemma 1.2.2.1 (Moser isotopy, compact)

Let M be a compact manifold, and ω_t a smooth family of symplectic forms on M which has an exact derivative:

$$\frac{d}{dt}\omega_t = d\sigma_t, \qquad \sigma_t \in \Omega^1(M).$$

Then there exists an isotopy $\psi_t \in \mathrm{Diff}(M)$ such that $\psi_t^*\omega_t = \omega_0$.

Proof

Instead of considering isotopies, we consider the time-dependent vector fields X_t which are associated to them. Because M is compact, this is a one-tone-one correspondence. So we have

$$\frac{d}{dt}\psi_t = X_t \circ \psi_t, \qquad \psi_0 = \mathrm{id}\,.$$

Now differentiate $\psi_t^*\omega_t$, we want it to be zero:

$$\frac{d}{dt}\psi_t^*\omega_t = \psi_t^*\left(\frac{d}{dt}\omega_t + \mathcal{L}_{X_t}\omega_t\right) = \psi_t^*(d\sigma_t + d\iota(X_t)\omega_t).$$

So it is sufficient to require that

$$\sigma_t + \iota(X_t)\omega_t = 0,$$

so $X_t = \widehat{\omega}_t^{-1}\sigma_t$ gives the time-dependent vector field. Taking its flow gives the desired isotopy. ■

We want to extend this result to non-compact manifolds. As-is we cannot do this, but the following lemma is sufficient.

Lemma 1.2.2.2 (Moser isotopy)

Let M be a manifold and $Q \subseteq M$ a compact submanifold. Suppose that $\omega_0, \omega_1 \in \Omega^2(M)$ are closed 2-forms such that ω_0 and ω_1 agree on Q and are nondegenerate at each point of q. Then there exist open neighborhoods U_0, U_1 of Q and a diffeomorphism $\psi\colon U_0 \to U_1$ such that

$$\psi|_Q = \mathrm{id}, \qquad \psi^*\omega_1 = \omega_0.$$

Proof

Note that if ω is nondegenerate at a point $p \in M$, then it is nondegenerate in a neighborhood of p. Indeed, relative to any local frame of p ω has nonsingular representation at p, and taking this representation is smooth, so the result follows because $\mathrm{GL}(n)$ is open in Mat_n. Thus ω_0, ω_1 are both nondegenerate in an open neighborhood U_0 of Q.

We claim the existence of a 1-form $\sigma \in \Omega^1(U_0)$ such that

$$\sigma|_Q = 0, \qquad d\sigma = \omega_1 - \omega_0.$$

Then define

$$\omega_t = \omega_0 + t(\omega_1 - \omega_0) = \omega_0 + t\,d\sigma.$$

This, as a convex combination of ω_0, ω_1, is nondegenerate on U_0 as well. Applying Moser's argument above to Q, we get diffeomorphisms $\psi_t\colon U_0 \to U_t$ satisying $\psi_t^*\omega_t = \omega_0$; take $\psi = \psi_1$ for the lemma. In more detail, define $X_t = \widehat{\omega}_t^{-1}\sigma$ in U_0, then this time-dependent vector field has global flow in a neighborhood of Q as it is compact. Shrink U_0 to be this neighborhood, so the ψ_t are diffeomorphisms from U_0 to a neighborhood of Q.

Thus all that remains is to find a $\sigma \in \Omega^1(U_0)$ which restricts to 0 on Q and whose exterior derivative is $\omega_1 - \omega_0$. Endow M with any Riemannian metric, then Q has a tubular neighborhood. We explain what this means. Let $\exp\colon \mathcal{E} \subseteq TM \to M$ be the exponential map, where $\exp(p, tv)$ is the geodesic starting at p with initial velocity v. Recall that $\mathcal{E}$ is an open subset of TM whose fibers are star-shaped about 0. Let $TQ^\perp \subseteq TM|_Q$ be the orthogonal subbundle of $TQ \subseteq TM|_Q$, then there is an open subset $W \subseteq TQ^\perp$ of the form

$$W = \left\{(q, v) \;\middle|\; q \in Q,\ v \in T_qQ^\perp,\ |v| < \varepsilon\right\}$$

such that exp forms a diffeomorphism from W to $\exp(W)$ which is also open. This $\exp(W)$ is called a **tubular neighborhood** of Q; its existence is nontrivial.

Define U_0 to be the tubular neighborhood $\exp(W)$. Then define for $0 \le t \le 1$,

$$\varphi_t\colon U_0 \to U_0, \qquad \varphi_t(\exp(q, v)) = \exp(q, tv),$$

which is a diffeomorphism for $t > 0$. And $\varphi_0(\exp(q,v)) = \exp(q,0) = q$ so that $\varphi_0(U_0) \subseteq Q$, $\varphi_1 = \mathrm{id}$, and $\varphi_t|_Q = \mathrm{id}$. The last claim is because $\exp(q,v) \in Q$ iff $v = 0$ because exp is a diffeomorphism from W (because $\exp(q,v) = \exp(\exp(q,v),0)$).

Let $\tau = \omega_1 - \omega_0$, then these facts mean

$$\varphi_0^*\tau = 0, \qquad \varphi_1^*\tau = \tau.$$

For $t > 0$ define

$$X_t = (\partial_t \varphi_t) \circ \varphi_t^{-1}.$$

Then we have

$$\frac{d}{dt}\varphi_t^*\tau = \varphi_t^*\mathcal{L}_{X_t}\tau = d(\varphi_t^*(\iota(X_t)\tau)),$$

so define $\sigma_t = \varphi_t^*(\iota(X_t)\tau)$. Then for $w \in T_xU_0$ we have

$$\sigma_t(x;w) = \tau(\varphi_t(x);\ X_t(\varphi_t(x)), d\varphi_t(x)(w)) = \tau(\varphi_t(x);\ \partial_t\varphi_t(x), d\varphi_t(x)(w)).$$

This is then smooth at $t = 0$, and for $x \in Q$ vanishes (because $\varphi_t(x) = x$ and $\tau(x;\bullet) = 0$ as ω_0, ω_1 agree on Q). Thus

$$\omega_1 - \omega_0 = \tau = \varphi_1^*\tau - \varphi_0^*\tau = \int_0^1 \frac{d}{dt}(\varphi_t^*\tau)\,dt = \int_0^1 d\sigma_t\,dt,$$

so define

$$\sigma = \int_0^1 \sigma_t\,dt$$

then we get

$$\omega_1 - \omega_0 = d\sigma.$$

And on Q, the σ_t vanish and thus so too does σ, so σ satisfies the conditions. ■

Theorem 1.2.2.3 (Darboux's)

Every symplectic form ω on M is locally symplectomorphic to the standard form ω_0 on $\mathbb{R}^{2n}$.

Proof

Choose a point $q \in M$, then ω has a symplectic basis on T_qM. Let (x_i, y_i) be a local coordinate chart on $q \in U$ such that $(\partial_i x_i, \partial_i y_i)$ is a symplectic basis of ω_0 on T_qM. Define the symplectic form on U,

$$\widetilde{\omega} = \sum_i dx_i \wedge dy_i.$$

By Moser isotopy, by taking $Q = \{q\}$, there is a diffeomorphism $\psi\colon U_0 \to U_1$, with $q \in U_0, U_1 \subseteq U$ open, such that $\omega = \psi^*\widetilde{\omega}$. So (U_0, ω) is symplectomorphic to $(U_1, \widetilde{\omega})$, which is clearly symplectomorphic to the standard Euclidean symplectic form ω_0. ■

Note:

In Riemannian geometry, not every metric is locally isomorphic to the Euclidean metric. The failure of this is measured by the curvature of the metric. In contrast to this, Darboux's theorem shows that all symplectic manifolds are locally isomorphic.

Thus there are no meaningful local invariants for symplectic manifolds, and so symplectic geometry focuses on global invariants.

Corollary 1.2.2.4

Let (M, ω) be a symplectic manifold of dimension $2n$. Then there exists an open cover $\{U_\alpha\}_\alpha$ of M with charts $\alpha: U_\alpha \to \alpha(U_\alpha) \subseteq \mathbb{R}^{2n}$ such that $\alpha^*\omega_0 = \omega$. The atlas has symplectic transition matrices:

$$d(\beta \circ \alpha^{-1})(x) \in \mathrm{Sp}(2n), \qquad x \in \alpha(U_\alpha \cap U_\beta).$$

Such charts are called **Darboux charts**.

Theorem 1.2.2.5 (Moser stability)

Let M be a closed manifold, and suppose ω_t is a smooth family of cohomologous symplectic forms on M. Then there is a smooth family of diffeomorphisms ψ_t of M such that

$$\psi_0 = \mathrm{id}, \qquad \psi_t^*\omega_t = \omega_0.$$

Proof

We apply lemma 1.2.2.1, so we need to find a smooth family of 1-forms σ_t such that

$$d\sigma_t = \tau_t = \frac{d}{dt}\omega_t.$$

Because ω_t are cohomologous, $\omega_t - \omega$ are exact and thus so too is τ_t. Thus there do exist 1-forms σ_t such that $d\sigma_t = \tau_t$. The only issue is ensuring σ_t is smooth, and this can be done by induction on the number of sets in a good cover of M.

TODO: This?

■

Definition 1.2.2.6

Two symplectic forms ω_0, ω_1 on a manifold M are **isotopic** if they can be connected by a smooth family of cohomologous symplectic forms ω_t.

They are said to be **strongly isotopic** if there is an isotopy ψ_t of M such that $\psi_1^*\omega_0 = \omega_1$.

In general being strongly isotopic implies being isotopic. This is because we can define $\omega_t = \psi_t^*\omega_0$. And as $\psi_\bullet$ is a homotopy, by the homotopy-invariance of de Rham cohomology, ψ_t^* induces the same map on homotopy for all t. Moser stability says that the two notions coincide when M is closed.

2.3 Isotopy extension theorems

Smooth isotopies are special kinds of homotopies, and as such we can define similar concepts.

Definition 1.2.3.1

In general an isotopy is a smooth map $\psi_\bullet: [0,1] \times X \to Y$ such that $\psi_t: X \to Y$ is a diffeomorphism for all t. We may also require only that ψ_t be an embedding for all t; the meaning will be made clear by context. If $A \subseteq X$ is an arbitrary subset, an **isotopy relative to** A is an isotopy $\psi_\bullet$ such that $\psi_t|_A = \psi_0|_A$ for all t. If $Q \subseteq X$ is a submanifold, an **isotopy of** Q is a smooth isotopy (of embeddings) ψ_t such that ψ_0 is the inclusion $Q \hookrightarrow X$.

All of these notions can be specialized to **symplectic isotopies**, where $\psi_t: Q \to M$ is a symplectomor-

phism.

In the following for real $r > 0$, $B(r)$ denotes the closed ball of radius r centered at 0 in $\mathbb{R}^{2n}$. This inherits the standard symplectic structure from $\mathbb{R}^{2n}$.

Theorem 1.2.3.2

Let (M, ω) be a connected symplectic manifold without boundary.

(1) Let $\lambda > 0$, then every symplectic embedding $\psi\colon B(r) \to M$ extends to a symplectic embedding $B(r) \hookrightarrow M$ for some $r > \lambda$.

(2) Let $0 \leq \lambda < r$ and $\psi_0, \psi_1\colon B(r) \to M$ be symplectic embeddings which are connected by a smooth family of symplectic embeddngs $\psi_t\colon B(\lambda) \to M$. Then there is a Hamiltonian isotopy of M φ_t and a constant $\lambda < \rho < r$ such that

$$\varphi_0 = \mathrm{id}, \qquad \varphi_1 \circ \psi_0|_{B(\rho)} = \psi_1|_{B(\rho)}, \qquad \varphi_t \circ \psi_0|_{B(\lambda)} = \psi_t.$$

Thus $\varphi_t \circ \psi_0$ is an extension of the isotopy ψ_t onto the larger ball $B(\rho)$.

We will first need the following lemma.

Lemma 1.2.3.3

Let $\psi_\bullet\colon [0,1] \times B(\lambda) \to M$ be a symplectic isotopy (of embeddings). Then there is a Hamiltonian isotopy $\varphi_\bullet\colon [0,1] \times M \to M$ such that

$$\varphi_0 = \mathrm{id}, \qquad \varphi_t \circ \psi_0 = \psi_t.$$

Proof

Let X_t be the time-dependent vector field generated by ψ_t. We have that $\psi_t^*\omega = \omega_0$ and thus we differentiate

$$0 = \frac{d}{dt}\psi_t^*\omega = \psi_t^*\mathcal{L}_{X_t}\omega = \psi_t^*\, d\iota(X_t)\omega = \psi_t^*\, d\omega(X_t, \bullet) = d\omega(\partial_t\psi_t, d\psi_t\bullet).$$

Thus we have a smooth family of closed 1-forms

$$\alpha_t = \omega(\partial_t\psi_t, d\psi_t\bullet) \in \Omega^1(B(\lambda)).$$

Because $B(\lambda)$ is contractible, this means α_t is exact for all t. So there is a smooth family of functions $h_t\colon B(\lambda) \to \mathbb{R}$ such that

$$dh_t = \alpha_t, \qquad h_t(0) = 0.$$

Note that we can extend ψ_t to an isotopy of embeddings $\psi_\bullet\colon [0,1] \times \mathbb{R}^{2n} \to M$. Then via this lift h_t to $H_t\colon M \to \mathbb{R}$ such that

$$H_t \circ \psi_t = h_t.$$

Then consider the Hamiltonian isotopy φ_t generated by H_t:

$$\frac{d}{dt}\varphi_t = X_{H_t} \circ \varphi_t, \qquad \varphi_0 = \mathrm{id}, \qquad \iota(X_{H_t})\omega = dH_t.$$

Now notice that

$$\omega(\partial_t(\varphi_t \circ \psi_0), d\psi_t\bullet) = \omega(X_{H_t} \circ \varphi_t \circ \psi_0, d\psi_t\bullet) = dH_t(d\psi_t\bullet) = d(H_t \circ \psi_t) = dh_t = \omega(\partial_t\psi_t, d\psi_t\bullet).$$

Because ψ_t is symplectic and ω nondegenerate, this means $\partial_t(\varphi_t \circ \psi_0) = \partial_t\psi_t$. And because they agree at $t = 0$ we have $\varphi_t \circ \psi_0 = \psi_t$. ∎

Proof (1.2.3.2 (1))

The symplectic embedding $\psi\colon B(\lambda) \to M$ extends to a smooth embedding $\psi\colon B(\lambda+\varepsilon) \to M$. Then the 2-form

$$\psi^*\omega \in \Omega^2(B(\lambda+\varepsilon))$$

agrees with ω_0 on $B(\lambda)$. Because $B(\lambda), B(\lambda+\varepsilon)$ are contractible, by homotopy invariance of de Rham cohomology, because $\omega_0 - \psi^*\omega$ is zero on $B(\lambda)$ and thus exact, $\omega_0 - \psi^*\omega$ is exact on $B(\lambda+\varepsilon)$. So let $\alpha \in \Omega^1(B(\lambda+\varepsilon))$ such that

$$\alpha|_{B(\lambda)} = 0, \qquad d\alpha = \psi^*\omega - \omega_0.$$

By potentially shrinking ε, we may assume $\omega_s = \omega_0 + s\,d\alpha$ is nondegenerate for all $0 \le s \le 1$. By lemma 1.2.2.1 there is an isotopy $\chi_t \in \mathrm{Diff}(B(\lambda+\varepsilon))$ such that

$$\chi_0 = \mathrm{id}, \qquad \chi_s|_{B(\lambda)} = \mathrm{id}, \qquad \chi_s^*\omega_s = \omega_0.$$

Thus $\chi_1^*\psi^*\omega = \omega_0$ and thus $\psi \circ \chi_1$ is the desired extension. ■

Proof (1.2.3.2 (2))

We first consider $\lambda > 0$.

TODO: This?
TODO: Continue this; just 3.3.2

2.4 Submanifolds in symplectic manifolds

In these notes a submanifold means an *embedded* submanifold. By the local k-slice a condition, a subspace $Q \subseteq M$ is an embedded k-manifold iff there is a cover of Q by open subsets of M such that Q's image in these charts is an open subset of $\mathbb{R}^k \subseteq \mathbb{R}^n$.

Definition 1.2.4.1

Let (M,ω) be a symplectic manifold. A submanifold $Q \subseteq M$ is **symplectic**, **isotropic**, **coisotropic**, or **Lagrangian** if for every $q \in Q$, $T_qQ \subseteq T_qM$ is a symplectic/isotropic/coisotropic/Lagrangian subspace.

It is clear then that Q is Lagrangian iff it is isotropic and of dimension $\frac{1}{2}\dim M$.

Example 1.2.4.2

Let $Q \subseteq L$ be an arbitrary submanifold. Then define its **annihilator** (or **conormal**)

$$TQ^\perp = \left\{(q,\varphi) \in T^*L \;\middle|\; q \in Q,\ \varphi|_{T_qQ} = 0\right\}.$$

This is a vector bundle over Q, and each fiber has dimension $\dim L - \dim Q$ (the dimension of $\mathrm{ann}(T_qQ) \subseteq \mathrm{ann}(T_qL)$). Thus $TQ^\perp$'s rank is $\dim L - \dim Q$, so its dimension is $\dim L$.

Notice that λ_{can} vanishes on $TQ^\perp$: $\lambda_{\mathsf{can}}(q,\varphi) = \varphi \circ d\pi_{(q,\varphi)}$, and $d\pi_{(q,\varphi)}$ lands in T_qQ which is annihilated by φ. Thus Q is isotropic and its dimension is half of T^*L's, so it is Lagrangian.

Taking $Q = L$, we get the zero section

$$TL^\perp = \{(q,0) \in T^*L \mid q \in L\},$$

and taking $q \in L$ and $Q = \{q\}$, we have $T_qQ = 0$ and thus

$$TL^\perp = T_q^*L.$$

So both the zero section and the fibers of T^*L are Lagrangian submanifolds.

Example 1.2.4.3

Let (M, ω) be a symplectic manifold, then its product $M \times M$ has a natural symplectic structure by $(-\omega) \oplus \omega$. Then $M \times *, * \times M$ are symplectic clearly. The diagonal is Lagrangian, as it is isotropic and of half of $M \times M$'s dimension.

Example 1.2.4.4

All 1-dimensional submanifolds are isotropic, as all 1-dimensional subspaces are isotropic. This means that all subspaces of codimension 1 are coisotropic, and thus all hypersurfaces (i.e. submanifolds of codimension 1) are coisotropic.

Proposition 1.2.4.5

Let $\sigma \in \Omega^1(M)$ be a 1-form. Then its graph $\Gamma_\sigma \subseteq T^*M$ is Lagrangian iff σ is closed.

Proof

The graph Γ_σ is the image of the embedding $\sigma\colon M \to T^*M$. Thus $-d\lambda_{\mathsf{can}}$ vanishes on Γ_σ iff

$$0 = \sigma^*(-d\lambda_{\mathsf{can}}) = d(\sigma^*\lambda_{\mathsf{can}}) = d\sigma.$$

The last equality is proven as follows. Write σ locally by $\sigma = \sum_i \sigma_i\, dx_i$, so $\sigma\colon M \to T^*M$ is the map $(x_1, \dots, x_n) \to (x_1, \dots, x_n, \sigma_1, \dots, \sigma_n)$. The pullback of λ_{can} along σ then is

$$\sigma^*\lambda_{\mathsf{can}}(\partial_i) = \lambda_{\mathsf{can}}\left(\frac{\partial}{\partial x_i} + \sum_j \frac{\partial \sigma_j}{\partial x_i}\frac{\partial}{\partial \sigma_j}\right),$$

and as $\lambda_{\mathsf{can}} = \sum_i \sigma_i\, dx_i$, this is equal to σ_i. Thus $\sigma^*\lambda = \sigma$ as desired. ■

Proposition 1.2.4.6

Let (M, ω) be symplectic and $\psi\colon M \to M$ a diffeomorphism. Then ψ is a symplectomorphism iff its graph

$$\Gamma_\psi = \{(q, \psi q) \mid q \in M\} \subseteq M \times M$$

is a Lagrangian submanifold of $(M, \times M, (-\omega) \oplus \omega)$.

Proof

By taking a coordinate chart $U \subseteq M$, then $U \times U$ is a coordinate chart of $M \times M$. We see then that $(U \times U) \cap \Gamma_\psi$ is the fiber of 0 under the map $U \times U \to U$ given by $(q, p) \mapsto \psi(q) - p$. This map has full rank and so we see that this defines an embedded manifold structure on Γ_ψ of dimension $\dim U = \dim M$. Furthermore the tangent space of Γ_ψ at a point of $U \times U$ is the kernel of this map, which is

$$T_{(q,\psi q)}\Gamma_\psi = \{(v, d\psi v) \mid v \in T_qM\} = \Gamma_{d_q\psi}.$$

Now $\psi\colon (W, \omega) \to (W, \omega)$ is a symplectomorphism iff $\Gamma_\psi \subseteq W \times W$ is Lagrangian. Indeed, it is Lagrangian iff it is isotopic iff $\omega(\psi w) - \omega(v) = 0$ for all $w \in W$, i.e. iff ψ is a symplectomorphism.

Applying this to the above computation ψ is a symplectomorphism if $d_q\psi$ is for all $q \in M$ iff $\Gamma_{d_q\psi}$ is Lagrangian for all $q \in Q$ iff $\Gamma_\psi \subseteq M \times M$ is Lagrangian. ■

If $F \hookrightarrow E$ is a subbundle, then we can form the quotient bundle E/F whose fibers are E_p/F_p. If $Q \subseteq M$ is a submanifold, the quotient bundle $TM|_Q/TQ$ is called the **normal bundle** ν_Q. When M is Riemannian, this is isomorphic to $TQ^\perp$ via the canonical isomorphism $W^\perp \cong V/W$ for finite-dimensional inner product spaces.

If M is symplectic and Q a symplectic submanifold then as in the Riemannian case ν_Q may be identified with TQ^ω and thus carries a symplectic structure. In the following theorem ν_Q refers to TQ^ω.

Theorem 1.2.4.7

Let (M_j, ω_j) be symplectic manifolds for $j = 0, 1$ and $Q_j \subseteq M_j$ be compact symplectic submanifolds. Furthermore suppose that there is an isomorphism

$$\Phi\colon \nu_{Q_0} \to \nu_{Q_1}$$

of the symplectic normal bundles which covers a symplectomorphism $\varphi\colon (Q_0, \omega_0) \to (Q_1, \omega_1)$. Then φ extends to a symplectomorphism $\psi\colon (U_0, \omega_0) \to (U_1, \omega_1)$ for open neighborhoods $Q_i \subseteq U_i$ such that $d\psi$ restricts to Φ on ν_{Q_0}.

Proof

By the tubular neighborhood theorem there are convex neighborhoods of the zero section of ν_{Q_i} which map diffeomorphically to a neighborhood U_i of Q_i. Let $\exp_i$ be these diffeomorphisms and define

$$\varphi' = \exp_1 \circ \Phi \circ \exp_0^{-1}.$$

On ν_{Q_0} we have that $d\varphi' = \Phi$ because on the zero section $\exp_i$'s differential is the identity. Now consider $\omega_1' = (\varphi')^*\omega_1$, so that on $TM|_Q$ we have

$$\omega_1' = \Phi^*\omega_1 = \omega_0.$$

So we can instead consider ω_0, ω_1'. These are two symplectic forms on U_0 which agree on the compact submanifold Q. The result then follows from Moser isotopy. ■

Note:

This result can be summarized as "the symplectic structure around a compact symplectic submanifold is determined by the symplectic structure of its symplectic normal bundle."

Lemma 1.2.4.8

Let $\varphi\colon M \to N$ be smooth between manifolds of the same dimension. Suppose that there is a compact submanifold $Q \subseteq M$ such that $\varphi|_Q$ is injective and for every $q \in Q$, φ has full rank. Then there is a neighborhood $Q \subseteq U$ such that $\varphi\colon U \to N$ is an embedding.

Proof

Because being of full rank is local, for every point $q \in Q$ there is a neighborhood on which φ has full rank. Potentially shrinking this neighborhood, φ is a diffeomorphism there. It is sufficient for φ then to be injective in a neighborhood of Q, as injective local diffeomorphisms are embeddings. Because smooth manifolds are metrizable (e.g. by a Riemannian distance function), the following claim proves the lemma.

Let $Q \subseteq M$ be a compact subset of a metric space, and $\varphi\colon M \to N$ a continuous function. Suppose that around every point of Q there is a neighborhood on which φ is injective, and that φ is injective on Q. Then

there is a neighborhood of Q on which φ is injective.

Define

$$\varepsilon = \inf\{d(x,Q) \mid \text{There is a } y \text{ such that } \varphi x = \varphi y \text{ and } d(y,Q) < d(x,Q)\}.$$

This ε is nonzero, otherwise there are sequences $x_n \neq y_n$ such that $d(x_n,Q), d(y_n,Q) \to 0$ and $\varphi(x_n) = \varphi(y_n)$. Because Q is compact there are $a_n, b_n \in Q$ such that $d(x_n,Q) = d(x_n,a_n)$ and $d(y_n,Q) = d(y_n,b_n)$. And again because Q is compact there are subsequences of a_n, b_n which converge, so we can assume that the sequences as a whole converge $a_n \to a$ and $b_n \to b$. Then $x_n \to a$ and $y_n \to b$ as well, so $\varphi(a) = \varphi(b)$, meaning $a = b$ as φ is injective on Q. But this means eventually x_n, y_n lie in the open neighborhood of a for which φ is injective, so $x_n = y_n$ a contradiction.

So $\varepsilon > 0$, and φ is injective on the open tube

$$U = \{x \in M \mid d(x,Q) < \varepsilon\}.$$

■

Theorem 1.2.4.9

Let $L \subseteq M$ be a compact Lagrangian submanifold. Then there exists a neighborhood $U_0 \subseteq T^*L$ of the zero section, a neighborhood $L \subseteq V$, and a symplectomorphism $\varphi\colon (U_0, \omega_{\mathsf{can}}) \to (V, \omega)$ which restricts to the identity on L (i.e. it maps the zero section to L).

Here $\omega_{\mathsf{can}} = -d\lambda_{\mathsf{can}}$ is the canonical symplectic structure on T^*L (see remark 1.2.1.15).

Proof

Let $J\colon TM \to TM$ be a complex structure compatible with ω (which exists by proposition). Denote the associated Riemannian metric by $g_J(v,u) = \omega(v, Ju)$. For $q \in Q$, $J_qT_qQ \subseteq T_qM$ is the orhogonal complement of T_qQ with respect to g_J and is Lagrangian. In general if $\Lambda \subseteq V$ is a Lagrangian subspace, then $J\Lambda$ and Λ have zero intersection as $\omega(v, Jv) > 0$ for $v \in \Lambda$, and $g_J(Jv,u) = \omega(Jv, Ju) = \omega(v,u) = 0$ for $v, u \in \Lambda$.

Define

$$\Phi_q\colon T_q^*L \to T_qL$$

to be the isomorphism induced by g_J, meaning

$$g_J(\Phi_q\varphi, \bullet) = \varphi(\bullet).$$

And consider $\varphi\colon T^*L \to M$ by

$$\varphi(q,\varphi) = \exp_q(J_q\Phi_q(\varphi)),$$

where $\exp\colon \mathcal{E} \subseteq TL \to M$ is the Riemannian exponential map associated to g_J. For $(v,\varphi) \in T_{(q,0)}T^*L = T_qL \oplus T_q^*L$ we have

$$d\varphi_{(q,0)}(v,\varphi) = v + J_q\Phi_q(\varphi).$$

And thus if $v = (v_0, \varphi_0)$ and $w = (v_1, \varphi_1)$ are tangent vectors in $T_{(q,0)}T^*L$ then because T_qL and J_qT_qL are Lagrangian

$$\begin{aligned}\varphi^*\omega_{(q,0)}(v,w) &= \omega_q(v_0 + J_q\Phi_q(\varphi_0), v_1 + J_q\Phi_q(\varphi_1)) = \omega_q(v_0, J_q\Phi_q(\varphi_1)) - \omega_q(v_1, J_q\Phi_q(\varphi_0)) \\ &= g_J(v_0, \Phi_q(\varphi_1)) - g_J(v_1, \Phi_q(\varphi_0)) = \varphi_1(v_0) - \varphi_0(v_1) = \omega_{\mathsf{can}}(v,w).\end{aligned}$$

Thus the 2-form $\varphi^*\omega$ agrees with the symplectic structure ω_{can} on the zero section of T^*L. Furthermore $\varphi^*\omega$ is closed as the pullback of a closed form.

So by Moser isotopy there is a diffeomorphism $\psi\colon U_0 \to U_1$ of neighborhoods of the zero section such that $\psi^*\varphi^*\omega = \omega_{\mathsf{can}}$ and $\psi|_L = \mathrm{id}$. Furthermore because $\varphi^*\omega$ is nondegenerate on the zero section, $d\varphi$ must have zero kernel on the zero section so that at points on the zero section φ has full rank. Because T^*L and M have the same dimension, by the above lemma φ is a diffeomorphism between from a neighborhood of the zero section to an open subset of M. Thus $\varphi \circ \psi$ is the desired diffeomorphism. ■

Note:
This result shows that the symplectic structure around a compact Lagrangian submanifold is entirely determined by the diffeomorphism class of L.

TODO: Whitney (C^k) topologies

Proposition 1.2.4.10

Let (M, ω) be symplectic, then a neighborhood of the identity in $\mathrm{Symp}(M)$ can be identified with a neighborhood of 0 in the vector space of closed 1-forms on M.

Proof

Consider the Lagrangian submanifold Δ of the product $(M \times M, (-\omega) \oplus \omega)$. There is a canonical diffeomorphism $\Delta \cong M$ and thus an isomorphism $T^*\Delta \cong T^*M$. By the above theorem there is a neighborhood $\mathcal{N}(\Delta) \subseteq M \times M$ of Δ, a convex neighborhood $\mathcal{N}(M_0) \subseteq T^*M$ of the zero section M_0, and a diffeomorphism $\Psi : \mathcal{N}(\Delta) \to \mathcal{N}(M_0)$ such that

$$\Psi(q, q) = (q, 0), \qquad \Psi^*(d\lambda_{\mathsf{can}}) = (-\omega) \oplus (\omega)$$

(the first condition is $\Psi|_{M_0} = \mathrm{id}$, under the identification $\Delta \cong M$).

Now suppose $\psi \in \mathrm{Symp}(M, \omega)$ is sufficiently C^1-close to the identity. By proposition 1.2.4.6 ψ's graph is a Lagrangian submanifold of $\mathcal{N}(\Delta)$. Thus $L = \Psi(\Gamma_\psi)$ is a Lagrangian submanifold of T^*M. Furthermore L is the image of $M \to T^*M$, $q \mapsto \Psi(q, \psi(q))$, which is C^1-close to the zero section. Thus L is the graph of a 1-form, and it is closed by proposition 1.2.4.5. ■

2.5 Distributions and foliations

Material from the book "Introduction to Smooth Manifolds" *by John M. Lee*

We take a small detour from our discussion on symplectic manifolds to discuss distributions and foliations. This is in preparation for the next section on contact structures.

Definition 1.2.5.1

A **distribution** of a manifold M is a (smooth) subbundle of TM. Its rank is its rank as a subbundle. An **integral manifold** of a distribution D is a submanifold $N \subseteq M$ such that $D|_N = TN$.

Example 1.2.5.2

If V is a nonvanishing vector field on M, then it spans a smooth rank-1 distribution. The image of any of its integral curves is an integral manifold.

Example 1.2.5.3

Not every distribution has integral manifolds. Take D to be the smooth distribution on $\mathbb{R}^3$ spanned by

$$X = \frac{\partial}{\partial x} + y\frac{\partial}{\partial z}, \qquad Y = \frac{\partial}{\partial y}.$$

If N is an integral manifold, then as X, Y are tangent to N their integral curves must lie in N for a short

while. The x-axis is an integral curve of X, and the y-axis is of Y. Thus N contains an open subset of the xy-plane, but for any point p on the xy-plane not on the x-axis, its tangent space is not equal to D_p.

Definition 1.2.5.4

A distribution D on M is **involutive** if the Lie bracket of any two local sections of D is another local section of D. And D is **integrable** if every point of M lies in an integral manifold of D.

Remark 1.2.5.5

It is sufficient to consider only global sections of D in determining if it is involutive. That is, D is involutive iff its space of global sections $\Gamma(D)$ is a Lie subalgebra of $\mathfrak{X}(M)$.

Clearly if D is involutive then this is true. Conversely suppose $\Gamma(D) \subseteq \mathfrak{X}(M)$ is a Lie subalgebra, and let $X, Y \in \Gamma(U, D)$ be local sections on $U \subseteq M$. Let $\psi \in C^\infty(M)$ be a smooth bump function which is 1 in a neighborhood of $p \in U$ and supported in U. Then $\psi X, \psi Y$ can be defined on all of M, so by assumption $[\psi X, \psi Y] \in \Gamma(D)$. Computing yields

$$[\psi X, \psi Y] = \psi^2[X, Y] + \psi(X\psi)Y - \psi(Y\psi)X,$$

and evaluating at p gives $[X, Y]_p$ (because ψ is constant around p, $X_p\psi = 0$). So $[X, Y]_p \in D_p$ meaning $[X, Y]$ is a local section of D as desired.

Proposition 1.2.5.6

Every integrable distribution is involutive.

Proof

If X, Y are local sections of $D \subseteq TM$ defined on $U \subseteq M$, let $p \in U$. Then p lies in an integral manifold N of D, and so $X_p, Y_p \in T_pN = D_p$. So X, Y are tangent to N at p, and thus so is $[X, Y]$, meaning $[X, Y]_p \in T_pN = D_p$ as desired. ■

Being involutive can be checked on a local frame of D: if $V_1, \ldots, V_k$ is a local frame for D, then D is involutive (in the neighborhood for which this is a frame) iff $[V_i, V_j]$ is a local section of D for all i, j. Indeed, let $X = X^iV_i$ and $Y = Y^jV_j$ be local sections of D, then

$$[X, Y] = X^iY^j[V_i, V_j] + X^i(V_iY^j)V_j - Y^j(V_jX^i)V_i$$

which is a local section of D as desired.

Lemma 1.2.5.7

Let M be n-dimensional, and $D \subseteq TM$ a not necessarily smooth distribution of rank k. Then D is smooth iff for every $p \in M$ there is a neighborhood $p \in U$ and smooth 1-forms $\omega^1, \ldots, \omega^{n-k}$ on U such that

$$D|_U = \ker \omega^1 \cap \cdots \cap \ker \omega^{n-k}.$$

Proof

If such 1-forms exist, then by dimensional considerations we must have that they are linearly independent. Complete them to a local coframe $\omega^1, \dots, \omega^n$ of T^*U, and let $E_1, \dots, E_n$ be their dual frame. Then D is spanned by $E_{n-k+1}, \dots, E_n$, and so around every point of M, D has a smooth local frame, and is thus a distribution.

Conversely if D is smooth, then let $Y_1, \dots, Y_k$ span D locally. We can complete these to a local frame of TM, $Y_1, \dots, Y_n$. Let $\omega^1, \dots, \omega^n$ be the dual coframe, then D is equal to $\ker \omega^{k+1} \cap \cdots \cap \ker \omega^n$. ■

Definition 1.2.5.8

If D is a rank-k distribution of M, then any local smooth 1-forms $\omega^1, \dots, \omega^{n-k}$ such that $D = \bigcap_i \ker \omega^i$ are called **local defining forms** for D. And a differential form $\omega \in \Omega^p(M)$ **annihilates** D if $\omega(X_1, \dots, X_p) = 0$ when $X_i \in D$.

Proposition 1.2.5.9

If $\omega^1, \dots, \omega^{n-k}$ are local defining forms for D, then $\eta \in \Omega^p(M)$ annihilates D iff

$$\eta = \sum_i \omega^i \wedge \beta^i,$$

where β^i are smooth $(p-1)$-forms.

Proof

An η of this form annihilates D as evaluating at $\overline{X} \in D$ will result in sums whose terms all have some $\omega^i(X_j) = 0$. Conversely if η annihilates D, extend ω^i to a local coframe for T^*M and let E_i be the associated local frame for TM. So $E_{n-k+1}, \dots, E_n$ is a local frame for D. Then η can be written as

$$\eta = \sum_I \eta_I \omega^I, \qquad \eta_I = \eta(E_{i_1}, \dots, E_{i_n}).$$

Thus η annihilates D iff $\eta_I = 0$ whenever I's values lie between $n-k+1$ and n. And so

$$\eta = \sum_{i_1 \le n-k} \eta_I \omega^I = \sum_{i_1=1}^{n-k} \omega^{i_1} \wedge \left(\sum_{I'} \eta_{i_1 I'} \omega^{I'} \right).$$

This is true in a neighborhood of each point of U (because extending to a local coframe of T^*M may require shrinking the neighborhood), and then patching together with a partition of unity gives us the desired result on all of U. ■

Proposition 1.2.5.10

Let D be a smooth distribution on M. Then D is involutive iff whenever $\eta \in \Omega^1(U)$ annihilates D so too does $d\eta \in \Omega^2(U)$.

Proof

If D is involutive, and suppose $\eta \in \Omega^1(U)$ annihilates D. Then for X, Y local sections of D,

$$d\eta(X,Y) = X(\eta(Y)) - Y(\eta(X)) - \eta([X,Y]) = 0$$

because D is involutive.

Conversely suppose the condition holds, and let ω^i be local defining sections for D. Then for local sections X, Y,

$$\omega^i([X,Y]) = X(\omega^i(Y)) - Y(\omega^i(X)) - d\omega^i(X,Y) = 0$$

by assumption, so $[X,Y]$ is a local section of D. ■

Note that if ω annihilates D and η is an arbitrary differential form, then $\omega \wedge \eta$ annihilates D. Thus D is involutive iff whenever $\omega \in \Omega^p(M)$ annihilates D so too does $d\omega$. This is an immediate consequence of the previous two results.

Definition 1.2.5.11

Suppose we have a rank-k distribution $D \subseteq TM$. A coordinate chart (U, φ) for M is **flat for** D if $\varphi(U) \subseteq \mathbb{R}^n$ is an open cube, and for every $p \in U$ D is spanned by $\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^k}$. In such a case, any slice $x^{k+1} = c^{k+1}, \ldots, x^n = c^n$ is an integral manifold of D. If every point of M lies in a flat chart for D, then D is said to be **completely integrable**.

Clearly a completely integrable distribution is integrable, and integrable distributions are involutive. The following theorem will show that these three concepts are equivalent.

Theorem 1.2.5.12

Every involutive distribution is completely integrable.

Proof

Suppose that D is locally spanned by commuting vector fields $V_1, \ldots, V_k$. Then by the canonical form for commuting vector fields $V_i = \frac{\partial}{\partial x^i}$ for a coordinate chart (U, φ); this chart is then flat for D. Thus we will show that if D is involutive, it is locally spanned by commuting vector fields.

Because our questions are local, we may assume M is an open subset $U \subseteq \mathbb{R}^n$. Let $X_1, \ldots, X_k$ be a local frame for D, and let $p \in U$. By potentially reordering coordinates, we may assume that D_p is complementary to the subspace spanned by $\frac{\partial}{\partial x^{k+1}}\big|_p, \ldots, \frac{\partial}{\partial x^n}\big|_p$. Let $\pi\colon \mathbb{R}^n \to \mathbb{R}^k$ be projection onto the first k coordinates, which induces $d\pi\colon T\mathbb{R}^n \to T\mathbb{R}^k$ by

$$d\pi\left(\sum_{i=1}^n v^i \left.\frac{\partial}{\partial x^i}\right|_q\right) = \sum_{i=1}^k v^i \left.\frac{\partial}{\partial x^i}\right|_{\pi q}.$$

Restricting $d\pi$ to D remains a smooth bundle morphism.

By choice of coordinates, $D_p \subseteq T_p\mathbb{R}^n$ is complementary to $\ker d\pi_p$. Thus $d\pi_p|_{D_p}$ is an isomorphism $D_p \to T_{\pi p}\mathbb{R}^k$. And so by continuity (e.g. of the determinant), so is $d\pi_q|_{D_q}$ for q in a neighborhood of p. Now define a new local frame of D in a neighborhood of p by

$$V_i|_q = (d\pi|_{D_q})^{-1} \left.\frac{\partial}{\partial x^i}\right|_{\pi q},$$

which is smooth as $d\pi$'s matrix entries are.

We claim that the V_i commute; this is sufficient to prove the theorem. By naturality of the Lie brackets,

$$d\pi_q[V_i, V_j]_q = \left[\left.\frac{\partial}{\partial x_i}\right|_{\pi q}, \left.\frac{\partial}{\partial x_j}\right|_{\pi q}\right] = 0,$$

and since D is involutive $[V_i, V_j]_q \in D_q$ on which $d\pi_q$ is an isomorphism, so $[V_i, V_j] = 0$ as desired. ■

Lemma 1.2.5.13

Let D be an involutive rank-k distribution, and $(U, (x^i))$ a flat chart for D. If H is any integral manifold of D, then $H \cap U$ is a countable union of disjoint open subsets of parallel k-slices of U. These parallel slices are open in H and embedded in M.

Proof

The final claim is immediate as being open in U implies being open in $U \cap H$ (as U is open), and slices of U are embedded in U and thus in M.

Let H be an integral manifold of D, and consider the connected components of $H \cap U$. Let V be such a connected component, we will show it is contained in a single slice. Because $dx^{k+1}, \dots, dx^n$ are local defining forms for D, it follows that the pullbacks of these forms to H and thus V are zero. Thus $x^{k+1}, \dots, x^n$ are all constant on V as it is connected, so V lies in a single slice S.

And so $H \cap U$ is the disjoint union of such slices of $x^{k+1}, \dots, x^n$, as desired. ■

A smooth manifold $H \subseteq M$ is **weakly embedded** if any smooth map $F\colon N \to M$ whose image lies in H is smooth as a map $F\colon N \to H$.

Theorem 1.2.5.14

Every integral manifold of an involutive distribution is weakly embedded.

Proof

Let H be an integral manifold of D, and $F\colon N \to M$ be smooth such that $F(N) \subseteq H$. Let $p \in N$ and $q = F(p) \in H$, so there are flat coordinates $(y^1, \dots, y^n)$ for D in a neighborhood $q \in U$. Let (x^i) be coordinates for N on a connected neighborhood B of p so that $F(B) \subseteq U$. We then write F in terms of these coordinates

$$(y^1, \dots, y^n) = (F^1(x), \dots, F^n(x)).$$

By the above lemma, as $F(B) \subseteq H \cap U$, $F^{k+1}, \dots, F^n$ takes only countably many values (as there are only countably many slices). As B is connected, the intermediate value theorem shows that these functions must be constant. Thus $F(B)$ lies in a single slice $S \subseteq U$. And since $H \cap S$ is open in H and embedded in M, $F|_B$ is smooth from $B \to S \cap H$, and thus smooth into H. ■

We can also view involutive distributions in another way, through foliations.

Definition 1.2.5.15

Let $\mathcal{F}$ be a collection of k-dimensional submanifolds of M. A smooth chart (U, φ) for M is **flat for** $\mathcal{F}$ if $\varphi(U) \subseteq \mathbb{R}^n$ is a cube and each manifold of $\mathcal{F}$ intersects U in a countable union of k-dimensional slices of the form $x^{k+1} = c^{k+1}, \dots, x^n = c^n$.

A k-dimensional **foliation** for M is a collection $\mathcal{F}$ of disjoint, connected, immersed k-dimensional submanifolds of M whose union is M such that each $p \in M$ lies in a smooth chart flat for $\mathcal{F}$. The manifolds of $\mathcal{F}$ are called the **leaves** of the foliation.

Proposition 1.2.5.16

Let $\mathcal{F}$ be a foliation on M, then the collection of tangent spaces to the leaves of $\mathcal{F}$ forms an involutive distribution.

Proof

Define D to be this distribution. Let $p \in M$ and U be flat for $\mathcal{F}$. Then for every leaf intersecting U, its tangent space is spanned by $\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^k}$. These then form a local frame of D, so it is smooth. Furthermore U is clearly flat for D, so it is completely integrable. ■

The converse of this proposition is also true, though we will not prove it.

Theorem 1.2.5.17 (Global Frobenius)

Let D be an involutive distribution. The collection of all maximal connected integral manifolds of D forms a foliation of M.

Thus there is a correspondence between involutive distributions and foliation.

2.6 Contact structures

Definition 1.2.6.1

If ξ is a distribution on M, an **integral submanifold** of ξ is an immersed manifold $L \subseteq M$ such that its tangent bundle is a subbundle of the distribution: $TL \subseteq \xi|_L$. This is in contrast to an integral manifold where this is an equality.

Contact geometry is the odd-dimensional analogue of symplectic geometry. In this section we consider corank-1 distributions, that is distributions of rank $\dim M - 1$. Recall that if ξ is a corank-1 distribution with local defining section α, then ξ is involutive (in this neighborhood) iff $d\alpha$ annihilates ξ (by proposition 1.2.5.10).

This is equivalent to $\alpha \wedge d\alpha = 0$. Indeed if $d\alpha$ does not annihilate $\ker \alpha$ then there are $X, Y \in \ker \alpha$ such that $d\alpha(X, Y) \neq 0$. Let $\alpha(Z) \neq 0$, then $(\alpha \wedge d\alpha)(X, Y, Z) = \alpha(Z)\, d\alpha(X, Y) \neq 0$. Conversely if $d\alpha$ annihilates $\ker \alpha$, then for any $X, Y \in \ker \alpha$ let $Z \notin \ker \alpha$ so it is linearly independent, then $(\alpha \wedge d\alpha)(X, Y, Z) = \alpha(Z)\, d\alpha(X, Y)$ is zero so that $d\alpha(X, Y) = 0$.

A contact structure is a distribution which is as far from being integrable as possible. Motivated by this α is a contact form iff $d\alpha$ is nondegenerate on ξ.

Definition 1.2.6.2

A **contact form** for a corank-1 distribution M is a local defining form α such that $d\alpha$ is nondegenerate on $\ker \alpha$. A **contact structure** is a corank-1 distribution such that every neighborhood has a contact form.

Proposition 1.2.6.3

Let M be a $(2n+1)$-dimensional manifold and $\xi \subseteq TM$ a distribution with local defining form α.

(1) α is a contact form iff $\alpha \wedge (d\alpha)^n \neq 0$.

(2) If α, α' are two local defining forms on the same open set, then α is a contact form iff α' is.

(3) If ξ is a contact structure, then the symplectic form on ξ induced by $d\alpha$ for a contact form α is independent on choice of α up to positive scaling function.

Proof

Because $\ker\alpha$ is $2n$-dimensional, $d\alpha$ is nondegenerate on it iff $(d\alpha)^n \neq 0$. As above, this is iff $\alpha\wedge(d\alpha)^n \neq 0$, proving (1).

If α, α' share a kernel, they are multiples of one another. This is because their kernels have corank-1, and so taking a vector X outside of the kernel gives the scaling factor: $\alpha = \alpha(X)/\alpha'(X)\alpha'$. So $\alpha' = f\alpha$, so $d\alpha' = f\,d\alpha + df\wedge\alpha$, which agrees with $f\,d\alpha$ on ξ, proving (2) and (3). ■

Suppose that α is a contact form for $\xi \subseteq TM$, and let $L \subseteq M$ be an integral submanifold for ξ. Then for $X, Y \in TL = \xi$, we also have $[X,Y] \in TL$,

$$d\alpha(X,Y) = X\alpha(Y) - Y\alpha(X) + \alpha([X,Y]) = 0$$

because α is zero on ξ. Thus for $q \in L$, T_qL is an isotropic subspace of $(\xi_q, d\alpha)$, and thus $\dim L \leq n$. If this is an equality, i.e. T_qL is a Lagrangian subspace of ξ_q, then L is called an **Legendrian** submanifold.

If α is a contact form for ξ, it defines a unique (local) vector field determined by

$$\iota(Y)\,d\alpha = 0, \qquad \alpha(Y) = 1.$$

Because $d\alpha$ is nondegenerate on ξ, $\ker d\alpha$ has dimension 1, thus Y is indeed unique. This is the **Reeb field** determined by α and is denoted by $Y = Y_\alpha$. Note that

$$\mathcal{L}_Y\alpha = d\iota(Y)\alpha + \iota(Y)\,d\alpha = 0,$$

so Y's flows preserves α, and hence ξ. That means that if θ_t is Y's flow then $\theta_t^*(\xi) = \xi$.

If f is a non-vanishing smooth real-valued function, then $f\alpha$ is another contact form, but $Y_{f\alpha}$ and Y_α differ, and their flows may have different properties.

Example 1.2.6.4

Consider $\mathbb{R}^{2n+1}$ with the standard coordinates $x_1,\dots,x_n,y_1,\dots,y_n,z$. Then

$$\alpha_0 = \sum_i y_i\,dx_i - dz$$

is a contact form on $\mathbb{R}^{2n+1}$, defining the **standard contact structure**. This is a contact form because

$$d\alpha_0 = \sum_i dy_i\wedge dx_i,$$

and thus

$$(d\alpha_0)^n\wedge\alpha_0 = n!(dy_1\wedge dx_1\wedge\cdots\wedge dy_n\wedge dx_n)\wedge\left(\sum_i y_i\,dx_i - dz\right) = -n!\,dy_1\wedge\cdots dx_n\wedge dz \neq 0.$$

Its Reeb field is $Y = -\frac{\partial}{\partial z}$: clearly $\alpha_0(-\frac{\partial}{\partial z}) = 1$ and $d\alpha_0(-\frac{\partial}{\partial z}, \bullet) = 0$.

Notice that α_0 is invariant under translations in the x_i and z directions. And submanifolds of the form $x_i = c_i, z = c$ (i.e. parallel to the y axis) are Legendrian submanifolds. They have dimension n and their tangent spaces are spanned by $\frac{\partial}{\partial y_i}$ which is annihilated by $d\alpha_0$.

Example 1.2.6.5

Let L be a manifold, then

$$J^1L = T^*L\times\mathbb{R}$$

is a contact manifold with contact form

$$\alpha = \lambda_{\mathsf{can}} - dz.$$

As before the Reeb vector field is $Y = -\frac{\partial}{\partial z}$. Then for any smooth function $S\colon L \to \mathbb{R}$,

$$L_S = \{(x, dS(x), S(x)) \mid x \in L\} \subseteq J^1 L$$

is Legendrian.

Definition 1.2.6.6

Let (M, ξ) be a contact manifold, then a diffeomorphism $\varphi\colon M \to M$ is a **contactomorphism** iff $\varphi_* \xi = \xi$, i.e. $d\varphi_p(\xi_p) = \xi_{\varphi p}$ for all $p \in M$.

If α is a contact form, then this is equivalent to $\ker \alpha = \ker \varphi^* \alpha$, so that $\varphi^* \alpha = f\alpha$ for a nowhere vanishing smooth function $f \in C^\infty(M)$.

Definition 1.2.6.7

A **contact isotopy** is an isotopy ψ_t such that each ψ_t is a contactomorphism.

For a contact isotopy ψ_t, so $\psi_t^* \alpha_t = f_t \alpha_t$, let X_t be the time-dependent vector field associated to the contact isotopy ψ_t, so

$$\frac{d}{dt}\psi_t = X_t \circ \psi_t,$$

then

$$\psi_t^* \mathcal{L}_{X_t} \alpha = \frac{d}{dt}\psi_t^* \alpha = \frac{d}{dt} f_t \alpha.$$

Now define

$$g_t = \frac{\frac{d}{dt} f_t}{f_t} \circ \psi_t^{-1},$$

which is smooth as f_t is nonzero, so that

$$\psi_t^*(g_t \alpha) = (g_t \circ \psi_t)\psi_t^* \alpha = \frac{\frac{d}{dt} f_t}{f_t} f_t \alpha = \frac{d}{dt} f_t \alpha,$$

thus we have

$$\psi_t^* \mathcal{L}_{X_t} \alpha = \psi_t^*(g_t \alpha) \implies \mathcal{L}_{X_t} \alpha = g_t \alpha.$$

Conversely if X_t satisfies the condition $\mathcal{L}_{X_t} \alpha = g_t \alpha$ then X_t's flow ψ_t determines a contact isotopy as follows. Define

$$h_t = \int_0^t g_s \circ \psi_s \, ds,$$

and then define $f_t = \exp(h_t)$, and this will satisfy the condition. We see this because $\psi_t^* \alpha$ satisfies (for all p) the ODE $\frac{d}{dt}\psi_t^* \alpha = (g_t \circ \psi_t)\psi_t^* \alpha$ which has this as its solution.

A vector field $X \in \mathfrak{X}(M)$ such that $\mathcal{L}_X \alpha = \alpha$ is called a **contact vector field**. Thus we have shown that the Lie algebra of the Lie group of contactomorphisms is the space of contact vector fields.

Proposition 1.2.6.8

Let α be a contact form with Reeb field Y.

(1) $X \in \mathfrak{X}(M)$ is a contact vector field iff there is a function $H\colon M \to \mathbb{R}$ such that

$$\iota(X)\alpha = H, \qquad \iota(X)\,d\alpha = (\iota(Y)\,dH)\alpha - dH.$$

(2) For every function $H\colon M \to \mathbb{R}$ there is a unique contact vector field X_H which satisfies the above identities.

Proof

By Cartan,

$$\mathcal{L}_X\alpha = \iota(X)\,d\alpha + d(\iota(X)\alpha),$$

so if the conditions hold, this is equal to $\iota(Y)\,dH\alpha$, so define $g = \iota(Y)\,dH = YH$. Conversely if $\mathcal{L}_X\alpha = g\alpha$, let $H = \iota(X)\alpha$ so that

$$\iota(X)\,d\alpha = \mathcal{L}_X\alpha - d(\iota(X)\alpha) = g\alpha - dH.$$

If we evaluate this at Y, because $\iota(Y)\,d\alpha = 0$, we get

$$g = g\iota(Y)\alpha = \iota(Y)\,dH$$

as desired.

For (2) given H there is a unique section Z of ξ such that

$$-\iota(Z)\,d\alpha|_\xi = dH|_\xi,$$

as $\iota(\bullet)\,d\alpha$ is an isomorphism $\Gamma(\xi) \to \Gamma^*(\xi)$. Then

$$X_H = HY + Z$$

satisfies $\alpha(X_H) = H\alpha(Y) = H$ and

$$\iota(X_H)\,d\alpha = \iota(Z)\,d\alpha = (\iota(Y)\,dH)\alpha - dH.$$

The last equality is because both sides agree on ξ, and vanish on Y. ∎

The above proposition shows, for example, that the constant function $H(p) = 1$ generates the Reeb vector field Y. Furthermore the only contact vector field which is everywhere tangent to ξ is the zero vector field: we have $H = \iota(X)\alpha = 0$ (by assumption) and thus $\iota(X)\,d\alpha = 0$, but since $X \in \xi$ and $d\alpha$ is nondegenerate this means $X = 0$.

Remark 1.2.6.9

If α is a global contact form on M, then it defines a **Poisson bracket** on $C^\infty(M)$ by

$$\{F, G\} = \alpha[X_F, X_G] = dF(X_G) - dG(X_F) + d\alpha(X_F, X_G) = dF(Y)\cdot G - dG(X_F).$$

Then it can be shown that $[X_F, X_G] = X_{\{F,G\}}$, so that $\{\bullet, \bullet\}$ forms a Lie algebra structure on $C^\infty(M)$.

TODO: Continue

II. Floer Homology

1 Morse Theory

Material from the article "A Quick View of Lagrangian Floer Homology" *by Andreś Pedroza and the book* "Morse Theory" *by J. Milnor*

1.1 Morse functions

Definition 2.1.1.1

Let $f: M \to \mathbb{R}$ be a smooth function, then a **critical point** of f is a point $p \in M$ such that $df_p = 0$. Denote by $\mathrm{Crit}(f)$ the set of critical points of f.

For $p \in \mathrm{Crit}(f)$ define the bilinear form $df_p^2: T_pM \times T_pM \to \mathbb{R}$ by

$$df_p^2(X, Y) = X(\widetilde{Y} f)$$

where $\widetilde{Y}$ is any extension of Y to a local vector field. Then p is said to be **nondegenerate** iff df_p^2 is a nondegenerate bilinear form. Its **index** is the number of negative eigenvalues of this bilinear form and is denoted $\mathrm{ind}(f; p)$.

Note df_p^2 is well-defined when p is critical: if $\widetilde{Y} = Y^i \frac{\partial}{\partial x^i}$ is any extension of Y, then

$$X(\widetilde{Y} f) = X\left(Y^i \frac{\partial f}{\partial x^i}\right) = X(Y^i) \left.\frac{\partial f}{\partial x^i}\right|_p + Y^i(p) X\left(\frac{\partial f}{\partial x^i}\right) = Y^i(p) X\left(\frac{\partial f}{\partial x^i}\right)$$

which is only dependent on $Y^i(p)$, i.e. Y. Relative to a coordinate chart $\frac{\partial}{\partial x^i}$, the bilinear form has the representation

$$df_p^2\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) = \frac{\partial^2 f}{\partial x^i x^j}(p),$$

i.e. it is the Hessian of f relative to this coordinate chart. Thus p is nondegenerate iff the Hessian is nonsingular, and its index is the number of negative eigenvalues of the Hessian.

Definition 2.1.1.2

A smooth function $f: M \to \mathbb{R}$ is called a **Morse function** if all of its critical points are nondegenerate.

Example 2.1.1.3

Consider $M = \mathbb{R}^2$ with $f(x, y) = x^2 + y^2$, $g(x, y) = x^2 - y^2$, and $h(x, y) = -x^2 - y^2$. The only critical point of these functions is the origin, and the Hessian of these are

$$H_f = 2I, \qquad H_g = \mathrm{diag}(2, -2), \qquad H_h = \mathrm{diag}(-2, -2),$$

thus

$$\mathrm{ind}(f; 0) = 0, \qquad \mathrm{ind}(g; 0) = 1, \qquad \mathrm{ind}(h; 0) = 2.$$

Theorem 2.1.1.4 (Morse Lemma)

Let $f: M \to \mathbb{R}$ be a smooth function and $p \in M$ a nondegenerate critical point of f with index λ. Then there is a coordinate chart (x_i) about p such that in this chart

$$f(x_1, \ldots, x_n) = f(p) - x_1^2 - \cdots - x_\lambda^2 + x_{\lambda+1}^2 + \cdots + x_n^2. \tag{1}$$

Proof

This is a local question so we may assume that $M = \mathbb{R}^n$ and $p = 0$. Firstly for any such coordinate chart, λ must be the index of p. Indeed in such a coordinate chart the Hessian of f has the form

$$\frac{\partial^2 f}{\partial x_i x_j}(p) = \begin{cases} -2 & i = j \leq \lambda \\ 2 & i = j > \lambda \\ 0 & \text{else} \end{cases},$$

which has λ negative eigenvalues and so $\lambda = \operatorname{ind}(f; p)$.

We may assume that $f(p) = f(0) = 0$, and thus we can write f via a Taylor series

$$f(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i g_i(x_1, \ldots, x_n), \qquad g_i(0) = \frac{\partial f}{\partial x_i}(0).$$

Because p is a critical point we have $g_i(0) = 0$ and so expanding these as a Taylor series

$$f(x_1, \ldots, x_n) = \sum_{i,j=1}^{n} x_i x_j h_{ij}(x_1, \ldots, x_n), \qquad h_{ij}(0) = \frac{\partial}{\partial^2 f} x_i x_j(0).$$

We may replace h_{ij} with $\frac{1}{2}(h_{ij} + h_{ji})$ so that $h_{ij} = h_{ji}$. Note that because the matrix $(h_{ij}(0))_{ij}$ is the Hessian of f at 0 which is nondegenerate, we may perform some linear change of coordinates and assume that $h_{11}(0) \neq 0$.

By induction we prove that there are coordinates $u_1, \ldots, u_n$ for which f has the representation

$$f = \pm u_1^2 \pm \cdots \pm u_{k-1}^2 + \sum_{i,j \geq k} u_i u_j H_{ij}(u_1, \ldots, u_n)$$

such that $(H_{ij}(u_1, \ldots, u_n))_{ij}$ is symmetric and $H_{kk}(0) \neq 0$. We have shown this for $k = 1$. Define

$$g(u_1, \ldots, u_n) = \sqrt{|H_{kk}(u_1, \ldots, u_n)|},$$

which is smooth and nonzero in some neighborhood of 0. Define now new coordinates $v_1, \ldots, v_n$ by

$$v_i = u_i \text{ for } i \neq k$$
$$v_k = g(u_1, \ldots, u_n)\left(u_k + \sum_{i>k} u_i H_{ik}(u_1, \ldots, u_n)/H_{kk}(u_1, \ldots, u_n)\right).$$

A computation shows that

$$\frac{\partial v_k}{\partial u_k}(0) = g(0) \neq 0,$$

so that there is a neighborhood in which the v_is form a coordinate chart. Then f has the representation

$$f = \sum_{i \leq k} \pm v_i^2 + \sum_{i,j>k} v_i v_j H'_{ij}(v_1, \ldots, v_n), \qquad H'_{ij}(u_1, \ldots, u_n) = H_{ij} - \frac{H_{ik}^2}{H_{kk}},$$

giving us the desired result. ■

Because smooth functions of the form (1) have a single critical point at the origin, an immediate corollary of this theorem is that nondegenerate critical points are isolated.

Corollary 2.1.1.5

Nondegenerate critical points of a smooth function are isolated.

We now show that every manifold posseses a Morse function.

Proposition 2.1.1.6

Let $M \subseteq \mathbb{R}^n$ be an embedded submanifold, then for almost every $p \in \mathbb{R}^n$ the map

$$f_p \colon M \to \mathbb{R}^n, \qquad x \mapsto |x - p|^2$$

is a Morse function on M.

Proof

The differential of f_p is

$$d_x f_p(v) = 2(x - p) \cdot v,$$

which is identically zero iff $x - p$ is orthogonal to $T_x M$. Let $\varphi = (u_i)$ be a local chart around x, so that in this chart

$$\frac{\partial f_p}{\partial u_i} = 2(x - p) \cdot \frac{\partial \varphi}{\partial u_i},$$

and thus

$$\frac{\partial^2 f_p}{\partial u_i u_j} = 2\left(\frac{\partial \varphi}{\partial u_j} \cdot \frac{\partial \varphi}{\partial u_i} + (x - p) \cdot \frac{\partial^2 \varphi}{\partial u_i u_j} \right).$$

So x is a nondegenerate critical point of f_p iff $x - p$ is orthogonal to $T_x M$ and the above matrix has full rank. We will show that the set of p for which f_p is not Morse are critical values of some smooth function; by Sard this will suffice.

Let $\nu_M \subseteq T\mathbb{R}^n$ be the normal bundle of M, i.e.

$$\nu_M = \left\{ (x, v) \in M \times \mathbb{R}^n \;\middle|\; v \in T_x M^{\perp} \right\} \subseteq M \times \mathbb{R}^n.$$

And define

$$E \colon \nu_M \to \mathbb{R}^n, \qquad (x, v) \mapsto x + v.$$

Then we claim that $p = x + v$ is a critical value of E iff the above matrix is singular. This can be shown. ■

By Whitney's embedding theorem, every smooth manifold can be embedded in some $\mathbb{R}^n$, so the above proposition holds for arbitrary M. Thus every manifold (of positive dimension) possesses a Morse function. In fact it can be shown that for a compact manifold M, the set of Morse functions is an open dense subset of $C^\infty(M, \mathbb{R})$.

1.2 Homotopy type in terms of critical values

Let f be a Morse function on a manifold M, then for $a \in \mathbb{R}$ we define the **level set**

$$M_a = f^{-1}(-\infty, a] = \{ p \in M \mid f(p) \leq a \}.$$

Because $(-\infty, a]$ has codimension 0 in $\mathbb{R}$, f is trivially transervsal to it, and so M_a is a smooth submanifold of M. It turns out that these level sets offer rich insight into M. For one, if M is compact then f attains a minimum a and a maxmimum b, so that

$$M_x = \emptyset \text{ when } x < a, \qquad M_y = M \text{ when } y \geq b.$$

Now a, b are critical values of f, and it turns out that these level sets have interesting behavior around level sets.

For a Riemannian metric g on M and $f \in C^\infty(M)$, define $\operatorname{grad} f \in \mathfrak{X}(M)$ to be the vector field determined by

$$\iota(\operatorname{grad} f) g = df.$$

Theorem 2.1.2.1

Let f be a smooth real-valued function on M, and let $a < b$ be real. Suppose that $f^{-1}[a,b] \subseteq M$ is compact and contains no critical points of f. Then M_a and M_b is diffeomorphic, and the inclusion $M_a \hookrightarrow M_b$ is a strong deformation retract.

Proof

Choose a Riemannian metric on M, and define $\rho: M \to \mathbb{R}$ to be a smooth function equal to $1/\langle \operatorname{grad} f, \operatorname{grad} f\rangle$ on the compact set $f^{-1}[a,b]$ which vanishes outside a compact neighborhood of $f^{-1}[a,b]$. Note that ρ is only-well defined by the assumption that f has no critical points in $f^{-1}[a,b]$. Then define $X = \rho \operatorname{grad}(f)$. This vector field is compactly supported and thus generates a global flow $\theta_t: M \to M$.

Fix a $p \in M$ and consider the map $t \mapsto f(\theta_t(p))$, which is smooth $\mathbb{R} \to \mathbb{R}$. Then

$$\frac{df(\theta_t(p))}{dt} = \left\langle \frac{d\theta_t(p)}{dt}, \operatorname{grad} f \right\rangle = \langle X, \operatorname{grad} f\rangle = \frac{\rho}{\operatorname{grad} f, \operatorname{grad} f}.$$

This is 1 when $a \le f(\theta_t(p)) \le b$, and otherwise we may assume it is bound by 1. Thus $t \mapsto f(\theta_t(p))$ is linear of the form $t + f(p)$ when $a \le f(\theta_t(p)) \le b$, and otherwise is less than $t + f(p)$.

Consider the diffeomorphism $\theta_{b-a}: M \to M$. Then if $f(p) \le a$ we have $f(\theta_{b-a}(p)) \le b - a + f(p) \le b$ so $\theta_{b-a}(M_a) \subseteq M_b$. Similarly we see that $\theta_{a-b} = \theta_{b-a}^{-1}$ maps M_b into M_a so that they form a diffeomorphism $M_a \cong M_b$.

Now define a homotopy $H: [0,1] \times M_b \to M_b$ by

$$H_t(p) = \begin{cases} p & \text{if } f(p) \le a \\ \varphi_{t(a-f(p))}(p) & \text{if } a \le f(p) \le b \end{cases}.$$

Both of these cases agree when $f(p) = a$, and thus this is continuous. Now H_0 is the identity, $H_t(p) = p$ when $p \in M_a$, and $H_1(p) = \varphi_{a-f(p)}(p) \in M_a$ for $p \notin M_a$ so that H is a strong deformation retract. ■

Theorem 2.1.2.2

Let $f: M \to \mathbb{R}$ be a smooth function and $p \in M$ a nondegenerate critical point with index λ and $f(p) = c$. If $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical points other than p then ε can be taken sufficiently small so that $M_{c+\varepsilon}$ is homotopy equivalent to $M_{c-\varepsilon}$ attached to a λ-cell.

Note:

By a λ-cell we mean a manifold diffeomorphic to the unit ball in $\mathbb{R}^\lambda$. Attaching it to a manifold means gluing it along some $g: \partial e_\lambda \to M$.

Proof

Take a neighborhood U around p such that f has the representation

$$f = c - u_1^2 - \cdots - u_\lambda^2 + u_{\lambda+1}^2 + \cdots u_n^2.$$

Take $\varepsilon > 0$ sufficiently small so that $f^{-1}[c-\varepsilon, c+\varepsilon]$ contains no critical points other than p and U contains the closed ball $\left\{ \overline{u} \;\middle|\; |u|^2 \le 2\varepsilon \right\}$. Define e_λ in U to be

$$e_\lambda = \left\{ (u_1, \ldots, u_\lambda, 0, \ldots, 0) \in U \;\middle|\; u_1^2 + \cdots + u_\lambda^2 \le \varepsilon \right\}.$$

Now, $\overline{u} \in e_\lambda \cap M_{c-\varepsilon}$ iff

$$u_{\lambda+1} = \cdots = u_n = 0, \qquad u_1^2 + \cdots + u_\lambda^2 \le \varepsilon, \qquad -u_1^2 + \cdots + u_\lambda^2 \le -\varepsilon,$$

i.e. $e_\lambda \cap M_{c-\varepsilon} = \partial e_\lambda$. Thus in U we see that $e_\lambda \cup M_{c-\varepsilon}$ is a gluing of e_λ and $M_{c-\varepsilon}$.
Let $\mu\colon \mathbb{R} \to \mathbb{R}$ be a smooth function such that

$$\mu(0) > \varepsilon, \qquad \mu(r) = 0 \text{ for } r \geq 2\varepsilon, \qquad -1 < \mu' \leq 0.$$

Define $F\colon M \to \mathbb{R}$ to be equal to f outside of U and within U,

$$F = f - \mu(u_1^2 + \cdots + u_\lambda^2 + 2u_{\lambda+1}^2 + \cdots + 2u_n^2).$$

This is a smooth function as it agrees on its overlaps. For convenience define

$$\xi, \eta\colon U \to [0, \infty), \qquad \xi = u_1^2 + \cdots + u_\lambda^2, \qquad \eta = u_{\lambda+1}^2 + \cdots + u_n^2$$

so that

$$f = c - \xi + \eta, \qquad F = c - \xi + \eta - \mu(\xi + 2\eta).$$

Now we make a few claims which will progress our proof.

(1) $F^{-1}(-\infty, c+\varepsilon] = M_{c+\varepsilon} = f^{-1}(-\infty, c+\varepsilon]$. Outside the ellipsoid $\xi + 2\eta \leq 2\varepsilon$ both f and F agree, so that these sets agree outside of this ellipsoid. Within the ellipsoid note that

$$F \leq f = c - \xi + \eta \leq c + \frac{1}{2}\xi + \eta \leq c + \varepsilon$$

and so the ellipsoid is contained in both sets.

(2) The critical points of F and f are the same. We have

$$\frac{\partial F}{\partial \xi} = -1 - \mu'(\xi + 2\eta) < 0, \qquad \frac{\partial F}{\partial \eta} = 1 - 2\mu'(\xi + 2\eta) \geq 1,$$

and so

$$dF = \frac{\partial F}{\partial \xi}\, d\xi + \frac{\partial F}{\partial \eta}\, d\eta,$$

and since $d\xi, d\eta$ are only every simultaneously zero at the origin, F's only critical point in U is the origin. And outside of U it agrees with f.

(3) $F^{-1}(-\infty, c-\varepsilon]$ is a deformation retract of $M_{c+\varepsilon}$. Now consider $F^{-1}[c-\varepsilon, c+\varepsilon] = F^{-1}[c-\varepsilon, \infty) \cap F^{-1}(-\infty, c+\varepsilon]$. By the first point, the second set here is $f^{-1}(-\infty, c+\varepsilon]$. And since $F \leq f$ in U, the first set is contained in $f^{-1}[c-\varepsilon, \infty)$. Putting this together we have

$$F^{-1}[c-\varepsilon, c+\varepsilon] \subseteq f^{-1}[c-\varepsilon, c+\varepsilon].$$

Thus it is compact, and by (2) the only critical point of F in it is potentially the origin. But

$$F(p) = c - \mu(0) < c - \varepsilon,$$

so that the origin is not in $F^{-1}[c-\varepsilon, c+\varepsilon]$. By theorem 2.1.2.1 this means that $F^{-1}(-\infty, c-\varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c+\varepsilon]$ which is $M_{c+\varepsilon}$ by (1).

Let us denote by H the closure of $F^{-1}(-\infty, c-\varepsilon] - M_{c-\varepsilon}$ so that $F^{-1}(-\infty, c-\varepsilon] = M_{c-\varepsilon} \cup H$. Now we have defined

$$e_\lambda = \{q \in U \mid \xi(q) \leq \varepsilon,\ \eta(q) = 0\}.$$

Because $\frac{\partial F}{\partial \xi} < 0$ we have that for $q \in e_\lambda$, $F(q) \leq F(p) < c - \varepsilon$ and $f(q) \geq c - \varepsilon$. Thus e_λ is contained in $F^{-1}(-\infty, c-\varepsilon] - M_{c-\varepsilon} \subseteq H$.
Finally we claim that $M_{c-\varepsilon} \cup e_\lambda$ is a deformation retract of $M_{c-\varepsilon} \cup H$. In lieu of (3) this proves the desired result. The deformation retract h_t is defined as follows:

(1) For $\xi \leq \varepsilon$ define

$$h_t(u_1, \ldots, u_n) = (u_1, \ldots, u_\lambda, tu_{\lambda+1}, \cdots, tu_n)$$

so that h_0 maps into e_λ, h_1 is the identity, and $h_t|_{e_\lambda}$ is the identity. This is well-defined, as in it maps $F^{-1}(-\infty, c-\varepsilon]$ to itself, because $\frac{\partial F}{\partial \eta} > 0$ so that $F(\xi, t\eta) \leq F(\xi, \eta) \leq c - \varepsilon$ for $t \leq 1$.

(2) For $\varepsilon \leq \xi \leq \eta + \varepsilon$, let

$$h_t(u_1, \ldots, u_n) = (u_1, \ldots, u_\lambda, s_t u_{\lambda+1}, \ldots, s_t u_n), \qquad s_t = t + (1-t)\left(\frac{\xi - \varepsilon}{\eta}\right)^{1/2}.$$

Here $s_t \in [0, 1]$ so that this is well-defined for the same reason as above. And $s_1 = 1$ so that h_1 is the identity, and h_0 maps into $f^{-1}(c-\varepsilon) \subseteq M_{c-\varepsilon}$. When $\xi = \varepsilon$ this agrees with the definition in (1), so it remains continuous.

(3) For $\eta + \varepsilon \leq \xi$, i.e. within $M_{c-\varepsilon}$, define h_t to be the identity. When $\xi = \eta + \varepsilon$ this corresponds to the definition in (2).

And so we have finished. ■

The following corollary can be obtained from topological considerations regarding, and is immediate in the case when M is compact (by induction).

Corollary 2.1.2.3

Let $f: M \to \mathbb{R}$ be a Morse function on M such that for every $a \in \mathbb{R}$, M_a is compact. Then M is homotopy-equivalent to a CW-complex with a cell of dimension λ for each critical point of index λ.

1.3 The Morse-Witten complex

Definition 2.1.3.1

Let $f: M \to \mathbb{R}$ be a Morse function. A **pseudo-gradient field** for f is a vector field $X \in \mathfrak{X}(M)$ such that

(1) $(Xf)(p) \leq 0$ with equality iff p is a critical point of f.

(2) There is a local Morse chart around every critical point of f in which X coincides with the negative gradient for f.

Explicitly point (2) means that if $p \in \operatorname{Crit} f$ then there is a Morse chart $(U, (u_i))$ centered at p such that X has the form

$$X = -\sum_i \frac{\partial f}{\partial u_i}\frac{\partial}{\partial u_i} = 2\left(-\sum_{i=1}^{\lambda} u_i \frac{\partial}{\partial u_i} + \sum_{i=\lambda+1}^{n} u_i \frac{\partial}{\partial u_i}\right). \tag{1}pgradloc$$

Every Morse function posseses a pseudo-gradient. Because critical points are isolated, there are only countably many. So let $\{c_i\}_{i\geq 1}$ enumerate f's critical points and let (U_i, φ_i) be disjoint Morse charts for each c_i. We can then add more charts to obtain an atlas $(U_i, \varphi_i)_i$ of M, and we may assume that each critical point lies in a single chart, and moreover lies in a neighborhood disjoint from all other U_js, call it V_i.

Now let X_j be a local vector field on U_j defined by pulling back $-\operatorname{grad} f \circ \varphi_i^{-1}$ on $\varphi_i(U_i) \subseteq \mathbb{R}^n$. Let $(\chi_j)_j$ be a partition of unity subordinate to this open cover and define

$$X = \sum_j \chi_j X_j,$$

where of course we extend $\chi_j X_j$ to all of M. Because $X_j f \leq 0$ with equality only at critical points we have $Xf \leq 0$ with equality iff $\chi_j X_j f = 0$ for all j, and taking a j where $\chi_j \neq 0$ we have $X_j f = 0$ so that the point is critical. For a critical point $c_i \in U_i$, X is equal to $X_i = -\operatorname{grad} f$ in the Morse chart V_i.

So we have proven the following.

Proposition 2.1.3.2

Every Morse function possesses a pseudo-gradient.

Definition 2.1.3.3

Let f be a Morse function and X a pseudo-gradient. Let θ_t be the flow of X, and $p \in \operatorname{Crit} f$ critical. Then we define its **stable** and **unstable manifolds** to be, respectively,

$$W_X^s(p) = \{x \in M \mid \lim_{t\to+\infty} \theta_t(x) = p\}, \qquad W_X^u(p) = \{x \in M \mid \lim_{t\to-\infty} \theta_t(x) = p\}.$$

Consider f in a Morse chart,

$$f(x_-, x_+) = c - |x_-|^2 + |x_+|^2.$$

Define the bilinear form Q on $\mathbb{R}^n$ by $Q(x_-, x_+) = -|x_-|^2 + |x_+|^2$. Now for $\varepsilon, \delta > 0$ define the set

$$U(\varepsilon, \delta) = \left\{ x \in \mathbb{R}^n \;\middle|\; -\varepsilon < Q(x) < \varepsilon,\ |x_-|^2 |x_+|^2 \leq \eta(\varepsilon + \eta) \right\}.$$

The Morse chart is diffeomorphic to some $U(\varepsilon, \eta)$ under some diffeomorphism φ. Then the boundary of $U = U(\varepsilon, \eta)$ can be written in three parts:

(1) Subsets of the sublevel sets of Q:

$$\partial_\pm U = \left\{ x \in \partial U \;\middle|\; Q(x) = \pm\varepsilon,\ |x_\mp|^2 \leq \eta \right\}.$$

(2) Trajectories of $\operatorname{grad} Q$:

$$\partial_0 U = \left\{ x \in \partial U \;\middle|\; |x_-|^2 |x_+|^2 = \eta(\varepsilon + \eta) \right\}.$$

Now suppose that $\Omega(p) = \varphi(U(\varepsilon, \eta))$ is a Morse chart about $p \in \operatorname{Crit} f$. Then define

$$\partial_\pm \Omega(p) = \varphi(\partial_\pm U), \qquad \partial_0 \Omega(p) = \varphi(\partial_0 U).$$

Note that in $\Omega(p)$, X has the form $(\,)$, whose flow lines are of the form

$$\theta(t, p) = (p_1 e^{-t}, \ldots, p_\lambda e^{-t}, p_{\lambda+1} e^{t}, \ldots, p_n e^{t}).$$

We see that as $t \to \infty$, this converges to 0 iff $p_i = 0$ for $i > \lambda$ and similarly for $-\infty$. Thus in U,

$$W^s(0) \cap U = U \cap V_+, \qquad W^u(0) \cap U = U \cap V_-,$$

where $V_+ = \{x \in \mathbb{R}^n \mid x_- = 0\}$ and similarly $V_- = \{x \in \mathbb{R}^n \mid x_+ = 0\}$. Define $W_0 = W^s(0) \cap U = U \cap V_+$ in M (i.e. its image under φ). Now if $x \in W^s(p)$ then $\theta(t, x) \in W_0$ for all $t > t_0$ for some t_0. If $\theta(t_0, x) \notin W_0$ then it has a coordinate $i > \lambda$ which is nonzero, but then by our explicit solution for θ in U we see that $\theta(t, x)$ cannot converge to zero, a contradiction.

These arguments can be seen to show the following. The dimensions come from the dimensions of V_+, V_-.

Proposition 2.1.3.4

The stable and unstable manifolds of $p \in \operatorname{Crit} f$ are submanifolds of M diffeomorphic to open balls, with dimensions

$$\dim W^u(p) = \operatorname{codim} W^s(p) = \operatorname{ind}(f; p).$$

Proposition 2.1.3.5

Suppose M is compact, and $\gamma\colon \mathbb{R} \to M$ an integral curve of a pseudo-gradient X. Then the following limits both exist and are critical points of f:

$$\lim_{t \to \infty} \gamma(t) = c, \qquad \lim_{t \to -\infty} \gamma(t) = d.$$

Proof

Let

$$\Omega = \bigcup_{c \in \operatorname{Crit} f} \Omega(c)$$

be the union of all the Morse neighborhoods, of which there are finitely many as M is compact.

We show the claim for c, the claim for d follows similarly or by considering $-f$. Suppose that c does not exist, which means that every time $\gamma(t)$ enters a Morse neighborhood it must then exit it. More than this, it must eventually leave it forever because $f \circ \gamma$ is decreasing. Thus γ must leave all of the Morse neighborhoods at some point, as there are only finitely many. So suppose that for every $t \geq t_0$, $\gamma(t_0) \notin \Omega$.

Because M, and thus $M - \Omega$, is compact, the map $x \mapsto (Xf)(x)$ has a maximum. And because $x \notin \Omega$ this maximum is negative. So there is an $\varepsilon_0 > 0$ such that $(Xf)(x) \leq -\varepsilon_0$ for all $x \notin \Omega$.

And so for every $t \geq t_0$ we have

$$f(\gamma(t)) - f(\gamma(t_0)) = \int_{t_0}^{t} \frac{d(f \circ \gamma)}{du}\, du = \int_{t_0}^{t} (Xf) \circ \gamma \leq -\varepsilon_0(t - t_0).$$

Thus as $t \to \infty$ we have that $f \circ \gamma(t) \to -\infty$ which is impossible as f has a minimum.

So γ remains in a Morse neighborhood forever, which is only if it converges to the critical point of that neighborhood. ■

Immediately we have the following corollary.

Corollary 2.1.3.6

Let M be compact, then it can be decomposed into

$$M = \bigsqcup_{p \in \operatorname{Crit} f} W^s(p) = \bigsqcup_{p \in \operatorname{Crit} f} W^u(f).$$

Definition 2.1.3.7

A pseudo-gradient X for a Morse function f satisfies the **Smale condition** if for every $p, q \in \operatorname{Crit} f$, $W^u(p)$ and $W^s(q)$ intersect transversally.

Notice that if $x \in W^u(p) \cap W^s(q)$ then

$$\lim_{t \to -\infty} f \circ \theta^x(t) = f(p), \qquad \lim_{t \to \infty} f \circ \theta^x(t) = f(q),$$

and since f decreases along X's flow, we must have $f(q) \leq f(p)$.
If X satisfies the Smale condition, then for $p, q \in \operatorname{Crit} f$ define

$$\mathcal{M}(p, q) = W^u(p) \cap W^s(q) = \{x \in M \mid \lim_{t \to -\infty} \theta(t, x) = p,\ \lim_{t \to \infty} \theta(t, x) = q\}.$$

This is the set of all of the flow lines coming from p which flow to q. We know that

$$\operatorname{codim} \mathcal{M}(p, q) = \operatorname{codim} W^u(p) + \operatorname{codim} W^s(q),$$

and thus

$$\dim \mathcal{M}(p, q) = \operatorname{ind}(f; p) - \operatorname{ind}(f; q).$$

Proposition 2.1.3.8

The group $\mathbb{R}$ acts smoothly on $\mathcal{M}(p, q)$ by translations:

$$t \cdot x = \theta(t, x).$$

This action is free and proper if $p \neq q$.

Proof

This is obviously a smooth group action. The only critical points in $\mathcal{M}(p,q)$ are those equal to both p and q (because they are invariant under the flow), so if $p \neq q$ then $\mathcal{M}(p,q)$ has no critical points. Thus any $x \in \mathcal{M}(p,q)$ is not critical and then $f \circ \theta^x$ is decreasing in t, is strictly so as none of the points $\theta^x(t)$ are critical, so if $\theta(t,x) = \theta(t',x)$ then $f(\theta(t,x)) = f(\theta(t',x))$ which implies $t = t'$.

To show that the action is proper, recall that $f(q) < f(p)$ because f decreases along the flow lines. For any $x \in \mathcal{M}(p,q)$ then we have $f(q) < f(x) < f(p)$, and because the limits of $f\theta^x(t)$ are $f(p), f(q)$, by the mean value theorem for any $f(q) < c < f(p)$ there is a unique t such that $f\theta^x(t) = c$. Choose any regular value c and define $S = \mathcal{M}(p,q) \cap f^{-1}c$, which is a submanifold. Then the map

$$\mathbb{R} \times S \to \mathcal{M}(p,q), \qquad (s,x) \mapsto \theta(s,x)$$

is a bijective smooth map, and is seen to be a diffeomorphism. Then $\mathbb{R}$'s action on $\mathcal{M}(p,q)$ has the form

$$t \cdot (s,x) = (t+s,x),$$

and so if $K \subseteq \mathcal{M}(p,q)^2 \cong (\mathbb{R} \times S)^2$ is compact, the preimage of $(t,s,x) \mapsto ((t+s,x),(s,x))$ is compact in S and bounded in $\mathbb{R}$, so compact as well. Thus the action is proper. ■

Thus for critical $p \neq q$, the quotient

$$\mathcal{L}(p,q) = \mathcal{M}(p,q)/\mathbb{R}$$

is a smooth manifold of dimension $\operatorname{ind}(f;p) - \operatorname{ind}(f;q) - 1$, called the **space of trajectories**. Thus $\mathcal{L}(p,q)$ is nonempty only when $\operatorname{ind}(p) > \operatorname{ind}(q)$. Considering the proof above we see that $\mathcal{L}(p,q)$ is diffeomorphic to $\mathcal{M}(p,q) \cap f^{-1}c$ for $f(q) \leq c \leq f(p)$ regular. It can be viewed as a smooth structure on the set of all (maximal) integral curves of X whose limits are p, q.

The existence of a pseudo-gradient satisfying the Smale condition is non-trivial and we omit the proof below.

Theorem 2.1.3.9

Every Morse function possesses a pseudo-gradient which satisfies Smale condition.

Definition 2.1.3.10

Let f be a Morse function and X a Smale pseudo-gradient. The **space of broken trajectories** is the set

$$\overline{\mathcal{L}}(p,q) = \bigcup_{p=p_0,\ldots,q=p_n \in \operatorname{Crit} f} \mathcal{L}(p_0,p_1) \times \cdots \times \mathcal{L}(p_{n-1},p_n).$$

Which will be endowed with a topology.

Let $\lambda = (\lambda_1, \ldots, \lambda_n)$ be a broken trajectory connecting $p_i \in \operatorname{Crit} f$. Thus λ_i exits $\Omega(p_{i-1})$ and enters $\Omega(p_i)$ (Morse neighborhoods) and $f \circ \lambda_i$ is decreasing. Let U_{i-1}^- be a neighborhood of the exit point from $\Omega(p_{i-1})$ which is contained in its (the exit point's) level set. Similarly U_i^+ is a neighborhood of the entrance point to $\Omega(p_i)$ contained in its level set. Let U^-, U^+ be the collections of U_i^- and U_i^+ respectively. And we define $\mathcal{W}(\lambda, U^-, U^+)$ to be the set of all broken trajectories $\mu = (\mu_1, \ldots, \mu_k)$ where there are integers $0 < i_0 < \cdots < i_k = n$ such that

(1) For all $j \leq k$, $\mu_j \in \mathcal{L}(p_{i_j}, p_{i_{j+1}})$.

(2) μ_j exits the charts $\Omega(p_{i_j}), \ldots, \Omega(p_{i_{j+1}-1})$ through the interiors of the corresponding U^-, and enters $\Omega(p_{i_j+1}), \ldots, \Omega(p_{i_{j+1}})$ through the corresponding charts in U^+.

Then this collection of $\mathcal{W}(\lambda, U^-, U^+)$ forms a basis for a topology on $\overline{\mathcal{L}}(p,q)$. Furthermore this topology agrees with the topology on $\mathcal{L}(p,q) \subseteq \overline{\mathcal{L}}(p,q)$.

Notice that if $\operatorname{ind}(p) = \operatorname{ind}(q) + 1$ then $\overline{L}(p,q) = \mathcal{L}(p,q)$ as the space of trajectories is only nonempty when the indices decrease. Thus for any other critical point c, one of $\mathcal{L}(p,c)$ or $\mathcal{L}(c,q)$ must be empty.

Proposition 2.1.3.11

The space $\overline{\mathcal{L}}(p,q)$ is compact.

Proof

Omitted.

Corollary 2.1.3.12

If $\operatorname{ind}(p) = \operatorname{ind}(q) + 1$ then $\mathcal{L}(p,q)$ is a finite set.

Proof

By the above discussion, $\mathcal{L}(p,q) = \overline{\mathcal{L}}(p,q)$ and is thus a compact manifold of dimension 0, and thus finite. ■

We now turn our focus to the Morse-Witten (or just Morse) complex of a Morse function (along with a Smale pseudo-gradient).

Important:
For the remainder of the discussion regarding Morse theory, the underlying manifold is assumed to be compact.

Definition 2.1.3.13

For a Morse function f and a Smale pseudo-gradient X, define $\operatorname{Crit}_k(f)$ to be the set of critical points of f of index k. Then define $C_k(f)$ to be the free $\mathbb{Z}/2$-module generated by $\operatorname{Crit}_k(f)$:

$$C_k(f) = \left\{ \sum_{p \in \operatorname{Crit}_k f} \alpha_p p \;\middle|\; \alpha_p \in \mathbb{Z}/2 \right\}.$$

These $C_k(f)$s form a chain complex with differential

$$\partial_X \colon C_k(f) \to C_{k-1}(f), \qquad \partial_X(p) = \sum_{q \in \operatorname{Crit}_{k-1} f} |\mathcal{L}(p,q)| q,$$

where $|\mathcal{L}(p,q)|$ denotes the (finite) cardinality of $\mathcal{L}(p,q)$ modulo $\mathbb{Z}/2$. This chain complex is called the **Morse-Witten complex** (or **Morse complex**).

The assumption that the manifold is compact ensures that ∂_X is well-defined, as $\operatorname{Crit}_k(f)$ is finitely-generated and thus all of the sums above are finite.

It is not a priori clear why this forms a chain complex. The double composition of ∂_X is, for $p \in \operatorname{Crit}_{k+1}(f)$,

$$\partial_X \circ \partial_X(p) = \sum_{q \in \operatorname{Crit}_{k-1} f,\ r \in \operatorname{Crit}_k f} |\mathcal{L}(p,r)||\mathcal{L}(r,q)| q.$$

The terms in this sum are subsets of $\overline{\mathcal{L}}(p,q)$, we can rewrite it as

$$= \sum_{q\in \mathrm{Crit}_{k-1} f} \left| \bigsqcup_{r\in \mathrm{Crit}_{k-1} f} \mathcal{L}(p,r)\times\mathcal{L}(r,q) \right| q,$$

this is $\overline{\mathcal{L}}(p,q)$ without $\mathcal{L}(p,q)$.

We will prove that this union is the boundary of $\overline{\mathcal{L}}(p,q)$:

$$\partial\overline{\mathcal{L}}(p,q) = \bigsqcup_{r\in \mathrm{Crit}_k f} \mathcal{L}(p,r)\times\mathcal{L}(r,q).$$

Furthermore we will show $\overline{\mathcal{L}}(p,q)$ is a compact manifold of dimension 1. Because all connected compact 1-manifolds are diffeomorphic to either $[0,1]$ or S^1, and thus all compact 1-manifolds are disjoint unions of copies of the unit interval and the 1-sphere, their boundaries have an even number of points. Thus if we can prove these two things we will have finished.

We just need to show that we can provide $\overline{\mathcal{L}}(p,q)$ with charts about points in $(\lambda_1,\lambda_2)\in\mathcal{L}(p,r)\times\mathcal{L}(r,q)$ which are diffeomorphic to the half-spaces $[0,\delta)$. So we will define a continuous embedding

$$\psi\colon [0,\delta)\to\overline{\mathcal{L}}(p,q)$$

such that

$$\psi(0)=(\lambda_1,\lambda_2), \qquad \psi(s)\in\mathcal{L}(p,q) \text{ for } s>0,$$

and ψ is smooth on $(0,\delta)$. This can be done.

Definition 2.1.3.14

The chain complex $(C_\bullet(f),\partial_X)$ gives rise to homology modules denoted

$$\mathrm{HM}_k(f,X;\mathbb{Z}/2) = H_k(C_\bullet(f),\partial_X) = \frac{\ker\partial_X}{\operatorname{im}\partial_X}.$$

These modules are called the manifold's **Morse homology**.

Remark 2.1.3.15

This construction can be altered, with some care, to give homology modules over $\mathbb{Z}$. First we discuss the notion of co-orientations. Recall that an orientation for a vector space is a choice of equivalence class of bases B under the equivalence $B\sim B'$ iff $\det_B B' > 0$ (which denotes the determinant of the transition matrix from B' to B). Now let $U\le V$ be a subspace and choose a basis B_0 for U. Define an equivalence on the bases of complements of U by

$$B\sim B' \iff \det_{(B,B_0)}(B',B_0)>0.$$

This equivalence is independent on choice of B_0, as the above determinant is equal to the determinant of the projection of B' onto B with respect to B. A choice of equivalence class is called a **co-orientation of** U. Note that a co-orientation is the same as an orientation of the quotient V/U.

Note that if $U\oplus W=V$, then an orientation of any two of the spaces defines an orientation of the third. If B_U,B_W are oriented bases of U,W, then (B_U,B_W) defines an orientation of V (in a consistent manner, independent of B_U,B_W). Similarly if B,B_U are oriented bases of V,U, then say B_W is oriented iff (B_U,B_W) is oriented relative to B. Thus an orientation and co-orientation of U determines an orientation of V.

If U,W are transversal (meaning $U+W=V$), U is oriented, and W is co-oriented, then $U\cap W$ is oriented. This is because $V=W\oplus U/U\cap W$, which means that $U/U\cap W$ is oriented, and because U is oriented as well, $U\cap W$ is too.

If $p\in\mathrm{Crit}\, f$ is critical, then $W^s(p)$ is diffeomorphic to a disk and thus orientable. Choose for each critical point an orientation for $W^s(p)$. This amounts to choosing a coorientation for $W^u(p)$, this is because $T_pW^s(p), T_pW^u(p)$ are complementary, so this co-orients $T_pW^u(p)$ and thus all of $W^u(p)$.

Thus $W^u(p) \cap W^s(q)$ is oriented, as the intersection of an oriented and co-oriented manifold. Note that a regular level set is co-oriented by any vector field which does not vanish outside of the level set. Thus a pseudo-gradient X co-orients a regular level set, and since $\mathcal{L}(p,q) = W^u(p) \cap f^{-1}c$ for c regular (between fp, fq) we see that $\mathcal{L}(p,q)$ is oriented as well.

Now we define $C_k(f)$ to be the $\mathbb{Z}$-module freely generated by $\mathrm{Crit}_k(f)$. Now when $\mathrm{ind}(p) = \mathrm{ind}(q) + 1$ we have that $\mathcal{L}(p,q)$ is an oriented compact manifold of dimension 0, thus it is a finite set of points each with a sign (an orientation). Let $N_X(p,q)$ be the sum of these signs, and define the differential

$$\partial_X \colon C_k(f) \to C_{k-1}(f), \qquad \partial_X(p) = \sum_{q \in \mathrm{Crit}_{k-1}(f)} N_X(p,q) q.$$

Note that modulo 2, $|\mathcal{L}(p,q)| \equiv N_X(p,q)$. To verify that $\partial_X^2 = 0$, we follow the above proof but considering orientations.